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Implied volatility is (almost) past-dependent: Linear vs non-linear models

Conghua Wen [a], Jia Zhai [b,*], Yinuo Wang [a], Yi Cao [a]

[a] Department of Financial and Actuarial Mathematics, Xi'an Jiaotong-Liverpool University, Wuzhong District, Suzhou, China
[b] International Business School, Xi'an Jiaotong-Liverpool University, Wuzhong District, Suzhou, China

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ABSTRACT

We explore and demonstrate a clear pattern of past-dependence in predicting implied volatility, which extends up to twenty days and is present in both linear and nonlinear models. Empirically, we find that linear models utilizing precedent values up to 20 days explain up to 80% of the variance in the S&P 500 index, while nonlinear models explain 78%. Furthermore, we observe that L2 regularization, as opposed to L1, enhances the performance of linear models and underscores the importance of past-dependence. As a result, a simple linear model with L2 regularization, incorporating precedent values up to the past 20 days, consistently outperforms both linear and nonlinear benchmark models when applied to the S&P 500 index option and SSE 50 ETF option over the past decades. Additionally, we uncover that the impact of all preceding values exists but consistently diminishes over time in all empirical studies involving past-dependence settings. Finally, we demonstrate that these forecasts yield profitable outcomes when implementing a simple long-short portfolio trading strategy. This study primarily investigates the 'path-dependency' of implied volatility in financial markets and further confirms that simple models can sometimes surpass complex nonlinear models in predicting volatility.

1. Introduction

Implied volatility, a central metric in options pricing, risk hedging, and broader portfolio construction, encapsulates critical dimensions of market sentiment and uncertainty. These underpin trading strategies, derivative valuation, and financial risk management. Over time, this fundamental concept has drawn substantial focus from both academic researchers and industry practitioners, reflecting its pivotal role across diverse aspects of the financial markets. The implied volatility forecasting model typically relies on two main strands: the stochastic volatility (SV) model, which considers the volatility as a stochastic process (Bates, 1996; Bollerslev & Wooldridge, 1992; Jacquier et al., 2002; Kim et al., 1998; Scott, 1987), and the economic variable model, which selects relevant predictors including the past volatility (Dunis et al., 2013), past return (Guyon & Lekeufack, 2023), the deterministic variables of moneyness, maturity time (Bernales & Guidolin, 2014), as well as macroeconomic and financial market related indicators (Li et al., 2023; Vrontos et al., 2021). These predictors are typically incorporated using deterministic linear model (Dunis et al., 2013) or quadratic formats (Bernales & Guidolin, 2014). Economic variable models not only acknowledge various sources of randomness impacting volatility but also explicitly recognize that predicted implied volatility depends on past underlying returns and volatility. This area of research has been relatively unexplored, with only a handful of studies delving into this domain. Among these studies, Konstantinidi et al. (2008) suggested that when examining implied volatility indices in the EU and USA, the vector autoregressive (VAR) model with up to ten lags consistently outperforms all other models, including the ARIMA and ARFIMA models. In a related investigation, Dunis et al. (2013) delved into the 48-lag dependence patterns within 30-minute high-frequency data of FX option implied volatility. They concluded that the VAR model, the ordinary least squares (OLS) model, and the linear regression model incorporating economic variables consistently exhibit superior forecasting performance across a range of maturities. Furthermore, while Niu et al. (2023) explored the pivotal role of geopolitical risk in predicting realized volatility, it remains evident that the significance of the most recent lags continues to overshadow the majority of predictors. Other demonstrations showing the predictive potential of lags have been examined using advanced models for forecasting either the value (Chalamandaris & Tsekrekos, 2014; Kearney et al., 2019) or the direction (Vrontos et al., 2021) of future implied volatility. These empirical studies demonstrate that implied volatility depends on the path followed by asset prices in the recent past, a characteristic that can be termed 'path-dependent' in financial markets. This aspect has been demonstrated in the studies conducted by Guyon (2014) and

* Corresponding author.
E-mail addresses: Conghua.Wen@xjtlu.edu.cn (C. Wen), jia.zhai@xjtlu.edu.cn (J. Zhai), yinuo.wang1902@student.xjtlu.edun.cn (Y. Wang), jason.caoyi@gmail.com (Y. Cao).

[1] See the works of path-dependent patterns in forecasting the realized volatility (Blanc et al., 2017; Chicheportiche & Bouchaud, 2014).

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Guyon and Lekeufack (2023).[1] In other words, the evolution of asset values is influenced not only by current market conditions but also by a complex connection to future volatility through a cumulative and feedback process, linking past actions to future fluctuations.

Expanding upon these findings, we identify two key issues that remain unexplored in the above research. First, does the lagged information of implied volatility itself possess significant predictive power? (Guyon & Lekeufack, 2023) explore how implied volatility depends on the path of past asset returns but does not address volatility itself. Option implied volatility encompasses both investor preferences and beliefs regarding the future fluctuations in the underlying asset's returns. In its essence, it encapsulates comprehensive information about expected market risk and investors' preferences concerning these risks. This makes the lagged implied volatility the potentially foremost predictor, encompassing all aspects of relevant risk that influence the future trajectory of implied volatility. Therefore, it is crucial to explore its path-dependency. Second, the process of predicting implied volatility is primarily driven by its inherent past dependency or the influence of a specific type of model. A model designed to capture inherent dependencies, which operates through a mechanism analogous to past dependence, has been effectively demonstrated by the Heterogeneous Autoregressive (HAR) model (Corsi, 2009). The HAR model demonstrates remarkable predictive powers by employing a straightforward linear structure to model dependencies across various volatility terms, including short, medium, and long. Such structure has been demonstrated to effectively capture the future realized volatility across a variety of applications (Bollerslev et al., 2016, 2018; Clements & Preve, 2021; Liang et al., 2023). The widely recognized HAR model underscores the potential of the simple dependence-capturing model as a powerful tool for capturing variations in implied volatility as well. Additionally, we aim to explore whether machine learning models, due to their ability to capture nonlinear relationships, perform better in forecasting than conventional linear models. Gu et al. (2020) demonstrated that neural networks far outperform linear models in predicting returns, yet (Guyon & Lekeufack, 2023) found the linear Lasso model to be optimal when studying implied volatility. Therefore, while exploring the path-dependency of implied volatility itself, we aim to determine whether linear or nonlinear models are more suitable for predicting implied volatility.

Building upon the natural research questions outlined above and the theoretical rationale behind them, the contribution of this paper is threefold. First, we demonstrate a consistent appearance of past-dependence in the forecasting of implied volatility through both linear and non-linear models. To assess the impact of lag lengths on dependence, we use historical implied volatilities for three distinct lengths: 5, 10, and 20 days, each serving as an individual input vector for our model. This approach seeks to elucidate the relationship between forecasting accuracy and the sequence of historical data used. Empirical results consistently demonstrate a persistent and monotonic pattern, wherein longer historical sequences lead to higher R^2 values. Simultaneously, we utilize the machine learning model XGBoost to explore feature importance. The model not only focuses on the statistical attributes of the features but also on their actual performance and contribution to constructing decision tree models. We ultimately find that the historical implied volatility from the previous one moment consistently emerges as the most important feature across different scenarios.

Importantly, this pattern consistently emerges in both linear and non-linear models. Furthermore, this dependence persists for up to 20 days and gradually decays over time. Secondly, we demonstrate that the linear model structure is sufficient to predict implied volatility when exploiting past dependence. Remarkably, a simple linear model with L2 regularization can explain up to 80% of the variance in implied volatility in out-of-sample forecast (varying depending on the index option and interpolation method). While advanced nonlinear models, such as deep neural networks, can also explain as much as 78% of the variance in out-of-sample data, their performance is comparable to, but slightly lower than, that of the linear model. Thirdly, we also demonstrate that of equal importance to both linear models and past dependence are the L2 regularization method applied to linear models and the linear interpolation approach used for implied volatility. The discovered path-dependence is relatively simple: the implied volatility is essentially a weighted average of its preceding values within the same term structure. This brings two critical considerations: the need for interpolating preceding values to construct a time series of implied volatility across the term structure, and the importance of penalizing model coefficients to enhance generalization. We demonstrate that linear interpolation significantly outperforms nonlinear interpolation methods across the results generated by nearly all linear and nonlinear models. Furthermore, we show that the L2 regularization method adeptly strikes a balance between bias and variance by progressively reducing the weights assigned to nearly all preceding volatilities over time. This approach significantly outperforms L1 regularization (as well as L1 & L2 mixed regularization) in out-of-sample, which reduces past-dependence by driving specific preceding weights to zero, consequently resulting in lower performance.

In summary, our empirical research leads to two important conclusions: compared to previous studies, we confirm that implied volatility is past-dependent, and increasing the length of historical data improves the final prediction results. Additionally, we find that simple linear models outperform complex machine learning and neural network models in predicting implied volatility. This analysis offers further perspective on the choice between straightforward, less precise linear models and complex, opaque nonlinear models. Despite the common belief that machine learning surpasses traditional models in forecasting accuracy, our research reveals the contrary: in financial markets, simpler models can surprisingly outperform in predicting certain assets. This pattern also aligns with recent discoveries in Bali et al. (2023) and Gu et al. (2020), wherein simpler structured neural networks exhibit enhanced performance in diverse asset pricing applications. Drawing upon these contributions, our findings propose a remarkably straightforward strategy for forecasting implied volatility. This approach includes linearly interpolating implied volatility across term structures when necessary, fitting a basic linear model with L2 regularization, and incorporating preceding volatilities with the same term structure, spanning at least 20 days. By following this approach, a notably high R^2 is highly likely to be achieved. This simple approach is a useful complement to the existing literature on implied volatility forecasting.

Our aforementioned contributions and findings stem from a comprehensive empirical study that utilizes the widely recognized S&P 500 index options over the past decade, spanning from 2012 to 2022. Additionally, we include the options of the Shanghai Stock Exchange 50 Exchange Traded Fund (SSE 50 ETF), one of the most liquid options in emerging markets, covering the period from its inception in 2015 to the end of 2022. This extensive selection of two options aims to provide a comprehensive evaluation of past-dependence across both a highly liquid and well-established market and a relatively illiquid and emerging market. Our findings for both options exhibit a consistent overall trend that strongly reinforces the primary contributions of this study, albeit with slight variations attributed to their distinct natures. Finally, we develop a trading strategy based on the implied volatility forecasted by the models. We trade options' contracts by comparing daily market implied volatilities with predicted values. Ultimately, trading based on Ridge model forecasts yields favorable Sharpe ratios and returns. This underscores the feasibility of using historical implied volatility as a predictive indicator and linear model forecasts, further proving the path-dependency of implied volatility.

The paper is structured as follows. In Section 3, we discuss the data and the constructed feature matrix. Section 4 presents the prediction models, while Section 5 offers insights from empirical studies. The significance of features is discussed in Section 6. The outcomes of the trading strategies are presented in Section 7, and finally, Section 8 provides the concluding remarks.

2. Relevant literature

The prediction of the implied volatility has been analyzed in various papers. Kim et al. (1998) enhance stochastic volatility prediction by efficiently combining offset mixture models with importance reweighting, demonstrating improved accuracy over traditional GARCH models. Jacquier et al. (2002) employ a cyclic Metropolis algorithm for stochastic volatility analysis in Markov chain simulations, achieving superior accuracy in parameter estimation and filtering compared to traditional methods, with applications to both daily and weekly data. Economic variable models emerge as an alternative approach for predicting implied volatility. Ahoniemi (2006) devises an economic indicator model incorporating both macro and micro variables, utilizing linear regression for forecasting implied volatility. Konstantinidi et al. (2008) enhances their analysis by integrating the previous day's historical implied volatility as a study variable, comparing the efficacy against VAR, PCA, ARIMA, and ARFIMA models, with VAR showing superior interval forecasting performance. Dunis et al. (2013) builds on this foundation, revealing that VAR and linear regression models, when including economic variables, adeptly capture 48-lag patterns in high-frequency data. Research in this field predominantly employs traditional linear econometric models, utilizing past implied volatility as a feature to demonstrate its impact on forecasting accuracy. However, specific investigations into this attribute are scarce, with a lack of emphasis on empirically studying the "past-dependence" characteristic of implied volatility. Guyon (2014) investigate path-dependent volatility models that merge the market-wide applicability of local volatility models with the intricate dynamics of stochastic volatility models, capturing a broad dynamic range and significant historical volatility patterns. Additionally, Guyon and Lekeufack (2023) delved into the 'path' between returns and implied volatility, applying both linear and nonlinear models to find that historical returns can explain up to 90% of the variance in implied volatility. They also introduce a continuous-time path-dependent volatility framework, effectively capturing volatility patterns closely aligned with market trajectories. Drawing on their insights, our study expands the investigation to ascertain whether implied volatility also exhibits 'past-dependence' on its past values.

In recent years, research on implied volatility has expanded, predominantly using historical information as a feature or extracting features from implied volatility for prediction. Initially, studies like those by Chalamandaris and Tsekrekos (2014) and Kearney et al. (2019) explored using linear models, adopting diverse approaches to extract factors from the implied volatility surface, and then employing varying lag lengths and linear time-series models for prediction. Moreover, Kearney et al. (2018) developed innovative Functional Time Series (FTS) techniques for analyzing and forecasting foreign exchange implied volatility, modeling surface evolution as univariate series and highlighting sequential observation dependencies. Shang and Kearney (2022) introduced dynamic, multivariate, and multilevel functional time-series extensions for modeling and forecasting currency option implied volatility, finding that dynamic univariate FTS methods more effectively captured the time-series momentum observed in daily implied volatility surface data. With the advancement of machine learning, complex nonlinear models have been adopted. Chen and Zhang (2019) treated discrete points of the implied volatility smile as a collective forecasting target, extracting daily, weekly, and monthly option implied volatilities as input features, establishing an LSTM-Attention model where the LSTM component effectively captured the long memory characteristic of volatility. Zhang et al. (2023) proposed a two-step framework for predicting temporal changes in the implied volatility surface without static arbitrage, initially extracting surface characteristics, followed by prediction using an LSTM model and construction of the forecasted surface with a DNN model. However, these studies have conducted empirical research on implied volatility from either a linear or nonlinear model perspective without comparing the two types. Testing the performance of linear and nonlinear models has become increasingly common in financial research in recent years. For instance, Gu et al. (2020) in the asset pricing domain have compared different linear and nonlinear models for predicting returns, finding that neural network models significantly outperform linear models. Additionally, Chen et al. (2024) confirmed that leveraging deep learning models to extract temporal features from macroeconomic time series significantly enhances prediction outcomes. However, Guyon and Lekeufack (2023) found that the Lasso linear model excels in predicting the relationship between returns and implied volatility. Based on these findings, our research employs both linear and nonlinear models to examine whether implied volatility exhibits 'past-dependence' characteristics and compares the predictive outcomes of these model types. We aim to determine which model demonstrates consistently superior performance in volatility prediction. Additionally, several studies employ machine learning and econometric models to examine how various factors contribute to predicting and understanding volatility. For instance, research by Guo et al. (2021) and Liang et al. (2024) has demonstrated that changes in market regulation, such as adjustments to price limits, impact asset volatility. Furthermore, He et al. (2023) has confirmed the relationship between economic fluctuations and systemic risks in the banking sector. Lastly, research by Luo et al. (2024), Wan et al. (2024), Wu et al. (2023), and Zhang et al. (2022) clarified how economic conditions and external factors influence the volatility of different sectors, including stocks and commodities like crude oil.

3. Data and feature matrix

3.1. Summary statistics

We obtain daily S&P500 and SSE 50 ETF options data from OptionMetrics IvyDB and Tonghuashun databases, respectively. The original datasets include option prices, option expiration dates, strike prices, and implied volatility calculated using the Black–Scholes formula. Our S&P500 sample covers the period from January 2012 to December 2022, while the SSE 50 ETF sample extends from February 2015 to December 2022. We filter the raw options data based on the work of Cao et al. (2020). In our initial step, we select options endowed with a comprehensive dataset, rigorously excluding those bereft of essential information such as implied volatility and critical Greek values. Following this precise selection criterion, we proceed to eliminate any options from our dataset that demonstrate a lack of market activity, specifically those with a trading volume of zero over the preceding seven calendar days. This step ensures that our analysis is grounded in options that exhibit tangible market engagement. Further refining our dataset, we apply a systematic exclusion criterion based on the option's remaining lifespan. Specifically, we remove options whose lifespan falls short of 14 days or overshoots 730 days, aligning our focus on options that are neither too fleeting in their market presence nor excessively distant in their maturity. This meticulous selection process results in a distilled dataset comprising 328,755 call options and 5,375,900 put options. We calculate two pivotal metrics for each option contract: m and τ. The metric m is calculated by dividing the option's strike price by its forward price, offering insight into the option's relative costliness. On the other hand, τ, representing the time to expiration normalized over a year, is calculated by dividing the option's remaining days by 365. This normalization allows us to compare the time value across options with differing expiration dates on a standardized basis. These calculated metrics, m and τ, are instrumental in furthering our understanding of the option's pricing dynamics and its temporal characteristics within the market.

Table 1 reports descriptive statistics for two samples. In both samples, put options outnumber call options, and the S&P500 sample size is significantly larger than that of SSE 50 ETF. In addition, Table 2 provides a systematic categorization of options based on term structure attributes and reports their respective sizes. The table is divided into

Table 1
The description of data samples. The table reports descriptive statistics for S&P500 and SSE 50 ETF options. Our original implied volatility data is sourced from the OptionMetrics IvyDB and Tonghuashun database. Our S&P500 sample covers the period from January 2012 to December 2022, while the SSE 50 ETF sample extends from February 2015 to December 2022. The data after filtering are presented in the table. m is calculated as the strike price divided by the forward price, while τ is determined by dividing the remaining days of the option by 365.

	Count	Mean	Std.	Min	25%	50%	75%	Max
Panel A: S&P500 option								
Call options								
Moneyness	3,285,755	1.0340	0.0978	0.0210	0.9926	1.0265	1.0689	2.3068
Time-to-maturity	3,285,755	0.2161	0.2621	0.0384	0.0767	0.1260	0.2384	2.0000
Implied volatility	3,285,755	0.1755	0.0915	0.0447	0.1185	0.1610	0.2112	2.9986
Put options								
Moneyness	5,375,900	0.8819	0.1476	0.0210	0.8230	0.9193	0.9791	2.4273
Time-to-maturity	5,375,900	0.2134	0.2483	0.0384	0.0767	0.1342	0.2411	2.0000
Implied volatility	5,375,900	0.2776	0.1516	0.0218	0.1813	0.2451	0.3259	2.9967
Panel: B SSE 50 ETF option								
	Count	Mean	Std.	Min	25%	50%	75%	Max
Call options								
Moneyness	90,017	1.0249	0.1167	0.6420	0.9507	1.0145	1.0870	1.9088
Time-to-maturity	90,017	0.2635	0.1776	0.0384	0.1096	0.2164	0.4055	0.6685
Implied volatility	90,017	0.2330	0.0997	0.0134	0.1764	0.2116	0.2664	1.3280
Put options								
Moneyness	104,336	1.0033	0.1189	0.6420	0.9240	0.9935	1.0704	1.9088
Time-to-maturity	104,336	0.2612	0.1765	0.0384	0.1096	0.2137	0.4000	0.6685
Implied volatility	104,336	0.2776	0.1300	0.0796	0.2031	0.2437	0.3130	2.4529

two sections: one for call options and another for put options. Within the m category, call options include DITM, ITM, ATM, OTM, DOTM, while put options consist of ITM, ATM, OTM, DOTM, and VDOTM. Regarding τ, both call and put options are categorized into short-term, medium-term, and long-term options.[2] Additionally, we categorize options based on their τ into short-term, medium-term, and long-term options. From the perspective of m, OTM options are the most numerous in both samples. For call options, DITM options are the least common. In the case of put options, VDOTM options are rarest, with none appearing in the original SSE 50 ETF sample. Additionally, short-term options are more prevalent than long-term ones.

3.2. Implied surface interpolate

On any day, implied volatilities are calculable only for specific pairs m and time-to-maturity τ, matching the options traded on that day. Given a set of implied volatilities $\{\sigma(m_i, \tau_i) : i = 1, \dots, n\}$ for a particular day, there are various methods to interpolate these data points to create a complete Implied Volatility Surface (IVS). Following the approach outlined in Zhang et al. (2023), we employ two interpolation methods for the IVS, encompassing both linear and nonlinear techniques.

Initially, we adopt a well-recognized parametric approach, commonly known as the deterministic volatility function (DVF) model, following the methodology of Dumas et al. (1998). This model hypothesizes that $\sigma(m, \tau) = \max(0.01, a_0 + a_1\, m + a_2\tau + a_3 m^2 + a_4\tau^2 + a_5 m\tau)$, ensuring that implied volatility does not fall below a minimum threshold of 0.01, thereby avoiding impractically low or negative figures. The model parameters $a_0, \dots, a_5$ are computed via regression analysis.

We also utilize a widely-used non-parametric approach for constructing the complete implied volatility surface, based on the NW estimator, as outlined in Härdle (1990). The formula for this estimator is given by:

$$\bar{\sigma}_t(m, \tau) = \frac{\sum_{i=1}^{n} \sigma_t(m_i, \tau_i) g(m - m_i, \tau - \tau_i)}{\sum_{i=1}^{n} g(m - m_i, \tau - \tau_i)},$$

Table 2
Option classification statistics. This table systematically categorizes two options based on term structure attributes and reports their size. Our original data is sourced from the OptionMetrics IvyDB database. Our original implied volatility data is sourced from the OptionMetrics IvyDB and Tonghuashun database. Our S&P500 sample covers the period from January 2012 to December 2022, while the SSE 50 ETF sample extends from February 2015 to December 2022. The data after filtering are presented in the table. m is calculated as the strike price divided by the forward price, while τ is determined by dividing the remaining days of the option by 365.

	Call options		Put options	
Panel A: S&P500 option				
Moneyness	DITM	43,626	ITM	589,298
	ITM	726,007	ATM	486,426
	ATM	475,607	OTM	3,151,822
	OTM	1,922,791	DOTM	857,933
	DOTM	131,578	VDOTM	290,421
Time-to-maturity	Short-term	2,488,308	Short-term	4,038,948
	Medium-term	510,736	Medium-term	901,887
	Long-term	286,711	Long-term	435,065
Panel B: SSE 50 ETF option				
	Call options		Put options	
Moneyness	DITM	1,815	ITM	46,056
	ITM	34,037	ATM	7,524
	ATM	7,387	OTM	47,659
	OTM	41,098	DOTM	3,097
	DOTM	6,375	VDOTM	0
Time-to-maturity	Short-term	48,081	Short-term	56,076
	Medium-term	28,007	Medium-term	32,695
	Long-term	13,929	Long-term	15,565

[2] For call options, these classifications encompass:
Deep In-The-Money (DITM) options with m values of 0.80 or less. In-The-Money (ITM) options falling within the m range of (0.80, 0.99]. At-The-Money (ATM) options with m values in the range (0.99, 1.01]. Out-of-The-Money (OTM) options with m values within the range (1.01, 1.21). Deep Out-of-The-Money (DOTM) options with m values exceeding 1.2. Similarly, for put options, our classifications include:
Deep In-The-Money (DITM) options with m values of 0.80 or less. In-The-Money (ITM) options falling within the m range of (0.80, 0.99]. At-The-Money (ATM) options with m values in the range (0.99, 1.01]. Out-of-The-Money (OTM) options with m values within the range (1.01, 1.21). Deep Out-of-The-Money (DOTM) options with m values exceeding 1.2. Additionally, we categorized options based on their τ as follows: Short-term options, with a remaining term of less than 90 days. Medium-term options, having a remaining term between 90 and 180 days. Long-term options, with a remaining term exceeding 180 days.

in which $g(x, y) = \frac{1}{2\pi} \exp\left(-\frac{x^2}{2h_1}\right) \exp\left(-\frac{y^2}{2h_2}\right)$ denotes the Gaussian kernel function. This method is controlled by two bandwidth parameters, h_1 and h_2, that are critical in determining the extent of smoothing. If these parameters are set too low, the resulting surface might appear uneven and overfit, while overly large values could lead to over-smoothing, potentially obscuring crucial patterns in the data. To optimize the values of (h_1, h_2), we implement a five-fold cross-validation process daily, choosing from a pre-established range of possible values.

3.3. Input feature matrix

We construct a feature matrix as the input for the hybrid model and assume the IV at day t is determined by m, τ, and historical sequence ($L = l$). More specifically, our approach to matrix selection involves integrating the Dynamic Volatility Factor (DVF) with traditional econometric volatility models, thereby constructing a novel matrix for predicting implied volatility. DVF, introduced by Dumas et al. (1998), posits that implied volatility linearly depends on m and τ. Furthermore, given the autocorrelation, clustering, and long memory properties of implied volatility, some studies leverage econometric models, typically reliant on structures with selected lags. Therefore we draw upon these studies and construct a vector $\boldsymbol{X}_t^{m_1,\tau_1} = \left[m_1, \tau_1, IV_{t-1}^{m_1,\tau_1}, \ldots, IV_{t-l}^{m_1,\tau_1}\right]$ for the implied volatility $\boldsymbol{Y}_t^{m_i,\tau_j}$ of m_1 and τ_1 at day t.

We then collect implied volatilities $\boldsymbol{Y}_t^{m_i,\tau_j}$ with all available m_i and $\tau_j (i = 1, \ldots, I, j = 1, \ldots, J)$ at day t. To investigate the dependence impact across varying lag lengths, we select the values for l equal to $l = 5, 10, 20$, representing the implied volatility values over the preceding 5, 10, and 20 days, separately. As a result, we generate three distinct input vectors, each denoting historical sequences in dimensions of 7, 12, and 22, respectively. Finally, we construct a dataset N such matrix from day t to T to estimate the implied volatility:

$$\begin{Bmatrix}
m_1, & \tau_1, & IV_{t-1}^{m_1,\tau_1}, & \ldots, & IV_{t-l}^{m_1,\tau_1} \\
m_2, & \tau_1, & IV_{t-1}^{m_2,\tau_1}, & \ldots, & IV_{t-l}^{m_2,\tau_1} \\
 & \vdots & & & \\
m_I, & \tau_1, & IV_{t-1}^{m_1,\tau_1}, & \ldots, & IV_{t-1}^{m_I,\tau_1} \\
m_1, & \tau_2, & IV_{t-1}^{m_1,\tau_2}, & \ldots, & IV_{t-l}^{m_1,\tau_2} \\
 & & \vdots & & \\
m_I, & \tau_2, & IV_{t-1}^{m_I,\tau_2}, & \ldots, & IV_{t-l}^{m_I,\tau_2} \\
 & & \vdots & & \\
m_I, & \tau_J, & IV_{t-1}^{m_I,\tau_J}, & \ldots, & IV_{t-l}^{m_I,\tau_J}
\end{Bmatrix}$$

4. Prediction models

In our research, we select a suite of six distinct models: Lasso, Ridge, Elastic Net, XGBoost, DNN, and LSTM. This combination strategically capitalizes on the strengths of linear, tree-based, and deep learning methodologies. Our objective is to bolster the precision and dependability of our predictions from multiple analytical angles. Linear models rely on straight-line fits and linear relationships, whereas nonlinear neural networks use activation functions to model complex nonlinearities in data. Particularly, deep learning models are adept at identifying intricate patterns and temporal sequences in data. This capability is supported by findings from Gu et al. (2020), which highlight the superior performance of neural networks over linear models in asset pricing. Moreover, Chen et al. (2024) show significant prediction improvements by applying deep learning to analyze temporal features in macroeconomic time series. The rationale behind our model selection lies in their demonstrated proficiency in navigating the complex dynamics of financial data. Nevertheless, Guyon and Lekeufack (2023) uncover that the Lasso linear model stands out in forecasting interactions between returns and implied volatility. Thus, our investigation intends to delve deeper into the comparative effectiveness of linear versus nonlinear models in predicting implied volatility. This aspect is critical in options pricing, where the interplay between market conditions, specific option characteristics, and, most importantly, implied volatility, is intricate. Implied volatility, a forward-looking metric, is crucial for estimating an option's premium.

4.1. Penalized linear models

In this study, we initially employ the linear regression model to forecast the implied volatility, denoted as $\boldsymbol{Y}_t^{m_i,\tau_j}$. Assuming the length of the historical sequence is l, the model is expressed as:

$$\boldsymbol{Y}_t^{m_i,\tau_j} = \beta_0 + \beta_1 m_i + \beta_2 \tau_j + \beta_3 IV_{t-1}^{m_i,\tau_j} + \cdots + \beta_{l+3} IV_{t-l}^{m_i,\tau_j} + \epsilon_t,$$

where $\boldsymbol{Y}_t^{m_i,\tau_j}$ represents the predicted implied volatility at time t for given m_i and time- τ_j. The error term, ϵ_t, accounts for the random fluctuations not explained by the model. The intercept term, β_0, indicates the baseline implied volatility when all other features are zero. The coefficients β_1, β_2, and $\beta_3, \ldots, \beta_{l+3}$ reflect the impact of m, τ, and historical implied volatilities on the current prediction, respectively.

Given the varying number of input features in our model, specifically 7, 12, and 22, there is a need to mitigate overfitting and enhance generalization. To this end, we apply regularization techniques and compare three distinct methods: Lasso (Tibshirani, 1996), Ridge (Hoerl & Kennard, 1970), and Elastic Net regressions (Zou & Hastie, 2005). Regularization effectively mitigates overfitting by penalizing large coefficients, thereby restraining model complexity. This approach minimizes the likelihood of the model capturing noise from the training data and places greater emphasis on learning fundamental patterns.

L1 Penalty The Lasso regression extends ordinary least squares by adding a penalty based on the absolute value of the parameters (L1 regularization). This approach aids in feature selection by reducing the coefficients of less significant features to zero, thus simplifying the model and decreasing the number of features. The formula can be explained as: minimize $\left(\frac{1}{2n}\|y - Xw\|_2^2 + \alpha\|w\|_1\right)$, where y represents the target variable, X the predictor variables (features), w the coefficient vector, and α the regularization strength, controlling the impact of L1 regularization.

L2 Penalty Moreover, Ridge regression incorporates a penalty based on the square of the parameters (L2 regularization), directly building upon ordinary least squares. This method is particularly useful in addressing multicollinearity among features, reducing the risk of overfitting by shrinking the magnitude of the coefficients. It aids in simplifying model complexity without discarding any features, which is ideal for cases where the preservation of all features is crucial. The formula can be explained as: minimize $\left(\frac{1}{2n}\|y - Xw\|_2^2 + \alpha\|w\|_2^2\right)$, where α represents the regularization strength, governing the influence of L2 regularization.

L1+L2 Penalty Furthermore, Elastic Net regression combines the penalties of Lasso and Ridge, incorporating both L1 and L2 regularizations. This method is well-suited for scenarios involving multiple correlated features, as it simultaneously facilitates feature selection and addresses multicollinearity. The formula can be explained as: minimize $\left(\frac{1}{2n}\|y - Xw\|_2^2 + \alpha\rho\|w\|_1 + \frac{\alpha(1-\rho)}{2}\|w\|_2^2\right)$, where ρ is the mixing parameter, determining the relative contribution of L1 and L2 regularization. When $\rho = 1$, the model is equivalent to Lasso; when $\rho = 0$, it corresponds to Ridge.

4.2. Nonlinear model

4.2.1. Extreme Gradient Boosting (XGBoost)

In our study, we choose XGBoost (Chen & Guestrin, 2016) as the first nonlinear model. This model, a tree-based ensemble learning method, falls under the category of ensemble models in machine learning. Its

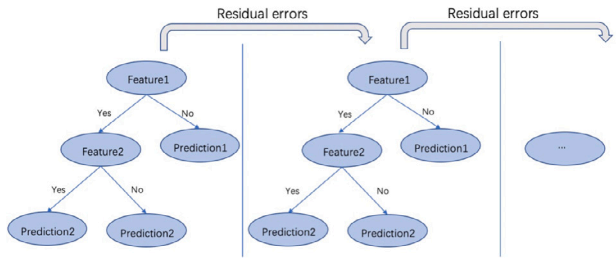


Fig. 1. The structure diagram of the XGBoost model. The figure depicts stages of an XGBoost model, which utilizes decision trees to make predictions. The left side shows an initial tree that bases its prediction on two features, leading to an outcome. The right side shows a subsequent tree that aims to correct the residual errors from the first, continuing the process to refine the predictions. Arrows leading to "Residual errors" indicate the model's iterative improvement by learning from previous mistakes.

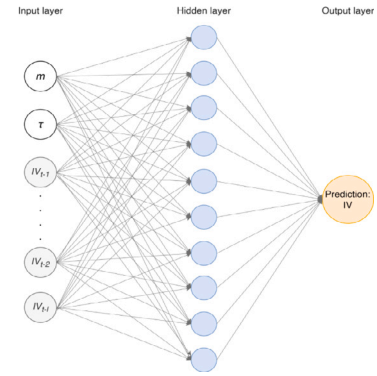


Fig. 2. The structure diagram of the DNN model. This figure illustrates a schematic of a simple neural network with a single hidden layer. The input layer, signified by circles, includes features such as m, τ, and historical implied volatility values (IV_{t-1}, …, IV_{t-l}). These feed into the hidden layer, which consists of ten nodes depicted by blue circles. Each connection between the input and hidden layer nodes represents a weighted parameter that adjusts the signal strength as it passes through the network. The hidden layer employs a nonlinear activation function to process the inputs before they are transmitted to the output layer. The output layer is represented by an orange circle labeled 'Prediction: IV', signifying the network's final prediction for implied volatility. This architecture enables the neural network to capture complex, nonlinear relationships within the data.

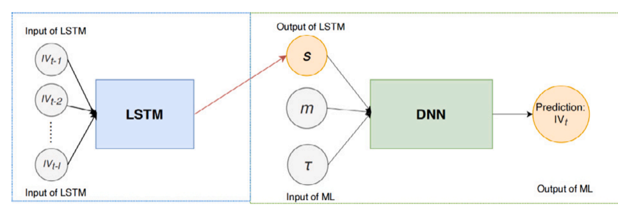


Fig. 3. The structure diagram of the new hybrid model. The hybrid model divides the input features into two components. The historical time series is fed into a double-layer LSTM model, which extracts a temporal feature(s). On the other hand, m and τ are processed separately using a machine learning model, which can be either a DNN or XGBoost, for prediction, resulting in the output implied volatility.

fundamental principle is gradient boosting, which involves the sequential addition of new trees, each aiming to correct the errors made by the previous tree. The enhancement at each step is guided by the gradient of the objective function, which typically comprises two parts: one reflecting the fit of the model to the training data and the other being a regularization term that controls the complexity of the model. Due to its capability to capture nonlinear relationships and complex patterns in data, XGBoost theoretically has the potential to predict implied volatility more accurately than linear models. Additionally, it is potentially more effective in handling complex interactions between features and can provide insights into the importance of features in the prediction process. Moreover, XGBoost is known for its robustness against outliers and noise, which might contribute to greater stability in the prediction of implied volatility. The architecture of our XGBoost model is depicted in Fig. 1.

4.2.2. Deep Neural Network (DNN)

We also employ deep neural networks as another nonlinear approach, widely considered among the most robust tools available in machine learning. DNNs offer an advantage over linear models by facilitating the estimation of complex non-linear functions with numerous parameters from extensive datasets. They incorporate multiple inputs known as features, and one or several outputs, referred to as targets. Drawing on the work of Gu et al. (2020), our focus lies on conventional neural networks.

In our Deep Neural Network (DNN) model, we incorporate three categories of features: m, τ, and the historical sequence ($L = l$). The model is designed with a singular objective: to predict implied volatility. These features form the input layer, while the prediction of implied volatility constitutes the output layer. To map the computations from the input to the output layer, we have devised a hidden layer consisting of ten neurons where intermediate computations are conducted. This structure is illustrated in Fig. 2. The leftmost layer in the figure represents the input layer, containing the values of the input variables, or features. The rightmost layer is the output layer, which holds the predicted values of implied volatility.

The DNN model is fundamentally a linear model that has been augmented with a series of nonlinear transformations. These transformations are executed through the deployment of activation functions, with the Rectified Linear Unit (ReLU) function being the chosen activator in our DNN model. Each layer within the network conducts a linear transformation followed by a nonlinear activation function. This amalgamation of linear and nonlinear processes endows DNNs with the capability to capture complex and sophisticated abstractions from the data, a feat that is beyond the reach of simple linear models. Consequently, by embracing depth and nonlinearity, DNNs are equipped to learn from a much more extensive class of functions compared to their linear counterparts.

4.2.3. A hybrid LSTM-DNN model

We introduce a hybrid model called LSTM-DNN to predict implied volatility and investigate the impact of long memory on implied volatility prediction. The proposed model integrates m and τ while also highlighting the crucial role of historical sequences in the forecasting process. The LSTM model is employed to capture the dynamic fluctuations in implied volatility and extract temporal features from historical sequences, which contain valuable hidden information. The temporal feature, along with m and τ of options, are included as individual parameters in a machine-learning model for forecasting implied volatility. This approach is inspired by prior research that applies LSTM models to extract key embedded information, represented as the hidden state of LSTM model, from macroeconomic variables in asset pricing, see the work of Cong et al. (2021). Their work claims a better explanatory power in using the extracted information instead of the raw macroeconomic variables. Following their work, we also anticipate capturing the temporal information from the sophisticated long-term dependencies in historical sequences by leveraging the mechanism within the LSTM model. More specifically, we utilize the vector $\boldsymbol{X}_t^{m_1,\tau_1}$ constructed in Section 3.3 as input and partition it into two components for the hybrid model. The historical sequence $\left[IV_{t-1}^{m_1,\tau_1},\ldots,IV_{t-l}^{m_1,\tau_1}\right]$ is fed into LSTM to generate a time feature, which is then combined with m and (τ) for the prediction task in the DNN model. The structure of our hybrid model is illustrated in Fig. 3:

The LSTM model (Hochreiter & Schmidhuber, 1997) is an improved version of Recurrent Neural Networks (RNNs) designed to overcome issues of gradient vanishing and exploding during lengthy sequence

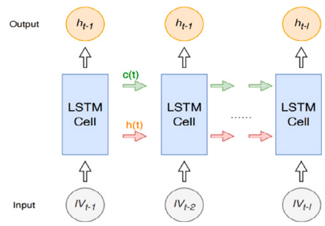


Fig. 4. The structure diagram of LSTM. The LSTM model is employed to predict implied volatility using the historical time series data. The model recurrently processes the input sequence across multiple time steps and propagates the hidden state information. It generates predictions for future time steps, representing the estimated values of implied volatility. Additionally, to incorporate the temporal aspect, the hidden state and memory cell value from the final time step, denoted as S, are extracted as a representative temporal feature.

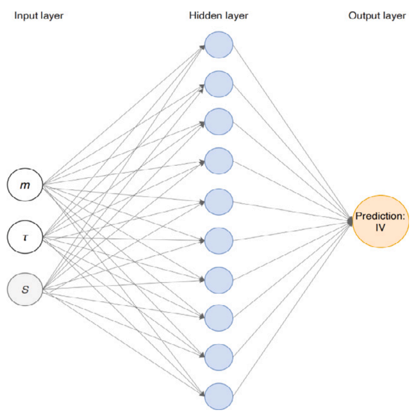


Fig. 5. The structure diagram of DNN in the hybrid model. This figure shows the structure of a DNN model in a hybrid model. The inputs of the input layer are the features extracted by the LSTM model, m and τ. An intermediate layer contains 10 hidden neurons. The final output is the predicted implied volatility result.

training. The architecture comprises interconnected memory blocks to store states, input gates to manage learning, forget gates to control forgetting, and output gates to govern content modification. Consequently, the LSTM model serves as a versatile hidden state space, proficient in capturing the dynamic patterns inherent in financial periodic data. Following Gu et al. (2020), we employ the last increment as a typical time feature to avoid modeling the entire historical series. This approach retains important dynamic information while reducing complexity. The LSTM structure is shown in Fig. 4:

The structure of the DNN model in LSTM-DNN is shown in Fig. 5:

In our nonlinear models, we employ strategies to curb overfitting, notably early stopping and learning rate adjustments. Additionally, the neural networks are designed with relatively simple hidden layers and neurons to avoid overfitting. For further hyperparameter details, refer to Appendix A.

5. Empirical study

This section provides an overview of the out-of-sample results obtained using the models described in the previous 4. The out-of-sample period for the S&P500 options spans from 2014 to 2022, while for the SSE 50 ETF options, it covers from 2017 to 2022.[3]

[3] The experiments are conducted using Python version 3.9 as the programming language, while PyTorch version 3.0 is employed to build neural networks. We employ a rolling window approach, training the model on two years of data and testing it on a separate one-year dataset.

5.1. Out-of-sample performance

We employ two metrics to evaluate the out-of-sample forecasting performance of two option indices. First, we utilize the R^2 statistic to assess the predictive power of six different models, as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}, \tag{1}$$

where y_i denotes the market implied volatility of the option, $\hat{y}_i$ denotes the predicted implied volatility of the option, and $\bar{y}$ denotes the mean over the n observations. The numerator $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ represents the sum of the squared differences between the market and the predicted implied volatility, which reflects the error of the prediction. The denominator $\sum_{i=1}^{n}(y_i - \bar{y})^2$ is the sum of the squared differences between the market and the predicted implied volatility, which captures the total variance of the implied volatility. The value of R^2 can range from negative infinity to one, where a value of one indicates perfect predictive accuracy. A value of zero means the model does not improve prediction beyond using the mean return as a forecast. Negative R^2 values indicate worse predictions compared to the simple mean return benchmark. Additionally, we calculate the standard deviation of out-of-sample R^2 values. Specifically, this measures the variation in daily model performance (in terms of explanatory power) relative to the model's average performance. A smaller standard deviation indicates consistent performance across different days, while a larger one suggests significant fluctuations in daily model performance.

Our method differs from Gu et al. (2020) by refining the R^2 formula, where we subtract the mean actual value in its denominator. This is crucial for linear models with an intercept, as it more accurately reflects the variance between model predictions and actual data. It also allows for a better comparison of the model's performance against historical averages. Our R^2 calculation, while possibly yielding lower values than those reported in Gu et al. (2020) under similar error conditions, represents a more stringent measure of model effectiveness. By excluding mean adjustment, we account for total data variability and avoid overstating model performance due to average fluctuations. This ensures our model's predictive accuracy is judged on its ability to detect true signals beyond historical averages.

Secondly, we adopt RMSE (Root Mean Square Error) as an evaluation metric to assess predictive effectiveness. The model's performance and the magnitude of prediction errors exhibit an inverse relationship with the error metrics, as defined by:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2} \tag{2}$$

where y_i denotes the market implied volatility for options, $\hat{y}_i$ denotes the predicted implied volatility for options, and n denotes the number of options. The RMSE metric measures the average magnitude of the error by calculating the square root of the average squared differences between the predicted values and actual values. A lower RMSE indicates a closer fit to the observed data and, therefore, a more effective model. Furthermore, we compute the standard deviation of out-of-sample RMSE values. This metric assesses the consistency or variability of the model's prediction errors across different days. A lower standard deviation indicates that the model's prediction errors are similar across days, signifying stable model performance. Conversely, a higher standard deviation reveals significant fluctuations in daily model performance, indicating lower stability.

Table 3 presents the out-of-sample RMSE and the R^2 results for S&P500 call and put options, spanning from January 2014 to December 2022. The interpolation method applied consists DVF in the first panel and Kernel interpolation in the second panel. The model input features are constructed as described in Section 3.3, using feature vectors with varying lag lengths of historical implied volatilities. Therefore, we evaluate the predictive performance of the six models in Section 4 using three distinct inputs, each consisting of historical sequences spanning 5, 10, and 20 days, respectively.

Three discernible patterns emerge from the findings presented in Table 3.

First, in examining forecasting outcomes using linear models, it is evident that the out-of-sample R^2 mean values generated through the Kernel method are comparable to those obtained through the DVF interpolation method across most models and historical sequence lengths. For instance, the Ridge model mean R^2 achieves values of 77.66%, 80.44%, and 81.01%, respectively, when employing the Kernel method, as opposed to 71.51%, 74.73%, and 75.47%, respectively, when utilizing the DVF method. The Ridge model consistently delivers strong results regardless of the interpolation method employed. However, in this sample, both Lasso and Elastic models consistently yield lower mean R^2 results. Notably, under the kernel interpolation method, these models produce negative R^2 values when the historical sequence equals 5. Additionally, the standard deviations of RMSE and R^2 consistently reveal that Ridge has the smallest values. This indicates that the model maintains similar prediction errors and consistent explanatory power across different days out-of-sample.

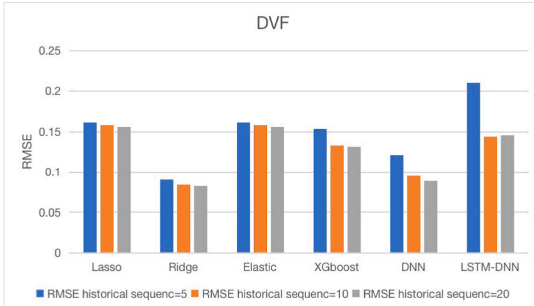
Secondly, it is noteworthy that nearly all nonlinear models produce positive mean R^2 coefficients. The sole exception to this trend is observed in the hybrid LSTM-DNN model, which yields a negative value of −32.43% when utilizing the DVF method with a 5-day historical sequence. In contrast, both the XGBoost and DNN models consistently achieve notably positive mean R^2 values, ranging from approximately 11% to 77%. For example, when 20 days of historical information is used with the DVF method, the DNN model can explain up to 77.82% of the volatility; while using 10 days of historical information with the Kernel method, the DNN model can explain a maximum of 49.91%. While these nonlinear models generally demonstrate strong performance in comparison to the linear models, they do not attain the highest R^2 values observed in the analysis. The highest R^2 coefficient of 81.01% is achieved by the Ridge model when employing the DVF method with a 20-day historical sequence. In reference to the LSTM-DNN model, despite its capacity to capture temporal features within the historical sequence, the empirical outcomes suggest that it does not lead to a discernible enhancement in predictive capability. Particularly noteworthy is the case of the DVF method with 5-day historical sequence, where the R^2 coefficient turns out to be negative. Moreover, the ensemble model XGBoost, while showing a marginal improvement over LSTM-DNN, consistently yields inferior R^2 coefficients across all three distinct historical sequences employed as input features when compared to the results produced by the DNN model. When comparing the performance of linear and nonlinear models under both interpolation methods, it becomes evident that the linear Ridge model consistently exhibits superior performance. It not only incurs the lowest prediction error but also yields the highest R^2 coefficients, indicating a greater explanatory power over the data. Among the machine learning models considered, the DNN model emerges as the top-performing choice.

Lastly, to investigate the impact of long memory on implied volatility prediction, we selected historical sequences of varying lengths as features to assess the model's predictive performance. Upon examining the collective results across all options, it becomes evident that the performance of the Lasso and Elastic models remains consistent across all three scenarios. In contrast, Ridge and DNN models exhibit decreasing RMSE mean values and increasing R^2 mean values as the length of the sequence expands under the DVF method. Our finding indicates that, for the best-performing linear and nonlinear models, Ridge and DNN respectively, using longer history data as features enhances predictive capability under this method. On the other hand, when utilizing the Kernel interpolation method, the Ridge model achieves the highest R^2 coefficient when the historical length is 5 days, while the DNN model performs the best when the historical length is 10 days. Consequently, it appears that the impact of longer historical sequence on the prediction process is not statistically significant when employing the Kernel interpolation method.

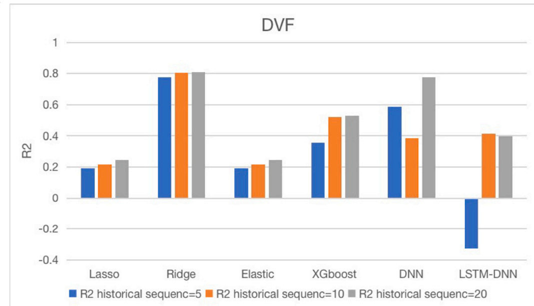
Table 3
S&P500 forecasting performance. This table presents the Root Mean Square Error (RMSE) and the R^2 for implied volatility forecasting results across six models. We report the mean and standard deviation of these two metrics out-of-sample. In each model's results, the regular numbers in the first line and the italicized numbers in the second line correspond to the mean and standard deviation, respectively. The input for these models consists of historical sequence lengths of five, ten, and twenty days. Two interpolation methods are employed in this evaluation: DVF in the upper panel and the Kernel method in the lower panel. The evaluation utilizes implied volatility data from S&P500 index options spanning from 2012 to 2022. Model training encompasses data ranging from 2012 to 2013, with the remaining data reserved for out-of-sample evaluation.

DVF						
Models	Input historical sequence					
	5		10		20	
	RMSE	R^2	RMSE	R^2	RMSE	R^2
Panel A Linear models						
Lasso	0.1613	0.1906	0.1584	0.2174	**0.1560**	**0.2427**
	0.0720	*0.6578*	*0.0720*	*0.6558*	***0.0717***	***0.6545***
Ridge	0.0905	0.7766	0.0843	0.8044	**0.0830**	**0.8101**
	0.0192	***0.0439***	***0.0190***	*0.0506*	*0.0198*	*0.0572*
Elastic	0.1613	0.1906	0.1584	0.2174	**0.1560**	**0.2427**
	0.0720	*0.6578*	*0.0720*	*0.6558*	***0.0717***	***0.6545***
Panel B Non-Linear models						
XGboost	0.1534	0.3567	0.1325	0.5200	**0.1310**	**0.5257**
	0.0480	*0.2453*	*0.0452*	*0.2115*	***0.0456***	***0.2344***
DNN	0.1206	0.5870	0.0952	0.7479	**0.0891**	**0.7782**
	0.0337	*0.1782*	*0.0257*	*0.0917*	***0.0218***	***0.0800***
LSTM-DNN	0.2099	−0.3243	0.1443	0.4135	**0.1455**	**0.3963**
	0.1020	*1.4042*	*0.0483*	*0.2892*	***0.0458***	***0.2872***
Kernel						
Models	Input historical sequence					
	5		10		20	
	RMSE	R^2	RMSE	R^2	RMSE	R^2
Panel A Linear models						
Lasso	0.1744	−0.9296	**0.1741**	−0.9215	0.1744	−0.9205
	0.0760	*1.2365*	***0.0764***	***1.2417***	*0.0772*	*1.2539*
Ridge	0.0654	0.7151	0.0620	0.7473	**0.0616**	**0.7547**
	0.0328	*0.2235*	*0.0300*	*0.1917*	***0.0296***	***0.1818***
Elastic	0.1744	−0.9296	**0.1741**	−0.9215	0.1744	−0.9205
	0.0760	*1.2365*	***0.0764***	***1.2417***	*0.0772*	*1.2539*
Panel B Non-Linear models						
XGboost	0.1187	0.1482	0.1078	0.3015	**0.0976**	**0.3770**
	0.0532	*0.5917*	***0.0485***	***0.4698***	*0.0521*	*0.6089*
DNN	0.1100	0.1792	**0.0887**	**0.4991**	0.0931	0.4247
	0.0560	*0.6634*	***0.0412***	***0.3610***	*0.0524*	*0.5255*
LSTM-DNN	**0.1110**	**0.2686**	0.1224	0.0648	0.1190	0.1519
	0.0353	***0.3269***	*0.0531*	*0.6075*	*0.0439*	*0.4560*

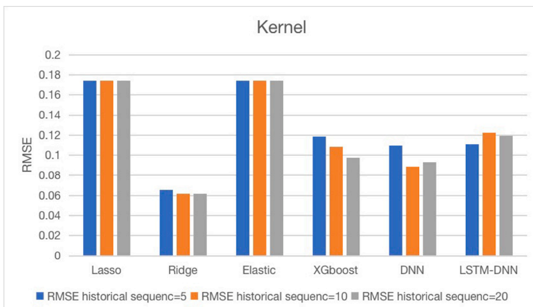
Figs. 6 and 7 provides a more dynamic and detailed perspective on the comparative performance of the models. Firstly, under DVF and kernel interpolation methods, the Ridge model consistently excels across all historical sequences, in terms of both the mean and standard deviation of performance metrics. However, the Lasso and Elastic models exhibit the lowest mean R^2 values among linear models under the DVF method, and persistently negative under the kernel method. This trend is also reflected in the standard deviation results, indicating these two models have the least stable out-of-sample explanatory power. This trend is more pronounced in the graphical representation. In nonlinear models, except for the LSTM-DNN model under the kernel method, which underperforms with only 5 days of historical sequence, models like XGBoost and DNN yield positive mean R^2 values. Additionally, under the DVF method with an input of 5, the LSTM-DNN model's standard deviations for RMSE and R^2 reach their peaks at approximately 0.1 and 1.4, respectively. This suggests that the hybrid model



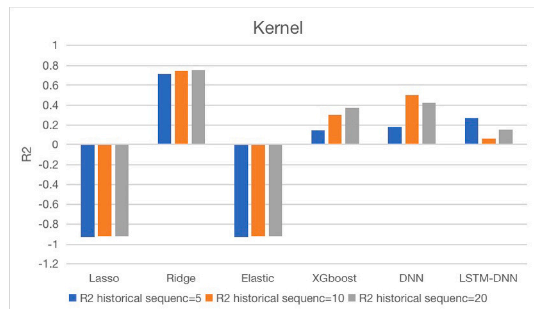
(a) DVF model RMSE (avg) results



(b) DVF model R^2 (avg) results



(c) Kernel model RMSE (avg) results



(d) Kernel model R^2 (avg) results

Fig. 6. The mean of S&P500 forecasting performance. This figure illustrates the mean Root Mean Square Error (RMSE) and the R^2 for implied volatility forecasting results obtained from the six models, utilizing the DVF and Kernel interpolation methods. The evaluation covers out-of-sample time periods spanning from 2014 to 2022. The models input includes the historical sequence of varying lengths of five, ten, and twenty days. We conduct an overall comparative analysis of the RMSE and R^2 across different historical sequence lengths for all out-of-sample forecast.

exhibits the least stable out-of-sample explanatory power and error in this scenario. Notably, DNN model under the DVF method explains up to 77.82% of volatility with 20-day data, as highlighted mean value in the figure. With Kernel interpolation, the Ridge model tops in R^2 with 20-day historical sequence, while DNN peaks with 10-day data.

Furthermore, upon overall examination, it becomes evident that the mean RMSE values associated with the Lasso and Elastic models exhibit relatively minor variations as the length of historical sequence used as input increases. In contrast, the mean RMSE values of the Ridge and DNN models exhibit a discernible trend of decrease, accompanied by corresponding increases in the mean R^2 values, as the input historical sequence length extends. This pattern suggests that longer historical sequence may have a beneficial impact on the predictive performance of these two models, especially when considering the DVF method. However, there are differences between the standard deviations and means of the performance metrics. With the Ridge model under the Kernel method, the lowest standard deviations for R^2 and RMSE occur at a sequence length of 20, indicating the most stable predictive explanatory power in this scenario. Under the DVF method, the lowest occurs at a sequence of 10. Furthermore, standard deviations are lower under the Kernel method, suggesting more stable model performance.

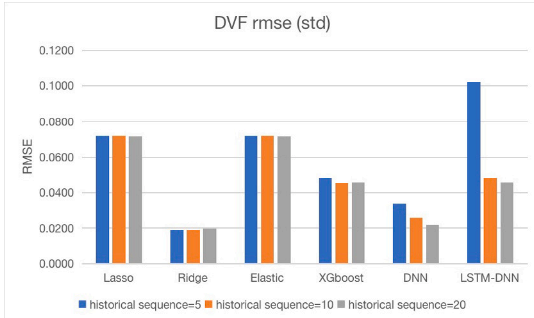
In addition to the full sample of S&P500 options implied volatility, we separately investigate the predictability of implied volatility for call and put options. Table 4 describes the predictive results of six models under the sample of call options. Under the DVF method, linear models such as Ridge show increasing R^2 mean values with longer historical sequence, peaking at 81.91% for 20-day historical sequence, indicating strong predictive power. On the contrary, the Lasso and Elastic models consistently exhibit the lowest R^2 mean values among all models, reflecting their limited utility in this scenario. Among the non-linear models, DNN stands out, especially at a 20-day historical sequence where it can explain up to 78.17% of the variance. Notably, compared to a 5-day historical sequence, the LSTM-DNN model shows improvement at a 10-day historical sequence, suggesting its performance varies with different historical sequence lengths.

Under the Kernel method, Ridge performs optimally at 20-day historical sequence, while DNN peaks at 10-day historical sequence. However, the Lasso and Elastic models demonstrate negative R^2 mean values, and XGBoost also performs poorly in comparison to its outcomes with the DVF method.

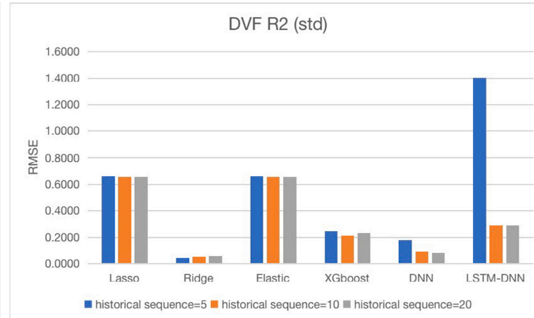
Table 5 describes the predictive results of six models under the sample of put options. Under the DVF method, we observe a trend similar to that of call options, where the Ridge and DNN models perform well, with Ridge reaching a peak R^2 mean value of 80.12% for the 20-day historical sequence. The R^2 mean values for the Lasso and Elastic models remain consistently negative. Non-linear models like XGBoost and DNN show positive R^2 mean values; DNN excels with 20-day historical sequence, reaching 77.47%. LSTM-DNN improves over longer series despite negative results for 5-day historical sequence. Under the Kernel interpolation method, the Ridge and DNN models show robust performance, with Ridge achieving the highest R^2 mean value of 93.26% for the 10-day historical sequence, while DNN maintains high R^2 mean values for the 20-day historical sequence. Conversely, the Lasso and Elastic models exhibit extremely negative R^2 values.

Comparing call and put options, we find that the Ridge model consistently shows strong performance for both, particularly under the DVF method with longer historical sequence. DNN also performs well. Lasso and Elastic models always demonstrate poor predictive ability in both scenarios. One notable difference is the performance of non-linear models; XGBoost and LSTM-DNN have better outcomes for put options than for call options, especially under the Kernel method. It is crucial to underscore that this study does not primarily aim to undertake a comparative analysis of interpolation methods. Instead, its central objective is directed toward evaluating the model that demonstrates superior performance and exploring the influence of historical sequence length as the input variable.

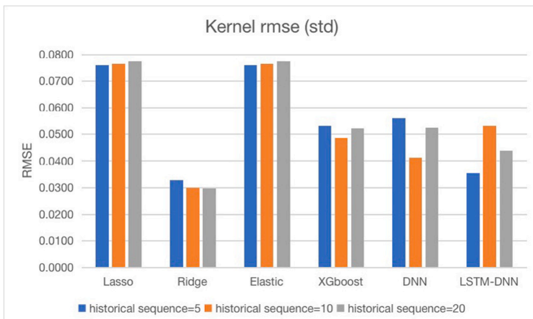
In alignment with the research objective, in brief, it can be summarized that for both call and put options, the utilization of longer historical sequences tends to augment the performance of the top-performing linear model, Ridge, and the non-linear model, DNN. The



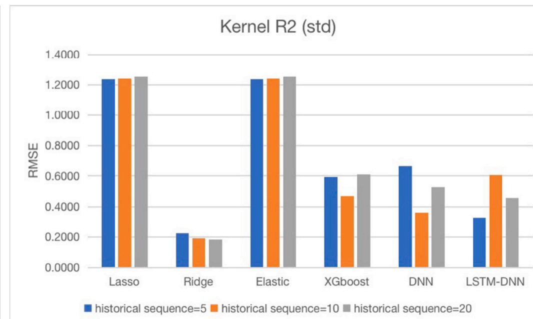
(a) DVF model RMSE (std) results



(b) DVF model R^2 (std) results



(c) Kernel model RMSE (std) results



(d) Kernel model R^2 (std) results

Fig. 7. The standard deviation of S&P500 forecasting performance. This figure illustrates the standard deviation Root Mean Square Error (RMSE) and the R^2 for implied volatility forecasting results obtained from the six models, utilizing the DVF and Kernel interpolation methods. The evaluation covers out-of-sample time periods spanning from 2014 to 2022. The models input includes the historical sequence of varying lengths of five, ten, and twenty days. We conduct an overall comparative analysis of the RMSE and R^2 across different historical sequence lengths for all out-of-sample forecasts.

DVF method accentuates this effect more prominently than the Kernel method. Consequently, the highest R^2 mean values attained are 81.01% and 77.82% for Ridge and DNN, respectively, when employing a 20-day historical sequence window as input.

Table 6, Figs. 8 and 9 present the out-of-sample RMSE and the R^2 results for SSE 50 ETF options, spanning from 2017 to 2022. Firstly, we observe that on the SSE 50 ETF sample, all models show higher R^2 mean results under the kernel interpolation method compared to the DVF method. This is especially true for the Lasso and LSTM-DNN models, where R^2 improves by approximately 40%. It is important to note that all RMSE mean values also significantly decrease. Additionally, consistent with findings on the S&P500, Ridge and DNN models are the best-performing linear and nonlinear models, respectively. Ridge outperforms DNN with no significant difference between the two interpolation methods. DNN shows a 26.92% improvement under the kernel method compared to DVF. Simultaneously, from the standard deviation results of R^2 and RMSE, we find that the Ridge model consistently yields the smallest values under both methods, followed by DNN. This demonstrates that these two models are the most stable for prediction.

Secondly, under the DVF method, when the historical sequence is 20, all linear models yield the highest R^2 mean values. For nonlinear models, LSTM-DNN achieves its peak R^2 at a sequence of 20, improving by 12.18% to 17.61% compared to shorter sequences. However, the DNN model performs best at a sequence of 5.

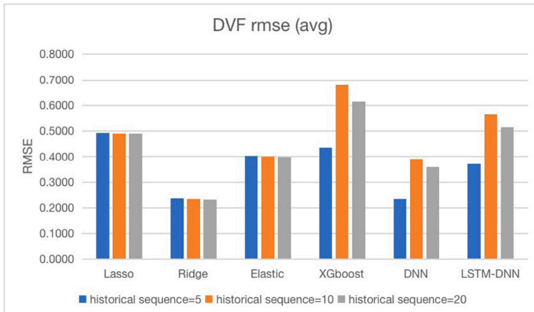
Thirdly, examining the kernel method results, both Ridge and DNN models reach their highest R^2 mean values, 88.32% and 87.75% respectively, at a sequence of 20. Other models peak at a sequence of 5.

In summary, compared to the S&P500 conclusions, some models do not consistently show their best predictive performance at a sequence of 20. However, similarly, the best model remains the linear Ridge at a sequence of 20, although in this sample, results at a sequence of 10 consistently lag behind those at 5.

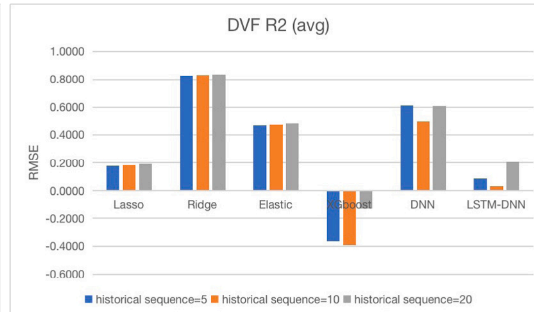
Then, we also study the predictability of implied volatility of SSE 50 ETF call and put options separately. The Tables 7 and 8 describe the predictions of each of the six models. Examining the call options sample, we first note that the Ridge model consistently achieves the highest R^2 mean values across different interpolation methods and input sequences. It shows a trend of increasing with sequence length, reaching up to 82.47% and 85% respectively. Secondly, the DNN model ranks second when the sequence is 20, but its performance is weakest at a sequence of 10. The results under the kernel method are significantly better than the DVF method, showing a 28.28% improvement. Under the DVF method, both Lasso and XGBoost yield negative R^2 mean values. Elastic and LSTM-DNN models also perform much worse compared to the kernel method. However, under the kernel method, all models demonstrate improved predictive ability. Lasso's R^2 mean value exceeds 76%, and other models show the capacity to explain over 80% of the outcomes.

In the put options sample, Ridge and DNN models remain the top two performers across all models. Except for the DNN model under the DVF method, the highest R^2 mean value is achieved at a sequence of 20. Particularly under the kernel method, both models demonstrate the ability to explain over 90% of the data. Besides, other models exhibit their best predictive performance at a sequence of 5 under this method. Additionally, nonlinear models outperform linear models, except for Ridge. XGBoost and LSTM-DNN achieve R^2 mean values of 84.64% and 90.26%, respectively.

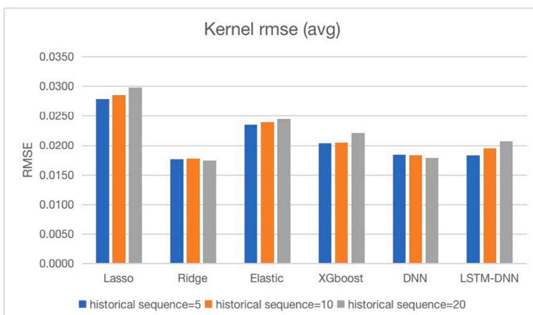
In conclusion, model performance comparison across two markets leads to the following insights. Firstly, analyses reveal that for S&P 500 and SSE 50 ETF options, Kernel's nonlinear method outperforms, except in S&P 500 call options where DVF linear interpolation excels. Secondly, the Ridge linear model exhibits excellent performance in both datasets. Ridge regression, incorporating an L2 penalty on coefficients, addresses volatility data's collinearity issue. Unlike Lasso that zeroes out coefficients, Ridge retains all features. This suggests Ridge's superior predictions indicate most variables are informative. Thirdly, within



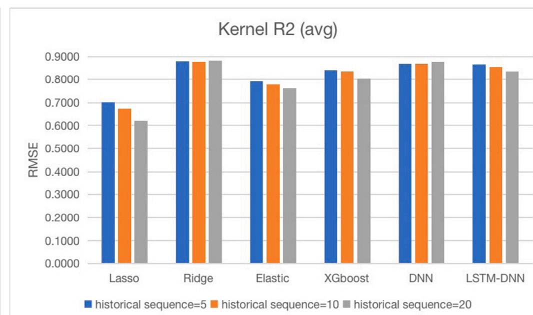
(a) DVF model RMSE (avg) results



(b) DVF model R^2 (avg) results

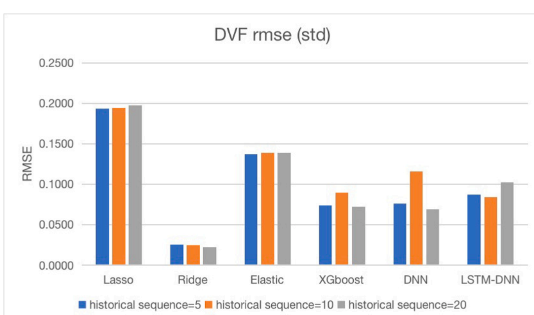


(c) Kernel model RMSE (avg) results

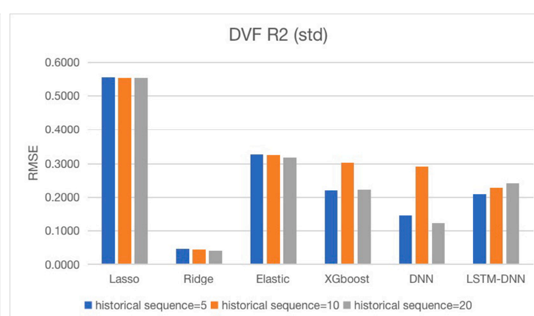


(d) Kernel model R^2 (avg) results

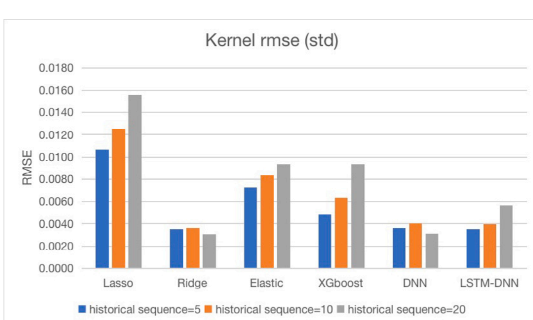
Fig. 8. The mean of SSE 50 ETF forecasting performance. This figure illustrates the mean Root Mean Square Error (RMSE) and the R^2 for implied volatility forecasting results obtained from the six models, utilizing the DVF and Kernel interpolation methods. The evaluation covers out-of-sample time periods spanning from 2017 to 2022. The models input includes the historical sequence of varying lengths of five, ten, and twenty days. We conduct an overall comparative analysis of the RMSE and R^2 across different historical sequence lengths for all out-of-sample forecast.



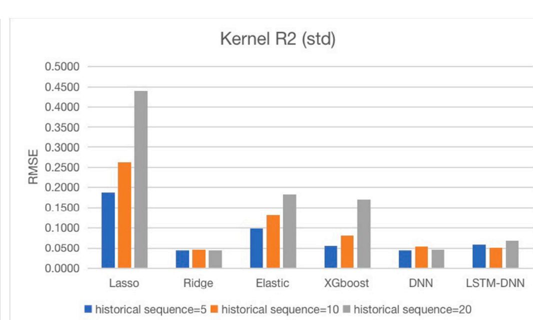
(a) DVF model RMSE (std) results



(b) DVF model R^2 (std) results



(c) Kernel model RMSE (std) results



(d) Kernel model R^2 (std) results

Fig. 9. The standard deviation of SSE 50 ETF forecasting performance. This figure illustrates the standard deviation Root Mean Square Error (RMSE) and the R^2 for implied volatility forecasting results obtained from the six models, utilizing the DVF and Kernel interpolation methods. The evaluation covers out-of-sample time periods spanning from 2017 to 2022. The models input includes the historical sequence of varying lengths of five, ten, and twenty days. We conduct an overall comparative analysis of the RMSE and R^2 across different historical sequence lengths for all out-of-sample forecasts.

Table 4
S&P500 forecasting performance of call options. This table presents the Root Mean Square Error (RMSE) and the R^2 for implied volatility of call options forecasting results across six models. We report the mean and standard deviation of these two metrics out-of-sample. In each model's results, the regular numbers in the first line and the italicized numbers in the second line correspond to the mean and standard deviation, respectively. The input for these models consists of historical sequence lengths of five, ten, and twenty days. Two interpolation methods are employed in this evaluation: the DVF in the upper panel and the Kernel method in the lower panel. The evaluation utilizes implied volatility data from S&P500 index options spanning from 2012 to 2022. Model training encompasses data ranging from 2012 to 2013, with the remaining data reserved for out-of-sample evaluation.

DVF						
Models	Input historical sequence					
	5		10		20	
	RMSE	R^2	RMSE	R^2	RMSE	R^2
Panel A Linear models						
Lasso	0.1336	0.2199	0.1315	0.2424	**0.1299**	**0.2624**
	0.0582	*0.6538*	*0.0584*	***0.6523***	*0.0585*	*0.6537*
Ridge	0.0731	0.7949	0.0690	0.8158	**0.0684**	**0.8191**
	0.0068	***0.0374***	*0.0092*	*0.0476*	*0.0108*	*0.0537*
Elastic	0.1336	0.2199	0.1315	0.2424	**0.1299**	**0.2624**
	0.0582	*0.6538*	*0.0584*	***0.6523***	*0.0585*	*0.6537*
Panel B Non-Linear models						
XGboost	0.1150	0.4915	**0.0964**	**0.6380**	0.0983	0.6258
	0.0177	*0.1273*	***0.0168***	***0.1153***	*0.0173*	*0.1184*
DNN	0.0992	0.6079	**0.0770**	0.7675	**0.0744**	**0.7817**
	0.0230	*0.1706*	***0.0148***	*0.0842*	*0.0149*	*0.0895*
LSTM-DNN	0.1285	0.3579	**0.1114**	**0.5025**	0.1211	0.4166
	0.0234	***0.2020***	*0.0255*	*0.2159*	*0.0305*	*0.2629*

Kernel						
Models	Input historical sequence					
	5		10		20	
	RMSE	R^2	RMSE	R^2	RMSE	R^2
Panel A Linear models						
Lasso	0.2125	−1.4822	**0.2122**	**−1.4634**	0.2127	−1.4475
	0.0273	***0.6923***	*0.0279*	*0.7019*	*0.0287*	*0.7108*
Ridge	0.0966	0.5022	0.0909	0.5620	**0.0900**	**0.5772**
	0.0128	*0.0930*	*0.0098*	*0.0661*	***0.0092***	***0.0508***
Elastic	0.2125	−1.4822	**0.2122**	**−1.4634**	0.2127	−1.4475
	0.0273	*0.6923*	*0.0279*	*0.7019*	*0.0287*	*0.7108*
Panel B Non-Linear models						
XGboost	0.1555	−0.2834	0.1429	−0.0697	**0.1393**	**−0.0596**
	0.0415	*0.4988*	***0.0338***	***0.3311***	*0.0405*	*0.5948*
DNN	0.1572	−0.3431	**0.1205**	**0.2351**	0.1320	0.0821
	0.0322	*0.4855*	***0.0315***	***0.2770***	*0.0444*	*0.5093*
LSTM-DNN	**0.1375**	**0.0043**	**0.1662**	−0.4487	0.1522	−0.2014
	0.0194	*0.1077*	*0.0346*	*0.3642*	*0.0282*	*0.2619*

Table 5
S&P500 forecasting performance of put options. This table presents the Root Mean Square Error (RMSE) and the R^2 for implied volatility of put options forecasting results across six models. We report the mean and standard deviation of these two metrics out-of-sample. In each model's results, the regular numbers in the first line and the italicized numbers in the second line correspond to the mean and standard deviation, respectively. The input for these models consists of historical sequence lengths of five, ten, and twenty days. Two interpolation methods are employed in this evaluation: the DVF in the upper panel and the Kernel method in the lower panel. The evaluation utilizes implied volatility data from S&P500 index options spanning from 2012 to 2022. Model training encompasses data ranging from 2012 to 2013, with the remaining data reserved for out-of-sample evaluation.

DVF						
Models	Input historical sequence					
	5		10		20	
	RMSE	R^2	RMSE	R^2	RMSE	R^2
Panel A Linear models						
Lasso	0.1889	0.1613	0.1852	0.1924	**0.1820**	**0.2230**
	0.0738	*0.6606*	*0.0743*	*0.6583*	*0.0743*	***0.6547***
Ridge	0.1078	0.7584	0.0997	0.7929	**0.0976**	**0.8012**
	0.0095	***0.0424***	*0.0129*	*0.0509*	*0.0155*	*0.0592*
Elastic	0.1889	0.1613	0.1852	0.1924	**0.1820**	**0.2230**
	0.0738	*0.6606*	*0.0743*	*0.6583*	*0.0743*	***0.6547***
Panel B Non-Linear models						
XGboost	0.1919	0.2219	0.1685	0.4019	**0.1637**	**0.4256**
	0.0367	***0.2604***	***0.0347***	*0.2197*	*0.0414*	*0.2754*
DNN	0.1420	0.5662	0.1134	0.7282	**0.1038**	**0.7747**
	0.0289	*0.1832*	*0.0209*	*0.0946*	***0.0173***	***0.0691***
LSTM-DNN	0.2914	−1.0065	0.1771	0.3246	**0.1699**	**0.3761**
	0.0837	*1.7240*	***0.0429***	*0.3237*	*0.0456*	***0.3083***

Kernel						
Models	Input historical sequence					
	5		10		20	
	RMSE	R^2	RMSE	R^2	RMSE	R^2
Panel A Linear models						
Lasso	0.1362	−0.3769	0.1361	−0.3797	**0.1361**	**−0.3935**
	0.0888	***1.4028***	*0.0894*	*1.4156*	*0.0904*	*1.4435*
Ridge	0.0343	0.9280	**0.0331**	**0.9326**	0.0332	0.9322
	0.0064	*0.0237*	***0.0059***	***0.0214***	*0.0073*	*0.0230*
Elastic	0.1362	−0.3769	0.1361	−0.3797	**0.1361**	**−0.3935**
	0.0888	***1.4028***	*0.0894*	*1.4156*	*0.0904*	*1.4435*
Panel B Non-Linear models						
XGboost	0.0820	0.5798	0.0726	0.6727	**0.0559**	**0.8136**
	0.0351	*0.2807*	*0.0332*	*0.2371*	***0.0178***	***0.0811***
DNN	0.0629	0.7015	0.0569	0.7630	**0.0542**	**0.7673**
	0.0282	*0.3143*	***0.0194***	*0.2112*	*0.0225*	*0.2413*
LSTM-DNN	0.0844	0.5330	0.0786	0.5783	**0.0859**	**0.5052**
	0.0265	***0.2497***	***0.0245***	*0.2794*	*0.0294*	*0.3126*

our nonlinear models, the DNN model performs best yet falls slightly short of the Ridge model. This suggests a strong linear relationship between changes in implied volatility and its features. In such cases, deep neural networks may overfit, capturing noise rather than actual signals. Additionally, using a hybrid LSTM-DNN model to input temporal features into the DNN does not enhance outcomes, indicating that overly complex models may not be necessary for predicting implied volatility in these markets. Fourthly, the best predictive R^2 results for the S&P 500 and SSE 50 ETF are 81.01% and 88.32%, respectively, with the Chinese market slightly outperforming by 7.31%. This might be due to the relatively late development of China's options market, potentially simplifying models and facilitating pattern recognition. However, with prediction results exceeding 80% in both markets, our feature matrix and models prove highly effective for implied volatility prediction across both emerging and mature options markets.

5.2. Diebold and Mariano (DM) test

We employ the DM test (Diebold & Mariano, 2002) to further compare the loss difference between the forecast results of the two models, ensuring covariance stationarity. Initially, we rearrange the matrix of the 3.3 and construct panel data based on various combinations of m and τ, where each combination corresponds to a set of predicted implied volatilities arranged in a time series. Subsequently, following the approach of Bali et al. (2023), we conduct the DM test considering only serial correlation and disregarding cross-sectional dependence. Thus, the formula for the DM test statistic between Model 1 and Model 2 is:

$$\mathrm{DM}^{(1,2)} = \frac{\bar{d}^{(1,2)}}{\hat{\sigma}_d^{(1,2)}}$$

where $\bar{d}^{(1,2)}$ and $\hat{\sigma}_d^{(1,2)}$ represent the time series average. Additionally, we utilize the method proposed by Newey and West (1986) to account for serial correlation and heteroscedasticity in the time series of cross-sectional averages. This is defined as:

$$d_{t+1}^{(1,2)} = \frac{1}{n_{\mathcal{T}_3}} \sum_{(i,t)\in\mathcal{T}_3} \left[\left(\hat{e}_{i,s,t+1}^{(1)}\right)^2 - \left(\hat{e}_{i,s,t+1}^{(2)}\right)^2 \right]$$

Table 6
SSE 50 ETF forecasting performance. This table presents the SSE 50 ETF Root Mean Square Error (RMSE) and the R^2 for implied volatility forecasting results across six models. We report the mean and standard deviation of these two metrics out-of-sample. In each model's results, the regular numbers in the first line and the italicized numbers in the second line correspond to the mean and standard deviation, respectively. The input for these models consists of historical sequence lengths of five, ten, and twenty days. Two interpolation methods are employed in this evaluation: DVF in the upper panel and the Kernel method in the lower panel. The evaluation utilizes implied volatility data from SSE 50 ETF index options spanning from 2015 to 2022. Model training encompasses data ranging from 2015 to 2016, with the remaining data reserved for out-of-sample evaluation.

DVF						
Models	Input historical sequence					
	5		10		20	
	RMSE	R^2	RMSE	R^2	RMSE	R^2
Panel A Linear models						
Lasso	0.4922	0.1789	0.4899	0.1853	**0.4903**	**0.1937**
	0.1940	*0.5561*	*0.1948*	***0.5547***	*0.1980*	*0.5550*
Ridge	0.2377	0.8260	0.2348	0.8307	**0.2311**	**0.8382**
	0.0259	*0.0462*	*0.0256*	*0.0432*	***0.0224***	***0.0407***
Elastic	0.4033	0.4678	0.4004	0.4743	**0.3996**	**0.4835**
	0.1380	*0.3271*	***0.1389***	*0.3256*	***0.1389***	***0.3187***
Panel B Non-linear models						
XGboost	0.6734	−0.3673	0.6787	−0.3947	**0.6149**	−0.1295
	0.0742	***0.2202***	*0.0901*	*0.3024*	***0.0729***	*0.2223*
DNN	**0.3521**	**0.6166**	0.3902	0.4980	0.3603	0.6083
	0.0767	*0.1445*	*0.1157*	*0.2913*	***0.0691***	***0.1222***
LSTM-DNN	0.5513	0.0864	0.5659	0.0321	**0.5158**	**0.2082**
	0.0881	***0.2097***	***0.0847***	*0.2277*	*0.1025*	*0.2406*
Kernel						
Models	Input historical sequence					
	5		10		20	
	RMSE	R^2	RMSE	R^2	RMSE	R^2
Panel A Linear models						
Lasso	**0.0278**	**0.7020**	0.0285	0.6759	0.0298	0.6198
	0.0107	***0.1879***	*0.0125*	*0.2629*	*0.0156*	*0.4393*
Ridge	0.0177	0.8789	0.0178	0.8777	**0.0174**	**0.8832**
	0.0035	*0.0441*	*0.0036*	*0.0444*	***0.0031***	***0.0436***
Elastic	**0.0235**	**0.7915**	0.0239	0.7793	0.0245	0.7623
	0.0073	***0.0981***	*0.0084*	*0.1317*	*0.0093*	*0.1836*
Panel B Non-Linear models						
XGboost	0.0204	0.8416	0.0205	0.8373	0.0221	0.8023
	0.0048	***0.0565***	*0.0064*	*0.0819*	*0.0093*	*0.1704*
DNN	0.0183	0.8702	0.0183	0.8690	**0.0178**	**0.8775**
	0.0036	***0.0435***	*0.0040*	*0.0549*	***0.0031***	*0.0454*
LSTM-DNN	**0.0183**	**0.8675**	0.0195	0.8541	0.0207	0.8371
	0.0035	*0.0592*	*0.0040*	***0.0515***	*0.0057*	*0.0684*

Table 7
SSE 50 ETF forecasting performance of call options. This table presents the SSE 50 ETF Root Mean Square Error (RMSE) and the R^2 for implied volatility forecasting results across six models. We report the mean and standard deviation of these two metrics out-of-sample. In each model's results, the regular numbers in the first line and the italicized numbers in the second line correspond to the mean and standard deviation, respectively. The input for these models consists of historical sequence lengths of five, ten, and twenty days. Two interpolation methods are employed in this evaluation: DVF in the upper panel and the Kernel method in the lower panel. The evaluation utilizes implied volatility data from SSE 50 ETF index options spanning from 2015 to 2022. Model training encompasses data ranging from 2015 to 2016, with the remaining data reserved for out-of-sample evaluation.

DVF						
Models	Input historical sequence					
	5		10		20	
	RMSE	R^2	RMSE	R^2	RMSE	R^2
Panel A Linear models						
Lasso	0.6041	−0.1258	**0.6013**	−0.1163	0.6022	−0.1041
	0.0678	*0.3078*	*0.0672*	*0.3039*	***0.0661***	***0.2999***
Ridge	0.2453	0.8132	0.2420	0.8182	**0.2396**	**0.8247**
	0.0294	*0.0535*	*0.0284*	*0.0516*	***0.0242***	***0.0474***
Elastic	0.4661	0.3316	**0.4644**	**0.3358**	0.4655	0.3416
	0.0451	*0.1617*	*0.0448*	*0.1605*	***0.0436***	***0.1602***
Panel B Non-Linear models						
XGboost	0.6881	−0.4186	0.7122	−0.5218	**0.6484**	−0.2409
	0.0600	*0.1389*	***0.0555***	***0.1315***	*0.0813*	*0.1860*
DNN	0.3982	0.5206	0.4834	0.2629	**0.3809**	**0.5673**
	0.0342	***0.0817***	*0.0759*	*0.2287*	*0.0628*	*0.1038*
LSTM-DNN	0.5641	0.0424	0.6082	−0.1146	**0.5279**	**0.1683**
	0.0874	***0.2170***	*0.0901*	*0.2309*	*0.1134*	*0.2763*
Kernel						
Models	Input historical sequence					
	5		10		20	
	RMSE	R^2	RMSE	R^2	RMSE	R^2
Panel A Linear models						
Lasso	**0.0224**	**0.7692**	0.0225	0.7673	0.0228	0.7637
	0.0027	***0.0971***	***0.0025***	*0.1008*	***0.0025***	*0.1097*
Ridge	**0.0182**	0.8539	0.0182	0.8546	0.0183	**0.8555**
	0.0027	*0.0431*	*0.0028*	***0.0428***	*0.0029*	*0.0441*
Elastic	**0.0206**	**0.8109**	0.0207	0.8103	0.0209	0.8092
	0.0028	***0.0620***	*0.0028*	*0.0642*	***0.0027***	*0.0689*
Panel B Non-Linear models						
XGboost	0.0192	0.8367	**0.0188**	0.8424	0.0190	**0.8434**
	0.0029	*0.0503*	***0.0029***	***0.0493***	*0.0032*	*0.0516*
DNN	0.0190	0.8436	0.0194	0.8342	**0.0186**	**0.8500**
	0.0037	***0.0406***	*0.0039*	*0.0546*	***0.0031***	*0.0475*
LSTM-DNN	**0.0192**	0.8324	0.0194	0.8324	0.0195	**0.8361**
	0.0024	*0.0617*	*0.0030*	***0.0544***	*0.0043*	*0.0573*

Additionally, to evaluate the similarity between forecast results of different models, we define 'forecast correlation' as follows:

$$\rho_{t+1}^{(1,2)} = \frac{\mathrm{Cov}\left(\hat{e}_{t+1}^{(1)}, \hat{e}_{t+1}^{(2)}\right)}{\sigma\left(\hat{e}_{t+1}^{(1)}\right)\sigma\left(\hat{e}_{t+1}^{(2)}\right)}$$

where $\mathrm{Cov}\left(\hat{e}_{t+1}^{(1)}, \hat{e}_{t+1}^{(2)}\right)$ represents covariance, and $\sigma\left(\hat{e}_{t+1}^{(1)}\right)\sigma\left(\hat{e}_{t+1}^{(2)}\right)$ represent standard deviations.

Table 9 provides the comparison results of various models within the S&P500 sample. For call options, results using the DVF interpolation method are shown, while for put options, the kernel method's results are displayed. Since Ridge and Elastic models yield equivalent results in the S&P500 sample, the latter is not listed here. Our study highlights a clear trend in which Ridge regression demonstrates superior performance over more complex models in call option forecasting, particularly across varying input historical sequences. For instance, with a historical sequence length of 5, Ridge significantly surpasses LSTM-DNN, as evidenced by a DM statistic of 24.3334, denoting statistical significance at the 5% level. Echoing the trends shown by R^2, the Lasso model remains uniformly inferior to all other linear and nonlinear models. Additionally, our hybrid model, LSTM-DNN, despite outperforming Lasso, does not match the efficacy of other models. Notably, the t-statistic for comparisons between LSTM-DNN and both DNN and Ridge consistently exceeds 20 in every scenario.

Aligned with our previous findings, Ridge and DNN emerge as the top-performing linear and nonlinear models, respectively. The results indicate a significantly better forecast accuracy of Ridge over DNN, especially when the historical sequence equals 5, where the t-statistic stands at 18.1944. Interestingly, this value diminishes as the length of the historical sequence increases, dropping to 8.6592 when the sequence length is 20. This pattern suggests that Ridge's comparative advantage over DNN is more pronounced in shorter sequences and decreases with longer data sequences. The results for put options are largely similar to those for calls, except that XGBoost's superiority over LSTM-DNN is not significant at sequences of 5 and 20.

The predictive correlation analysis in Panel B gives us some new insights. In our study, we observe a consistent pattern across three different input historical sequences. Firstly, XGBoost and Ridge models

Table 8
SSE 50 ETF forecasting performance of put options. This table presents the SSE 50 ETF Root Mean Square Error (RMSE) and the R^2 for implied volatility forecasting results across six models. We report the mean and standard deviation of these two metrics out-of-sample. In each model's results, the regular numbers in the first line and the italicized numbers in the second line correspond to the mean and standard deviation, respectively. The input for these models consists of historical sequence lengths of five, ten, and twenty days. Two interpolation methods are employed in this evaluation: DVF in the upper panel and the Kernel method in the lower panel. The evaluation utilizes implied volatility data from SSE 50 ETF index options spanning from 2015 to 2022. Model training encompasses data ranging from 2015 to 2016, with the remaining data reserved for out-of-sample evaluation.

DVF						
Models	Input historical sequence					
	5		10		20	
	RMSE	R^2	RMSE	R^2	RMSE	R^2
Panel A Linear models						
Lasso	0.3803	0.4836	0.3786	0.4868	**0.3783**	**0.4915**
	0.2137	***0.5815***	*0.2159*	*0.5840*	*0.2212*	*0.5906*
Ridge	0.2302	0.8388	0.2275	0.8432	**0.2226**	**0.8518**
	0.0191	*0.0327*	*0.0199*	*0.0277*	***0.0166***	***0.0266***
Elastic	0.3406	0.6041	0.3364	0.6129	**0.3337**	**0.6253**
	0.1678	*0.3883*	*0.1685*	*0.3845*	***0.1673***	***0.3704***
Panel B Non-Linear models						
XGboost	0.6587	−0.3161	0.6453	−0.2676	**0.5814**	−0.0181
	0.0834	*0.2691*	*0.1045*	*0.3651*	***0.0421***	***0.1985***
DNN	0.3060	0.7126	**0.2971**	**0.7331**	0.3397	0.6494
	0.0796	*0.1291*	***0.0606***	***0.0827***	*0.0691*	*0.1253*
LSTM-DNN	0.5385	0.1304	0.5236	0.1788	**0.5036**	**0.2481**
	0.0868	*0.1922*	***0.0514***	***0.0853***	*0.0887*	*0.1905*
Kernel						
Models	Input historical sequence					
	5		10		20	
	RMSE	R^2	RMSE	R^2	RMSE	R^2
Panel A Linear models						
Lasso	**0.0332**	**0.6348**	0.0346	0.5845	0.0367	0.4759
	0.0128	***0.2284***	*0.0153*	*0.3337*	*0.0195*	*0.5766*
Ridge	0.0172	0.9040	0.0173	0.9009	**0.0164**	**0.9109**
	0.0041	*0.0280*	*0.0042*	*0.0322*	***0.0029***	***0.0182***
Elastic	**0.0263**	**0.7721**	0.0272	0.7483	0.0280	0.7154
	0.0090	*0.1210*	*0.0105*	*0.1692*	*0.0119*	*0.2414*
Panel B Non-Linear models						
XGboost	**0.0216**	**0.8465**	0.0222	0.8321	0.0251	0.7613
	0.0059	***0.0617***	*0.0082*	*0.1045*	*0.0121*	*0.2282*
DNN	0.0176	0.8969	0.0172	0.9037	**0.0170**	**0.9050**
	0.0034	*0.0269*	*0.0038*	*0.0254*	***0.0030***	***0.0185***
LSTM-DNN	**0.0174**	**0.9026**	0.0195	0.8759	0.0220	0.8381
	0.0042	***0.0271***	*0.0048*	*0.0376*	*0.0065*	*0.0779*

demonstrate a correlation, with p-values exceeding 0.8. This trend is also evident between LSTM-DNN and DNN models, particularly when the historical sequence is 5, where large values with a cutoff at 90% are noted. Secondly, we find a correlation between the DNN and Ridge models when the sequence is 10 and 20, while at a sequence of 5, it is XGBoost and Lasso that show a correlation. Additionally, there are fewer instances of significant correlations between models in put options. Only XGBoost and linear models, as well as LSTM-DNN and DNN at a sequence of 5, show correlations.

Table 10 shows the results of inter-model comparison of SSE 50 ETF option samples under kernel interpolation method. On call options, we observe that Ridge is significantly superior to all other models, particularly in comparison with Lasso, where the statistical values are around 9.7. However, at a sequence of 20, Ridge's superiority over DNN and LSTM-DNN is not statistically significant. Furthermore, Lasso and Elastic models are consistently outperformed in predictive performance comparisons with other models. The DNN model mostly ranks just below Ridge but shows inconsistency. For instance, XGBoost significantly outperforms DNN when the sequence is 10. The pattern is similar in put option samples. Notably, at a sequence of 5, the statistical value between Ridge and Lasso reaches 12.7472, indicating Ridge's significant advantage. Additionally, DNN is consistently superior to other models in all cases, except when compared to Ridge. Additionally, regarding correlation, on the SSE 50 ETF sample, except for the lack of significant correlation between DNN and LSTM-DNN with either Lasso or Elastic, most models exhibit significant correlations with each other. Furthermore, the correlation among DNN, XGBoost, and Ridge consistently exceeds 90%, indicating a significant relationship.

By comparing analyses of the S&P 500 and SSE 50 ETF option samples, we observe several key similarities and differences. Firstly, the Ridge model demonstrates superior predictive performance in both samples, especially evident in shorter historical sequences. However, there are differences between the market outcomes. In the S&P 500 sample, the LSTM-DNN model performs less satisfactorily, particularly in comparison with Ridge and DNN models. In contrast, this performance gap is not significant in the SSE 50 ETF option sample over longer historical sequences. Additionally, while both samples show model correlations, the specific combinations of models and the degrees of their correlations differ. These observations highlight the Ridge model's general advantage and reveal the unique aspects of complex interactions between different market data and models.

5.3. Forecast performance across years

To further investigate the performance variation of the best two models, Ridge and DNN, we show their performance across the out-of-sample periods spanning from 2014 to 2022 in S&P500 sample. We first compare the mean R^2 of these two models over the out-of-sample periods in Fig. 10. For the sake of clarity, we only include the results generated using the interpolation methods with the highest mean R^2 values for each model. Furthermore, we opt to analyze the linear and nonlinear models that exhibit the best predictive performance, namely Ridge and DNN. For our S&P500 put options Ridge sample, we use the kernel interpolation method; for all other analyses, the DVF method is employed.[4]

In the detailed assessment of the Ridge model, we observe a prevailing tendency for R^2 values to ascend with the extension of historical sequence length. A case in point is the year 2021, where the model inaugurates with an R^2 of 86.59% for the 5-day historical sequence and culminates at 91.03% for the 20-day historical sequence, signifying the augmentation of predictive performance with protracted historical sequence. Nevertheless, an irregularity in this pattern surfaces in certain years such as 2014, where the R^2 value exhibits a slight decrement from 73.74% in the 5-day historical sequence to 73.51% in the 20-day historical sequence. In the period from 2017 to 2022, there is a consistent revelation that the 20-day historical sequence outperforms the 5-day historical sequence in terms of R^2 values, corroborating the merit of lengthier time horizons in deciphering the intricate patterns of the market.

Turning to the DNN model, there is a marked enhancement in R^2 values across the years, especially pronounced with the 20-day historical sequence, which escalates from 60.43% in 2014 to a zenith of 90.54% in 2021. This uptrend robustly suggests that the use of extended historical sequence considerably fortifies the model's predictive accuracy. In particular, during the years 2015 and 2016, the R^2 values for both the 10-day and 20-day substantially exceed those of the 5-day historical sequence, with the former series improving by over 30 percentage points relative to the latter in the said years. Furthermore, in 2021, while the DNN model manifests a high R^2 value of 84.63% for the 5-day historical sequence, it continues to exhibit an amplification in predictive capability with an R^2 of 90.54% for the 20-day historical sequence. This underscores the model's adeptness at leveraging a more

[4] We have all other results in the internet Appendix.

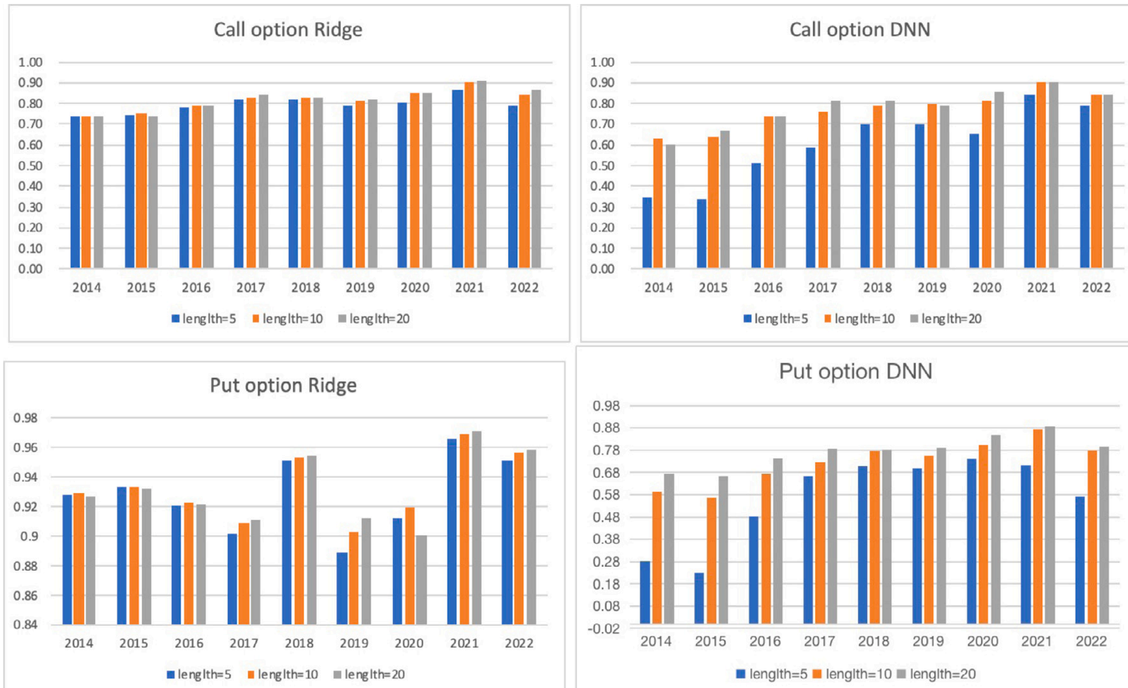


Fig. 10. Forecasting the mean S&P500 R^2 performance across years. This figure presents the R^2 of the forecasting performance of implied volatility for Call and Put options on the S&P 500 Index. The R^2 values are based on the two most proficient models, namely Ridge and DNN. The evaluation encompasses out-of-sample time periods spanning from 2014 to 2022. The models are fed with historical sequence of varying durations, specifically five, ten, and twenty days, with the DVF interpolation method. A comparative examination of R^2 values is conducted across diverse historical sequence lengths, with a specific focus on implied volatility for Call and Put options over the specified time frame.

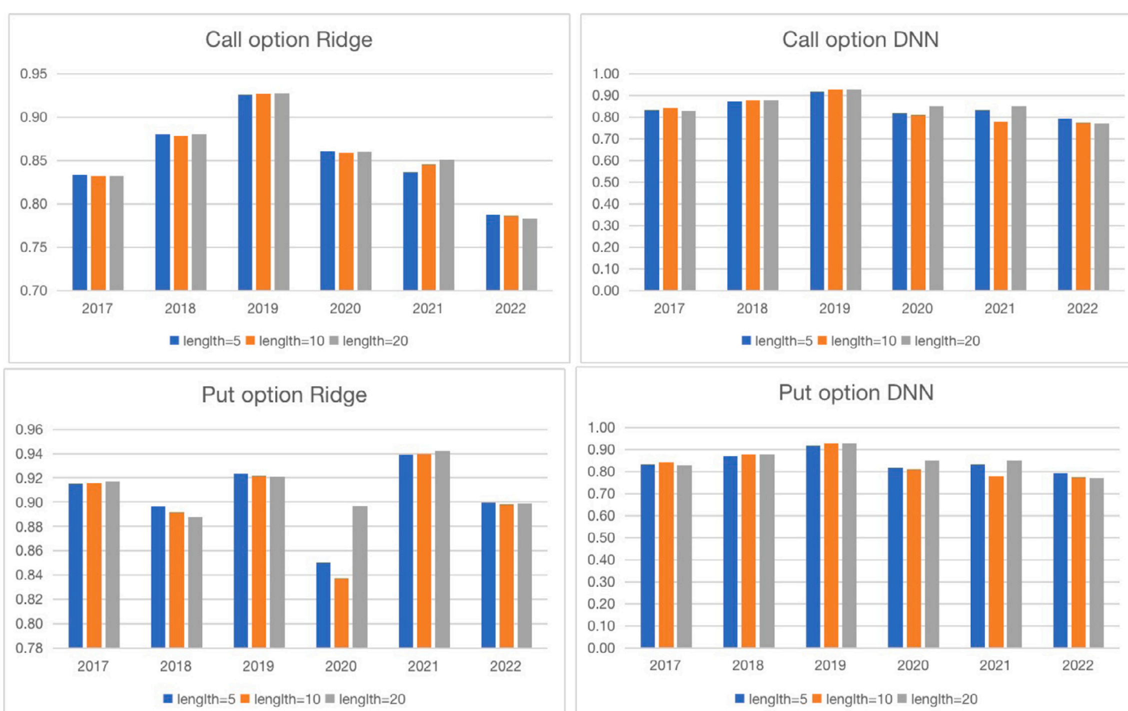


Fig. 11. Forecasting the mean SSE 50 ETF R^2 performance across years. This figure presents the R^2 of the forecasting performance of implied volatility for Call and Put options on the SSE 50 ETF Index. The R^2 values are based on the two most proficient models, namely Ridge and DNN. The evaluation encompasses out-of-sample time periods spanning from 2017 to 2022. The models are fed with historical sequence of varying durations, specifically five, ten, and twenty days, with the kernel interpolation method. A comparative examination of R^2 values is conducted across diverse historical sequence lengths, with a specific focus on implied volatility for Call and Put options over the specified time frame.

Table 9
Forecast comparison of S&P500 option. Panel A displays the Diebold–Mariano (DM) test results among models on the S&P500 option sample. Positive numbers indicate that the model in the row outperforms the model in the column, and the presented results are statistically significant at the 1% level (5% level). Additionally, Panel B illustrates the forecast correlation between models. In the table, light blue and blue markings respectively represent large values with a cutoff at 90% (70%).

SP500 Call option

	Lasso	XGBoost	DNN	LSTM-DNN
Panel A: Diebold and Mariano (1995) forecast comparison				
Input historical sequence = 5				
Ridge	**12.5986**	**21.2655**	**18.1944**	**24.3334**
Lasso		−7.5555	−8.2915	−3.6034
XGBoost			−6.4497	**7.3322**
DNN				**22.6444**
Input historical sequence = 10				
Ridge	**12.3525**	**16.5401**	**12.3692**	**21.8741**
Lasso		−10.0170	−11.1736	−5.5099
XGBoost			−11.3350	**7.9314**
DNN				**22.9349**
Input historical sequence = 20				
Ridge	**11.7960**	**16.1829**	**8.6592**	**21.5222**
Lasso		−9.2325	−10.8202	−3.7655
XGBoost			−12.0973	**10.7627**
DNN				**21.6309**
Panel B: Forecast correlation				
Input historical sequence = 5				
Ridge	0.6925	0.8830	0.5552	0.5771
Lasso		0.8351	0.0516	0.0877
XGBoost			0.3649	0.4433
DNN				0.9254
Input historical sequence = 10				
Ridge	0.5724	0.8598	0.8230	0.6988
Lasso		0.6949	0.2043	0.1149
XGBoost			0.6047	0.5811
DNN				0.8593
Input historical sequence = 20				
Ridge	0.5014	0.8419	0.8927	0.6832
Lasso		0.6325	0.2534	0.1088
XGBoost			0.6887	0.5249
DNN				0.8165

SP500 Put option

	Lasso	XGBoost	DNN	LSTM-DNN
Panel A: Diebold and Mariano (1995) forecast comparison				
Input historical sequence = 5				
Ridge	**13.3694**	**11.5620**	**11.3533**	**17.9592**
Lasso		−13.3454	−11.2535	−10.6635
XGBoost			−4.1973	1.0365
DNN				**10.9815**
Input historical sequence = 10				
Ridge	**13.1880**	**9.7272**	**13.3196**	**17.0438**
Lasso		−13.7037	−11.6552	−10.5994
XGBoost			−4.2318	**1.1609**
DNN				**14.5015**
Input historical sequence = 20				
Ridge	**12.8287**	**12.1620**	**9.0033**	**19.7160**
Lasso		−12.6005	−11.3754	−9.3564
XGBoost			0.9944	14.1317
DNN				**14.3857**
Panel B: Forecast correlation				
Input historical sequence = 5				
Ridge	0.5498	0.6447	0.3569	0.4643
Lasso		0.8220	−0.1059	0.1017
XGBoost			0.0118	0.3398
DNN				0.8276
Input historical sequence = 10				
Ridge	0.4125	0.6232	0.4882	0.4111
Lasso		0.7929	−0.2668	−0.1087
XGBoost			0.1000	0.7671
DNN				
Input historical sequence = 20				
Ridge	0.2592	0.7687	0.5642	0.3481
Lasso		0.3903	−0.2662	−0.1259
XGBoost			0.3350	0.2515
DNN				0.3190

comprehensive span of historical sequence to enhance its predictive accuracy.

In our put option sample, we employ the kernel interpolation method for Ridge results and the DVF method for DNN outcomes. In the corresponding analysis of the put options, the Ridge model displays a comparable trajectory to that seen with call options. The R^2 values usually increase as we look at longer periods of historical sequence, but there are some slight variations. For example, in 2021, the Ridge model's R^2 for put options starts at 81.38% with 5-day historical sequence and goes up to 89.42% with 20-day historical sequence. These numbers are a bit lower than what we see for call options, suggesting that the model's ability to predict put options is strong but varies a bit with more data. Yet, we see exceptions like in 2014, where the R^2 for put options actually drops a little, from 70.95% to 71.46%, when we compare 5-day to 20-day historical sequence. This decrease is different from the steady increase we see with call options and shows that the put options market might have unique traits that the model does not pick up as well with more data. Between 2017 and 2022, however, the Ridge model consistently shows that 20-day historical sequence gives us more accurate predictions than 5-day historical sequence for put options, even though this difference is not as big as it is with call options. This tells us that using data from longer time frames is important, but it also suggests there are some fundamental differences in how call and put options behave in the market. Furthermore, results from the DNN model reveal a more pronounced trend of increasing R^2 values with the lengthening of the input historical sequence. This is particularly noticeable in 2014 and 2015, where sequences equal to 20 show an approximate 35% increase in R^2 compared to those with a sequence of 5. Additionally, between 2020 and 2021, R^2 values reach over 80%.

In S&P500 sample, both models demonstrate a general trend where longer historical sequence improves predictive accuracy for put options, aligning with the observations from call options. However, the extent of improvement and the peak R^2 values differ slightly, indicating that the characteristics of the put options market may affect the models' performance.

Fig. 11 shows their performance across the out-of-sample periods spanning from 2014 to 2022 in SSE 50 ETF sample. We select the R^2 results of Ridge and DNN models under the kernel interpolation method. In the call option sample, the R^2 values of the two models show minor differences across different input sequences. Both models reach their peak R^2 in 2019, with the highest values occurring at a sequence of 20, exceeding 92%. Additionally, for the Ridge model, the highest R^2 at a sequence of 20 is observed in 2018 and 2021. In other years, the peak is at a sequence of 5. For the DNN model, except in 2017 and 2022, the best predictive performance occurs at a sequence of 20. Looking at the results for put options, the Ridge model achieves

Table 10
Forecast comparison of SSE 50 ETF option. Panel A displays the Diebold–Mariano (DM) test results among models on the SSE 50 ETF option sample. Positive numbers indicate that the model in the row outperforms the model in the column, and the presented results are statistically significant at the 1% level (5% level). Additionally, Panel B illustrates the forecast correlation between models. In the table, light blue and blue markings respectively represent large values with a cutoff at 90% (70%).

SSE 50 ETF call option					
Panel A: Diebold and Mariano (1995) forecast comparison					
Input historical sequence = 5					
	Lasso	Elastic	XGBoost	DNN	LSTM-DNN
Ridge	**9.7810**	**7.9872**	**5.0521**	**3.1497**	**4.1292**
Lasso		−6.273131	−7.3259447	−6.7186	−7.2048
Elastic			−5.6025	−5.4365	−5.2689
XGBoost				−0.4902	−0.4631
DNN					0.1650
Input historical sequence = 10					
	Lasso	Elastic	XGBoost	DNN	LSTM-DNN
Ridge	**9.7771**	**7.8377**	**3.0548**	**3.8291**	**4.2348**
Lasso		−6.3104	−7.5569	−5.2764	−5.9009
Elastic			−5.4426	−3.0949	−3.4192
XGBoost				**2.2117**	1.5614
DNN					−0.6476
Input historical sequence = 20					
	Lasso	Elastic	XGBoost	DNN	LSTM-DNN
Ridge	**9.8217**	**7.7303**	**3.3054**	1.3526	1.6922
Lasso		−6.5659	−7.5800	−8.2521	−3.7072
Elastic			−5.3210	−6.6302	−1.9329
XGBoost				−2.6818	0.8142
DNN					1.6550
Panel B: Forecast correlation					
	Lasso	Elastic	XGBoost	DNN	LSTM-DNN
Ridge	0.7766	0.8712	0.9392	0.9650	0.9338
Lasso		0.9492	0.7172	0.6910	0.6768
Elastic			0.7586	0.7727	0.7421
XGBoost				0.9446	0.9311
DNN					0.9599
Input historical sequence = 10					
	Lasso	Elastic	XGBoost	DNN	LSTM-DNN
Ridge	0.7734	0.8678	0.9412	0.9354	0.8939
Lasso		0.9487	0.7098	0.7133	0.6104
Elastic			0.7683	0.7948	0.6820
XGBoost				0.9110	0.9220
DNN					0.91047
Input historical sequence = 20					
	Lasso	Elastic	XGBoost	DNN	LSTM-DNN
Ridge	0.7707	0.8636	0.9337	0.9464	0.8972
Lasso		0.9502	0.7042	0.6789	0.6386
Elastic			0.7692	0.7546	0.7306
XGBoost				0.9279	0.8857
DNN					0.9393

SSE 50 ETF put option					
Panel A: Diebold and Mariano (1995) forecast comparison					
Input historical sequence = 5					
	Lasso	Elastic	XGBoost	DNN	LSTM-DNN
Ridge	**12.7472**	**11.5810**	**4.9353**	1.0158	0.6260
Lasso		−12.670865	−10.0796	−12.3569	−12.6618
Elastic			−5.7118	−10.7406	−11.1531
XGBoost				−4.6245	−4.9052
DNN					−0.9088
Input historical sequence = 10					
	Lasso	Elastic	XGBoost	DNN	LSTM-DNN
Ridge	**11.1933**	**10.5083**	**6.7477**	−1.0466	**3.3975**
Lasso		−10.9886	−11.3342	−11.2175	−10.1648
Elastic			−8.3684	−10.4712	−7.9908
XGBoost				−6.9926	−3.8514
DNN					**4.1583**
Input historical sequence = 20					
	Lasso	Elastic	XGBoost	DNN	LSTM-DNN
Ridge	**9.2816**	**9.3402**	**6.1019**	**4.9333**	**6.7706**
Lasso		−8.8700	−9.5701	−9.2159	−7.6948
Elastic			−3.3058	−9.1935	−5.3737
XGBoost				−5.9432	−2.9299
DNN					**6.3758**
Panel B: Forecast correlation					
	Lasso	Elastic	XGBoost	DNN	LSTM-DNN
Ridge	0.6919	0.7785	0.9465	0.9513	0.9468
Lasso		0.9523	0.6541	0.5508	0.5925
Elastic			0.7403	0.6662	0.7009
XGBoost				0.9282	0.9299
DNN					0.9646
Input historical sequence = 10					
	Lasso	Elastic	XGBoost	DNN	LSTM-DNN
Ridge	0.6570	0.7542	0.9388	0.9707	0.9210
Lasso		0.9546	0.7101	0.5960	0.6060
Elastic			0.7855	0.7128	0.6882
XGBoost				0.9238	0.8969
DNN					0.93087
Input historical sequence = 20					
	Lasso	Elastic	XGBoost	DNN	LSTM-DNN
Ridge	0.5366	0.6722	0.8541	0.9752	0.8275
Lasso		0.9592	0.7566	0.5349	0.5145
Elastic			0.8198	0.6715	0.5993
XGBoost				0.8465	0.7492
DNN					0.8536

its peak R^2 in 2021, explaining over 94% of the outcomes at a sequence of 20. Additionally, in 2020, the model's R^2 at a sequence of 20 is 5.9% higher compared to a sequence of 10. The DNN model also maintains its highest R^2 at a sequence of 20 from 2019 to 2021.

Through analyzing S&P 500 and SSE 50 ETF option samples, we identify several commonalities and distinctions. Both samples indicate that Ridge and DNN models perform notably well with longer historical sequences, suggesting that extensive historical data enhances the precision of implied volatility forecasts. However, there are clear differences in the specific performances of models and the impact of market characteristics on their efficacy. In the S&P 500 sample, the trends in R^2 values for both put and call options are similar, albeit with slight variations. In contrast, the SSE 50 ETF sample reveals more distinct differences. For instance, the Ridge model accounted for over 94% of the outcomes in put option predictions in 2021, illustrating variance in predictive abilities for different option types within this sample.

5.4. Term-structure analysis

This section presents a comprehensive analysis of the Ridge and DNN models' abilities to predict implied volatility across different m and τ categories for S&P500 index options. We first compare the mean R^2 of these two models over the out-of-sample periods in Tables 11 and 12. For the sake of clarity, we only include the results generated using the interpolation methods with the highest mean R^2 values for each model. Furthermore, we opt to analyze the linear and nonlinear models that exhibit the best predictive performance, namely Ridge and DNN. For our S&P500 put option Ridge sample, we use the kernel interpolation method; for all other analyses, the DVF method is employed.[5]

Table 11 presents the analysis of the S&P500 call options. First, the Ridge model exhibits a range of mean R^2 outcomes, with the

[5] We have all other results in the internet Appendix.

Table 11
Forecasting the mean S&P500 R^2 performance of call option term structure. This table gives R^2 of the implied volatility predictions of Ridge and DNN models under different m and τ combinations. The input for these models consists of historical sequence lengths of five, ten, and twenty days. The interpolation method DVF is employed in this evaluation. The evaluation utilizes implied volatility data from S&P500 index call options spanning from 2012 to 2022. Model training encompasses data ranging from 2012 to 2013, with the remaining data reserved for out-of-sample evaluation.

Input historical sequences	Models	m	Short-term	Medium-term	Long-term
5	Ridge	ATM	0.1204	−0.2982	0.2959
		DITM	0.3978	0.5683	0.5858
		DOTM	0.0409	−0.1026	0.1492
		ITM	0.2240	0.2099	0.4330
		OTM	0.0328	−0.0615	0.4246
	DNN	ATM	−0.4154	−0.7302	0.0364
		DITM	−0.1569	−0.1369	−0.0065
		DOTM	−0.0843	0.0445	0.0432
		ITM	−0.2619	−0.1166	0.2690
		OTM	−0.0266	0.2859	0.4681
10	Ridge	ATM	0.2630	0.2579	0.4871
		DITM	0.4361	0.6281	0.6363
		DOTM	0.1918	0.1593	0.2504
		ITM	0.3147	0.5256	0.5699
		OTM	0.2038	0.3073	0.5162
	DNN	ATM	0.0937	0.1080	0.3396
		DITM	0.2895	0.4532	0.4764
		DOTM	0.2032	0.2959	0.2977
		ITM	0.1655	0.2813	0.4207
		OTM	0.1998	0.5284	0.5723
20	Ridge	ATM	0.2935	0.4609	0.5501
		DITM	0.4394	0.6442	0.6492
		DOTM	0.2256	0.2166	0.2477
		ITM	0.3299	0.6340	0.6144
		OTM	0.2445	0.4077	0.5192
	DNN	ATM	0.3230	0.3618	0.4562
		DITM	0.3327	0.5280	0.5145
		DOTM	0.2106	0.2428	0.2199
		ITM	0.3366	0.5271	0.5091
		OTM	0.2755	0.4591	0.5239

Table 12
Forecasting the mean S&P500 R^2 performance of put option term structure. This table gives R^2 of the implied volatility predictions of Ridge and DNN models under different term-structure. The input for these models consists of historical sequence lengths of five, ten, and twenty days. The interpolation method Kernel is employed in this evaluation. The evaluation utilizes implied volatility data from S&P500 index put options spanning from 2012 to 2022. Model training encompasses data ranging from 2012 to 2013, with the remaining data reserved for out-of-sample evaluation.

Input historical sequence length	Models	m	Short-term	Medium-term	Long-term
5	Ridge	ATM	0.2596	0.5475	0.6879
		ITM	0.3048	0.3451	0.1677
		OTM	0.3990	0.6614	0.7164
		DOTM	0.6858	0.8164	0.8791
		VDOTM	0.7067	0.8076	0.7112
	DNN	ATM	−1.6935	−1.5378	−0.2112
		ITM	−0.8442	0.2634	0.3213
		OTM	−1.1085	−0.7755	−0.0593
		DOTM	−0.5157	−0.4773	−0.1936
		VDOTM	−0.5140	−0.6036	−0.4457
10	Ridge	ATM	0.4929	0.6523	0.7027
		ITM	0.3908	0.3910	0.1793
		OTM	0.5900	0.7396	0.7278
		DOTM	0.7775	0.8656	0.8881
		VDOTM	0.7544	0.8409	0.7246
	DNN	ATM	−0.2617	−0.4581	0.1586
		ITM	0.1590	0.2431	0.3029
		OTM	−0.0530	−0.0614	0.2325
		DOTM	0.0811	0.0245	0.1511
		VDOTM	0.0755	0.0423	0.0991
20	Ridge	ATM	0.6259	0.6916	0.6945
		ITM	0.4299	0.3860	0.1625
		OTM	0.6944	0.7701	0.7203
		DOTM	0.8239	0.8845	0.8862
		VDOTM	0.7745	0.8549	0.7276
	DNN	ATM	0.2606	0.4428	0.5216
		ITM	0.2377	0.2573	0.1571
		OTM	0.2811	0.6025	0.5730
		DOTM	0.2226	0.4190	0.4137
		VDOTM	0.1589	0.2592	0.2715

most notable performance seen in the DITM category across all term structures. The model's efficacy is highlighted by its consistent positive mean R^2 values, indicating robust predictability, particularly in the medium and long term where the mean R^2 values are highest. Secondly, a contrasting pattern is observed in the DNN model, which shows negative mean R^2 values in the ATM category for the 5-day historical sequence, suggesting challenges in capturing the volatility dynamics in the short term. However, there is a notable improvement in the medium and long term, especially in the OTM category, where R^2 becomes positive and reaches the highest for the 20-day data length, emphasizing the model's learning curve and adaptability over extended historical periods. Lastly, when comparing the two models' performance for 10-day and 20-day data lengths, both models generally show an increase in R^2 values as the term structure matures, with the Ridge model often outperforming the DNN model in the short term but both models achieving closer R^2 values in the long term. This suggests that while the Ridge model may provide more stable short-term forecasts, the DNN model catches up and even exceeds in certain m categories as the option's term to maturity lengthens.

Table 12 provides the predicted performance in the S&P500 put option. We reveal several key differences between the models in call and put options. For the Ridge model, put options show a positive trend across all m categories, especially in the DOTM and VDOTM categories. This is a notable deviation from the call options, where these categories did not consistently demonstrate such high R^2 values. Moreover, put options' R^2 values generally rise with longer historical sequence. This trend is also seen in call options. However, for puts, the growth in R^2 is more marked in the DOTM and VDOTM categories. Additionally, the DNN model displays a marked contrast between put and call options, particularly in the short-term m categories. Negative R^2 values are significantly more pronounced for put options, suggesting that the DNN model struggles more with the short-term predictions of put options compared to calls. In the medium and long term, the DNN model shows some improvement in the R^2 values for put options, but the values remain more volatile and generally lower than those for call options. Furthermore, in the 20-day historical sequence, the Ridge model maintains a consistent and positive R^2 performance across most m categories for put options, much like call options. The DNN model shows gains from the short to long-term. It posts negative R^2 values in certain m categories for puts. This suggests a less steady prediction rate than with calls.

In summary, the predictive performance differences between S&P500 put and call options are evident in both models. The Ridge model shows a more stable and robust performance in predicting put options across all term structures, particularly for DOTM and VDOTM categories. Conversely, the DNN model's performance is less consistent, with its predictive ability for put options appearing to be more challenging, especially in the short-term and ATM categories.

Tables 13 and 14 provide the predicted performance for the SSE 50 ETF option. We select the R^2 results of Ridge and DNN models under the kernel interpolation method. Looking at call options, all models display similar patterns, with the highest R^2 predominantly observed in ITM options, followed by ATM options, most exceeding 90%. For the Ridge model, except for DITM options in short-term with a sequence of 5 and 10, which achieve the highest R^2, the best results generally occur in medium-term options. Long-term outcomes are consistently the

Table 13
Forecasting the mean SSE 50 ETF R^2 performance of call option term structure. This table gives R^2 of the implied volatility predictions of Ridge and DNN models under different term-structure. The input for these models consists of historical sequence lengths of five, ten, and twenty days. The interpolation method Kernel is employed in this evaluation. The evaluation utilizes implied volatility data from SSE 50 ETF index put options spanning from 2015 to 2022. Model training encompasses data ranging from 2015 to 2016, with the remaining data reserved for out-of-sample evaluation.

Input historical sequence length	Models	m	Short-term	Medium-term	Long-term
5	Ridge	ATM	0.9052	0.9221	0.8159
		OTM	0.8937	0.9165	0.8386
		ITM	0.9114	0.9254	0.7870
		DOTM	0.8955	0.9143	0.8423
		DITM	0.9032	0.8936	0.5635
	DNN	ATM	0.8978	0.9233	0.8123
		OTM	0.8855	0.9179	0.8380
		ITM	0.9031	0.9263	0.7765
		DOTM	0.8821	0.9145	0.8443
		DITM	0.8882	0.8900	0.5231
10	Ridge	ATM	0.9035	0.9202	0.8113
		OTM	0.8923	0.9147	0.8340
		ITM	0.9096	0.9235	0.7832
		DOTM	0.8944	0.9126	0.8404
		DITM	0.8999	0.8898	0.5660
	DNN	ATM	0.7893	0.9221	0.8068
		OTM	0.7649	0.9164	0.8318
		ITM	0.8074	0.9251	0.7732
		DOTM	0.7435	0.9144	0.8460
		DITM	0.8436	0.8839	0.5372
20	Ridge	ATM	0.9002	0.9197	0.8082
		OTM	0.8885	0.9148	0.8328
		ITM	0.9068	0.9231	0.7811
		DOTM	0.8918	0.9139	0.8407
		DITM	0.8982	0.8910	0.5654
	DNN	ATM	0.9053	0.9212	0.8023
		OTM	0.8946	0.9161	0.8299
		ITM	0.9093	0.9238	0.7697
		DOTM	0.8924	0.9141	0.8422
		DITM	0.8897	0.8816	0.5357

Table 14
Forecasting the mean SSE 50 ETF R^2 performance of put option term structure. This table gives R^2 of the implied volatility predictions of Ridge and DNN models under different term-structure. The input for these models consists of historical sequence lengths of five, ten, and twenty days. The interpolation method Kernel is employed in this evaluation. The evaluation utilizes implied volatility data from SSE 50 ETF index put options spanning from 2015 to 2022. Model training encompasses data ranging from 2015 to 2016, with the remaining data reserved for out-of-sample evaluation.

Input historical sequence length	Models	m	Short-term	Medium-term	Long-term
5	Ridge	ATM	0.9033	0.9030	0.7949
		ITM	0.9005	0.9223	0.8197
		OTM	0.8990	0.8993	0.8131
		DOTM	0.8912	0.8981	0.8499
		VDOTM	0.8896	0.9015	0.8391
	DNN	ATM	0.8381	0.9119	0.8010
		ITM	0.7848	0.9243	0.8259
		OTM	0.8393	0.9086	0.8212
		DOTM	0.8347	0.9092	0.8604
		VDOTM	0.8009	0.9142	0.8511
10	Ridge	ATM	0.8952	0.8954	0.7875
		ITM	0.8962	0.9177	0.8170
		OTM	0.8902	0.8909	0.8054
		DOTM	0.8810	0.8884	0.8437
		VDOTM	0.8793	0.8922	0.8343
	DNN	ATM	0.8873	0.9062	0.7970
		ITM	0.8789	0.9209	0.8185
		OTM	0.8856	0.9028	0.8162
		DOTM	0.8860	0.9035	0.8547
		VDOTM	0.8781	0.9069	0.8428
20	Ridge	ATM	0.8970	0.9012	0.8183
		ITM	0.9020	0.9259	0.8361
		OTM	0.8930	0.8982	0.8335
		DOTM	0.8858	0.8969	0.8531
		VDOTM	0.8845	0.8990	0.8386
	DNN	ATM	0.8341	0.9070	0.8175
		ITM	0.8554	0.9255	0.8371
		OTM	0.8267	0.9048	0.8342
		DOTM	0.8078	0.9065	0.8596
		VDOTM	0.7942	0.9101	0.8452

poorest. Similarly, the DNN model follows a comparable trend, except for a sequence of 10, where ATM, OTM, and DOTM options in short-term have the lowest R^2. Long-term predictions are usually the least accurate, particularly for DITM options in both models, with results around 55%. In the put option sample, both models exhibit similar trends. Except for the Ridge ATM at a sequence of 5, which achieves the highest R^2 in short-term, the R^2 values are generally highest in medium-term options, consistently exceeding 90%. Moreover, except at a sequence of 10 where the DNN model predicts VDOTM options with the highest R^2, ITM options usually have the highest R^2 values. Overall, on the SSE 50 ETF sample, these two models demonstrate some common patterns: ITM options and medium-term options tend to yield the highest R^2.

Analyzing the predictive performances of S&P 500 and SSE 50 ETF option samples reveals both similarities and differences between them. A commonality is that Ridge and DNN models tend to deliver the highest R^2 values for mid-term options, particularly ITM options in the SSE 50 ETF sample. This suggests that a longer historical sequence and a mid-term time structure are crucial for enhancing prediction accuracy, regardless of market conditions. However, notable differences exist between the two samples. For instance, in the S&P 500 sample, put options in the DOTM and VDOTM categories show a more positive trend, while in the SSE 50 ETF sample, ITM options across all categories consistently exhibit the highest R^2 values. Additionally, the S&P 500 sample reveals significant challenges for the DNN model in predicting short-term put options. In contrast, the SSE 50 ETF sample analysis does not specifically highlight this issue with short-term forecasts but instead emphasizes that long-term predictions are generally the least accurate, especially for DITM options.

6. Function of features

In this section, we explore the role of different features in the prediction of implied volatility.

6.1. Feature importance

Initially, we utilize the XGBoost model, as mentioned in Section 4.2.1, to examine feature importance. We input features such as m, τ, and the historical sequence ($L = l$) into the XGBoost model, which then constructs numerous decision trees during the training phase. Throughout this construction process, the model calculates a "gain" for each feature at every branching point of the trees. This gain quantifies the extent to which a particular feature enhances the model's predictive capability for the target variable; fundamentally, it reflects the usefulness of a feature in making accurate predictions. Subsequently, the model aggregates these gains across all trees for each feature, providing a measure of each feature's overall importance within the model. Subsequently, the model assigns a rank to each feature according to their aggregate gains, where a higher sum of gains suggests a more significant role of the feature in the predictive accuracy of the model. The feature importance results for S&P500 call and put options generated during the XGBoost model training process are depicted in Fig. 12. We predict the implied volatility on day t, and the historical sequence assumes that there are l days. We use $t-1$ through $t-l$ to represent the features. For example, $t-1$ represents the implied volatility feature for one day before day t.

Fig. 12 displays the feature importance scores generated by XG-Boost on the S&P500 sample. For call options, results using the DVF

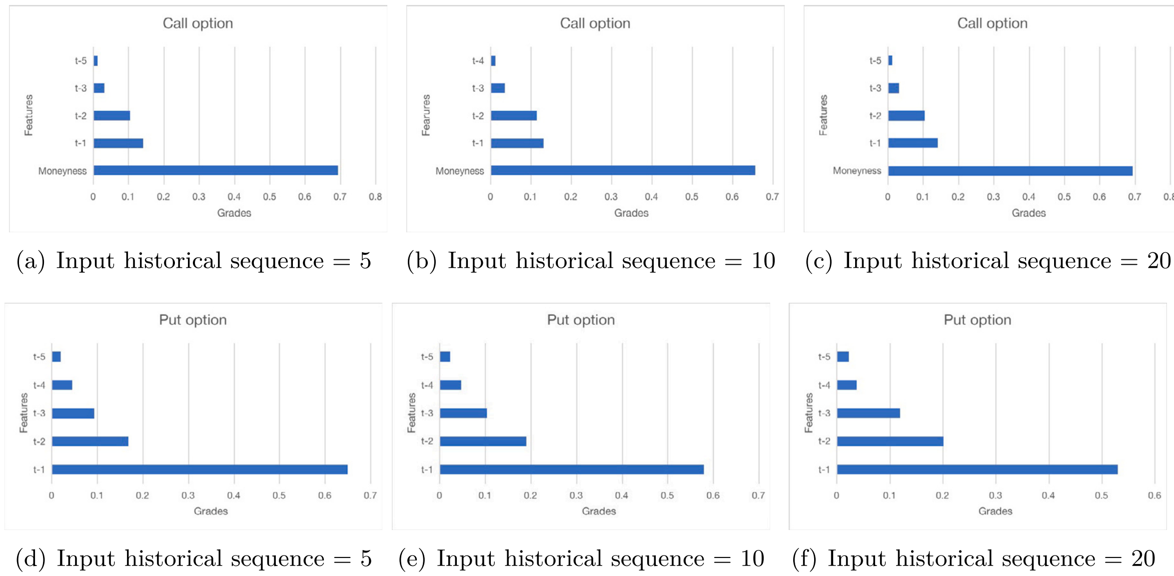


Fig. 12. S&P500 XGBoost model features importance.This figure displays the feature importance for call and put options derived from an XGBoost model. We predict the implied volatility on day t, and the historical sequence assumes there are l days. We use $t-1$ through $t-l$ to represent the features. For example, $t-1$ represents the implied volatility feature for one day before day t. The length of each bar reflects the feature's importance, indicating its influence on the forecasted implied volatility. The longer the bar of a feature, the more significant its role in aligning with the model's current implied volatility prediction. We list the top 5 features on the final results.

interpolation method are shown, while for put options, the kernel method's results are displayed. Figure (a) to (c) illustrates the results of an XGBoost model's feature importance analysis for S&P500 call options, focusing on the top five features for historical time series of 5, 10, and 20 days. In this analysis, call options utilize results from the DVF interpolation method, while put options employ outcomes derived using the kernel method. Across all three tables, 'Moneyness' consistently emerges as the most influential feature, commanding the highest importance scores of 0.693, 0.656, and 0.618 respectively. This indicates a strong correlation between the moneyness of an option and its implied volatility.

The historical implied volatility features, denoted by ($t-1$, $t-2$, $t-3$), and $t-5$ or $t-4$ (depending on the time series length), show a declining trend in importance as we move further back in time from the current day. For a historical sequence of 5 days, the immediate past day's volatility ($t-1$) holds considerable sway, with an importance score of 0.140, suggesting a significant impact of short-term volatility on the model's predictions. This pattern persists across longer historical sequences, with the $t-1$ feature maintaining a prominent role in the model's decision-making process.

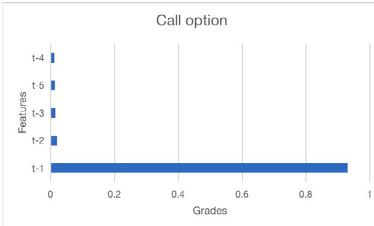
Notably, as the length of the historical sequence increases, the importance of features corresponding to more recent volatility ($t-1$ and $t-2$) remains relatively stable. In contrast, the influence of older volatility data (t-3 and t-4) marginally increases. However, none of the historical volatility features approach the preeminence of moneyness. The absence of τ as a top feature in these tables may indicate that it does not play as critical a role in the model's predictions as moneyness and recent historical volatilities do. This could imply that, within the scope of the data analyzed, moneyness and recent historical volatilities are better predictors of implied volatility for call options on the S&P 500, or it may suggest that the model is capturing the effects of τ indirectly through other features.

Fig. 12(d) to (f) illustrates the results of an XGBoost model's feature importance analysis for put options samples. Across the tables for historical sequences of 5, 10, and 20 days, the absence of 'Moneyness' as a leading feature is noticeable. Instead, the most recent historical volatility feature, $t-1$, dominates the importance rankings with scores of 0.650, 0.579, and 0.530, respectively. This suggests that, for put options, the feature heavily weighs recent market volatility. Another distinction is the relative importance of the subsequent historical volatilities ($t-2, t-3, t-4, t-5$). Unlike the call options, where the importance of historical volatilities decreases more significantly with time, the put options show a less dramatic decrease. The scores indicate a more gradual decline, with $t-2$ and $t-3$ maintaining a substantial influence on the model's predictions.

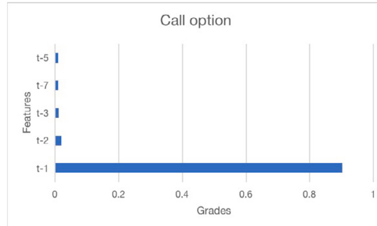
These observations imply that for put options, the model may be more responsive to the immediate past market conditions rather than the intrinsic value represented by moneyness. The consistency in the importance of recent historical volatilities across different time sequences also suggests that the model finds these short-term trends crucial in predicting the implied volatility for put options.

Fig. 13 illustrates the results of an XGBoost model's feature importance analysis for SSE 50 ETF call options, focusing on the top five features for historical time series of 5, 10, and 20 days. In this analysis, call and put options all utilize results from the kernel interpolation method. In our analysis of SSE 50 ETF samples, we observe patterns similar to those found in SP500 samples. Notably, features like Moneyness and Maturity continue to have no significant impact on the forecast results. Moreover, the feature $t-1$ plays a crucial role in the model's predictive process, a trend more pronounced in the SSE 50 ETF dataset. For both call and put options, $t-1$ demonstrates an influence exceeding 0.8 when the historical sequence is 5 and 10, and this influence surpasses 0.9 with a sequence of 20. Additionally, in call option samples, we note that features ($t-2, t-3, t-5, t-7$), despite having scores around 0.01, impact the results under various conditions. Conversely, in put option samples, the prior five historical implied volatilities, from $t-1$ to $t-5$, consistently emerge as the top five influencing factors on the outcomes.

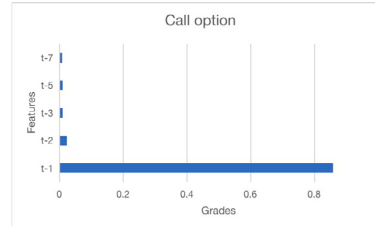
In summary, while moneyness stands out as a primary feature for S&P500 call options, it does not hold the same level of significance for other options within the scope of the analyzed data. Furthermore, the τ feature does not appear as a top-ranking feature in either call or put options, which could indicate its lesser relevance in the model's consideration for implied volatility predictions. Subsequently, we identify a consistent pattern across all samples: the $t-1$ feature plays a vital role. Particularly in the SSE 50 ETF samples, this feature significantly outperforms other characteristics in terms of its influence. Furthermore, historical features closer to the forecasted implied volatility gain importance, as evidenced in the put option tests of both samples, where $t-1$ to $t-5$ consistently emerge as the five most crucial features.



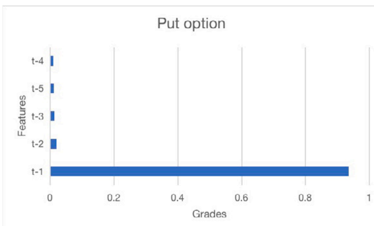
(a) Input historical sequence = 5



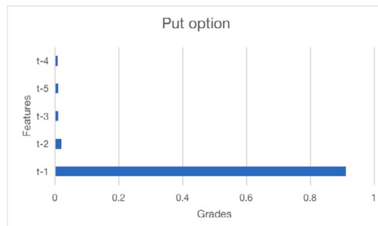
(b) Input historical sequence = 10



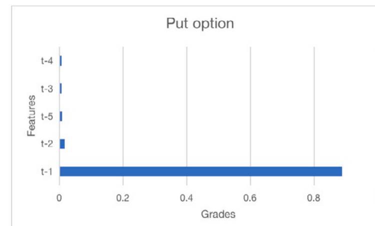
(c) Input historical sequence = 20



(d) Input historical sequence = 5



(e) Input historical sequence = 10



(f) Input historical sequence = 20

Fig. 13. SSE 50 ETF XGBoost model features importance.This figure displays the feature importance for call and put options derived from an XGBoost model. We predict the implied volatility on day t, and the historical sequence assumes there are l days. We use $t-1$ through $t-l$ to represent the features. For example, $t-1$ represents the implied volatility feature for one day before day t. The length of each bar reflects the feature's importance, indicating its influence on the forecasted implied volatility. The longer the bar of a feature, the more significant its role in aligning with the model's current implied volatility prediction. We list the top 5 features on the final results.

6.2. Features analysis of linear models

Next, we utilize the Ridge and Lasso models to analyze the impact of input features on the final prediction. In the process of fitting regularized linear models, each feature is assigned a corresponding coefficient to measure its influence on the target variable. These two models aim to minimize the absolute values of coefficients during the data-fitting process, thereby preventing overfitting. Therefore, we report these coefficients to observe the contribution of each feature to the model. The magnitude of the coefficient indicates the importance of the feature for prediction, with a positive sign denoting a positive correlation and a negative sign denoting a negative correlation.

We analyze the coefficients of the Ridge and Lasso models on the call sample under different historical sequences to understand the impact on the predicted results. We observe the coefficients of the input features and their magnitudes on the S&P500 call option under DVF method, as shown in Fig. 14: From the results of the Ridge model, it is evident that across all historical sequences, $t-1$ consistently stands out as the most crucial feature, possessing a relatively large coefficient. Its sustain importance indicates a robust influence on the prediction outcome. Furthermore, with an increase in the length of historical sequences, the coefficients of time-series features $(t-2, t-3, t-4, t-5)$ gradually diminish. This suggests that while these features maintain significance, their impact gradually weakens with longer historical data.

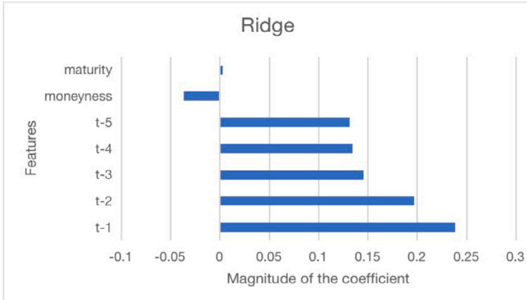
Due to the introduction of sparsity by Lasso, certain coefficients become zero, implying that some features are entirely disregarded in the prediction process. Similar to Ridge, the $t-1$ feature remains pivotal. However, the importance of $t-1$ fluctuates with the length of historical sequences. Interestingly, in Lasso model results, $moneyness$ exhibits significant influence, manifested as relatively large negative coefficients. This indicates that $moneyness$ has an impact on predicting implied volatility for call options in these scenarios. Additionally, the coefficient of $maturity$ is reduced to zero in all historical sequence cases. In the cases where the historical sequence is 10 and 20, the Lasso model sets the coefficients of both $t-6$ and $t-7$ features to zero. Moreover, in the scenario with a historical sequence of 20, the historical implied volatilities from $t-11$ to $t-20$ are all reduced to zero.

In summary, both models emphasize the importance of time-series features $(t-1, t-2, t-3, t-4)$ in predicting implied volatility for call options. Despite varying coefficients, these features retain significance across different historical sequences.

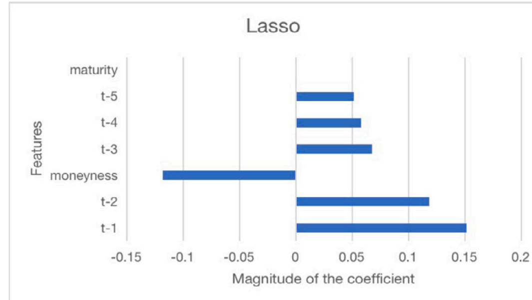
We observe the coefficients of the input features and their magnitudes on the S&P500 put option under the kernel method, as shown in Fig. 15: We discover patterns on put samples similar to those on call samples. For instance, in any historical sequence, both models exhibit a significant coefficient for $t-1$, and in the ridge model, the $t-2$ feature plays a crucial role. However, there are notable differences in the results of the Lasso model compared to the call samples. For example, maturity is not reduced to zero in all three sequence scenarios, instead showing a slight negative correlation with the results. Additionally, there are distinct patterns in features reduced to zero during the Lasso model prediction process on put samples. For instance, when the historical sequence is 5, the coefficients of the $t-5$ feature are equal to 0. In cases where the historical sequence is 10 and 20, the model discards all historical implied volatilities preceding $t-5$.

We observe the coefficients of the input features and their magnitudes on the SSE 50 ETF option, as shown in Figs. 16 and 17: In our analysis of this sample, we observe patterns akin to those in the S&P500. Notably, the $t-1$ feature consistently plays the most pivotal role. In both call and put options, the magnitude of this feature's coefficient in the Ridge model exceeds 0.7, while in the Lasso model, it surpasses 0.6, indicating greater significance in the Ridge model. Additionally, the $t-2$ feature consistently ranks as the second most influential in determining the outcomes. Furthermore, both maturity and moneyness continue to exert minimal, almost negligible, impact on the forecast results. In the Lasso model's forecast for call options, only four features are retained: $t-1$, $t-2$, moneyness, and maturity, with others reduced to zero. Conversely, in put options, only $t-1$ and $t-2$ features are retained.

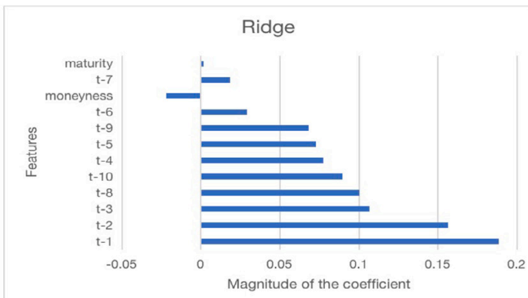
From the feature importance results of the linear model and XGBoost model, we observe similar patterns. The historical volatility of the day before the predicted implied volatility $(t-1)$ plays a crucial role in the prediction, while maturity consistently lacks significance. This pattern is particularly pronounced in the SSE 50 ETF sample, where the importance scores and coefficients calculated for this feature significantly surpass those of other features. Additionally, the t-2 feature consistently emerges as the second most important. Moreover, we observe that historical features closer to the predicted value are increasingly significant.



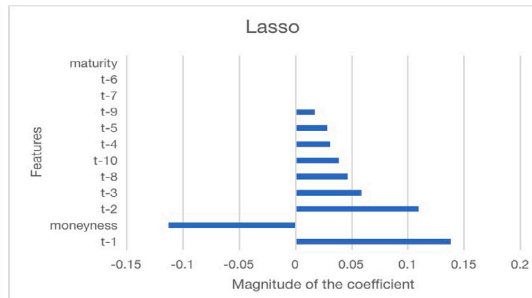
(a) Input historical sequence = 5



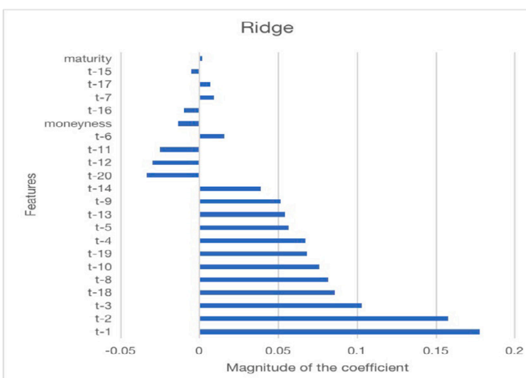
(b) Input historical sequence = 5



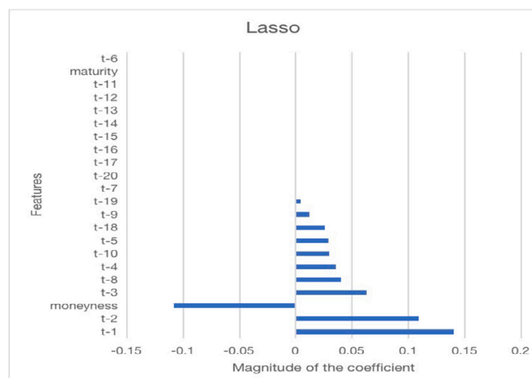
(c) Input historical sequence = 10



(d) Input historical sequence = 10



(e) Input historical sequence = 20



(f) Input historical sequence = 20

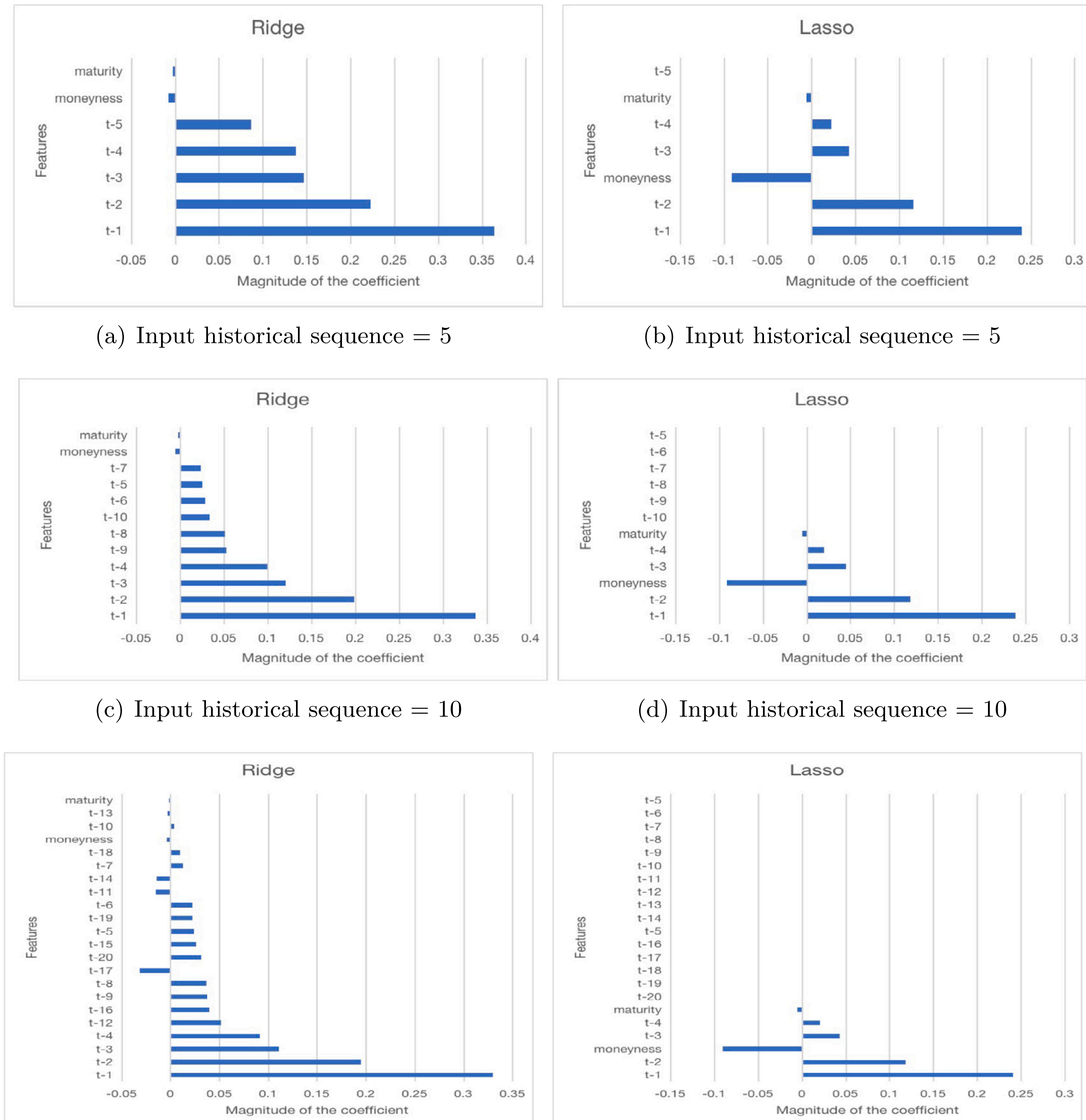
Fig. 14. S&P500 call option magnitude of the coefficient. This figure displays the magnitude of the coefficient for call options derived from Ridge and Lasso models. We predict the implied volatility on day t, and the historical sequence assumes there are l days. We use $t-1$ through $t-l$ to represent the features. For example, $t-1$ represents the implied volatility feature for one day before day t. The length of each bar reflects the magnitude of the coefficient, indicating its influence on the forecasted implied volatility. The longer the bar of a feature, the more significant its role in aligning with the model's current implied volatility prediction. If there is no corresponding bar for a feature in the figure, it indicates that the Lasso model has reduced the coefficient of that feature to 0.

Consequently, we further substantiate that implied volatility is predominantly past-dependence, showing greater reliance on recent historical implied volatilities. This is particularly evident in the results of put options in both samples, where features from t-1 to t-5 markedly influence the final forecast, demonstrating a specific 'past path' that affects the prediction outcome. Moreover, the influence of historical implied volatility diminishes over time, aligning with the concept of fading memory in path-dependent systems. This suggests that while historical data remains pertinent, its impact on current conditions weakens as it ages, indicating a 'decay' in the influence of past volatility. However, the Lasso model, in its forecasting process, reduces the coefficients of some historical implied volatilities to zero within the historical sequence. In contrast, the Ridge model's results are superior in both samples. This leads us to infer that, even if certain features minimally impact the final prediction, their exclusion still lessens the model's predictive capability. Therefore, in these option markets, recent implied volatility likely reflects the latest market information or sentiment.

7. Trading strategies

Initially, we consolidate all S&P 500 call and put contracts. The top three contracts are then selected based on the largest differences between their predicted IV and the current Black–Scholes (B–S) IV for one day. If the predicted IV is expected to be higher than its current value, investors take a long position by buying volatility contracts. Conversely, a short position is adopted if the predicted IV is lower (Konstantinidi et al., 2008). The following day, we reassess and choose the top three contracts again. If a contract continues to be among these top three, it is hold; otherwise, the position is liquidated.

The Fig. 18 reflects the positive performance of trading strategies based on Ridge and DNN models across different years. The Annual Return from the Ridge model varies significantly, ranging from 28.53% in 2014 to 617.10% in 2022, with a notable negative return of −112.05% in 2021. Similarly, the DNN model demonstrates variable performance, outperforming Ridge in certain years like 2015 and 2022, but also showing a substantial negative return of −179.80% in 2021. Regarding



(a) Input historical sequence = 5 (b) Input historical sequence = 5

(c) Input historical sequence = 10 (d) Input historical sequence = 10

(e) Input historical sequence = 20 (f) Input historical sequence = 20

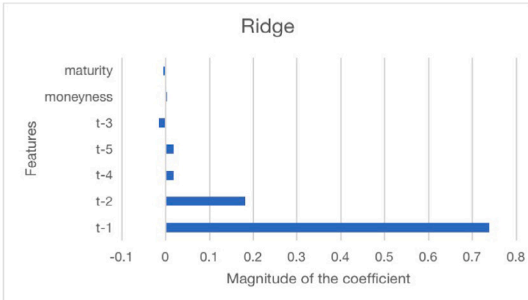
Fig. 15. S&P500 put option magnitude of the coefficient. This figure displays the magnitude of the coefficient for put options derived from Ridge and Lasso models. We predict the implied volatility on day t, and the historical sequence assumes there are l days. We use $t-1$ through $t-l$ to represent the features. For example, $t-1$ represents the implied volatility feature for one day before day t. The length of each bar reflects the magnitude of the coefficient, indicating its influence on the forecasted implied volatility. The longer the bar of a feature, the more significant its role in aligning with the model's current implied volatility prediction. If there is no corresponding bar for a feature in the figure, it indicates that the Lasso model has reduced the coefficient of that feature to 0.

the Sharpe ratio, the Ridge model shows an average value of 0.1098, oscillating between positive and negative values. It notably outperforms the DNN model in 2015–2016, 2018, and 2020–2021, with a peak of 0.2804 in 2015. Overall, both models show comparatively better performance in 2020 and 2022, but poor results in 2021, not only failing to exceed the risk-free rate but also incurring losses. Our strategy results reveal that, in comparison to the work of Bernales and Guidolin (2014) and Konstantinidi et al. (2008), we achieve a relatively substantial Sharpe ratio.

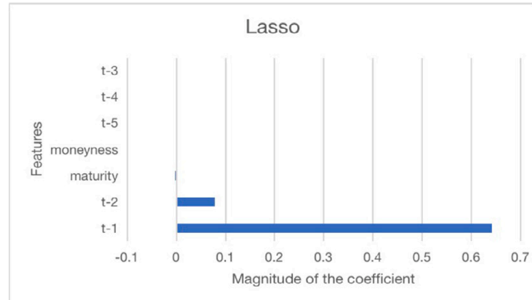
8. Conclusion

This study explores and predicts the implied volatility of the S&P500 and SSE 50 ETF options, conducting empirical analyses to address two key questions: first, whether the lagged information in implied volatility substantially influences its predictive power. Second, whether the capability of machine learning models to identify nonlinear relationships improves prediction accuracy, or if the prediction mainly hinges on the inherent past-dependence of the data. We analyze three linear models: Lasso, Ridge, and Elastic, each utilizing a different regularization approach. Complementing these, we also explore three nonlinear models - XGBoost, DNN, and the hybrid LSTM-DNN, specifically designed for extracting temporal features from historical data. The models' out-of-sample predictions are assessed using R^2 and RMSE metrics. For comparative analysis of their predictive abilities, we apply the DM method. Additionally, we analyze the models' dynamic performance across various years and term structures. Finally, we examine the importance of features and construct a trading strategy to test the efficacy of our predictive results.

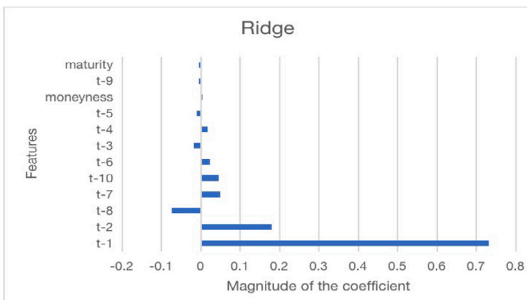
Our study confirms the presence of past-dependence in implied volatility predictions within both linear and nonlinear models. Initially, we explore the impact of historical implied volatility over the preceding 5, 10, and 20 days on prediction outcomes. Empirical results from out-of-sample testing reveal that in most models, longer historical sequences lead to higher R^2 values. Moreover, our results show that



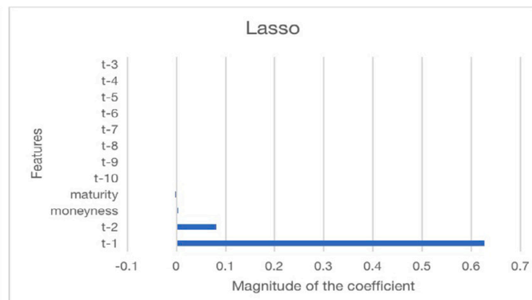
(a) Input historical sequence = 5



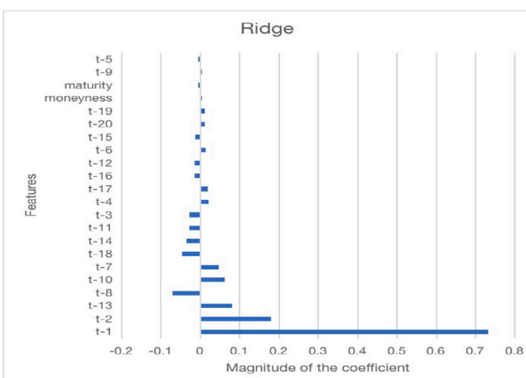
(b) Input historical sequence = 5



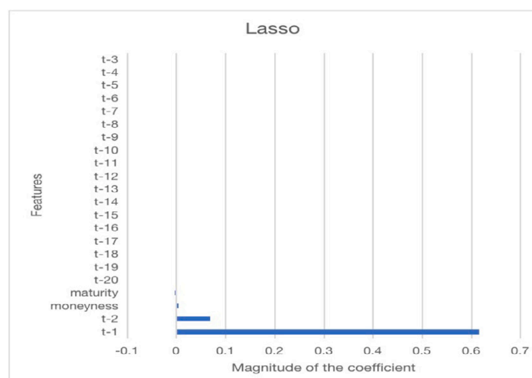
(c) Input historical sequence = 10



(d) Input historical sequence = 10



(e) Input historical sequence = 20



(f) Input historical sequence = 20

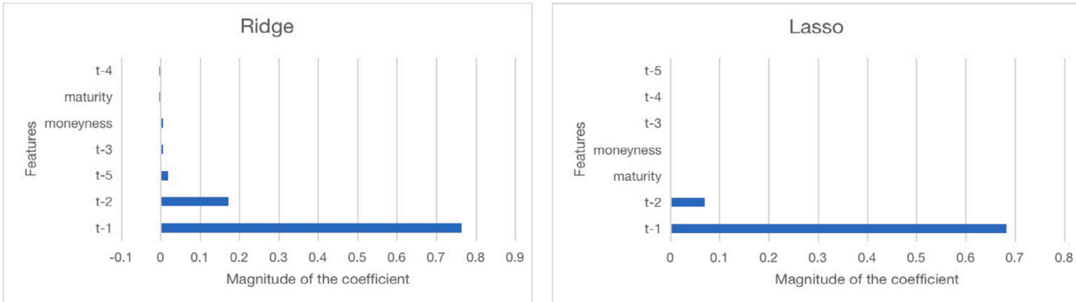
Fig. 16. SSE 50 ETF call option magnitude of the coefficient. This figure displays the magnitude of the coefficient for call options derived from Ridge and Lasso models. We predict the implied volatility on day t, and the historical sequence assumes there are l days. We use $t-1$ through $t-l$ to represent the features. For example, $t-1$ represents the implied volatility feature for one day before day t. The length of each bar reflects the magnitude of the coefficient, indicating its influence on the forecasted implied volatility. The longer the bar of a feature, the more significant its role in aligning with the model's current implied volatility prediction. If there is no corresponding bar for a feature in the figure, it indicates that the Lasso model has reduced the coefficient of that feature to 0.

the impact of all preceding values exists but steadily diminishes over time.

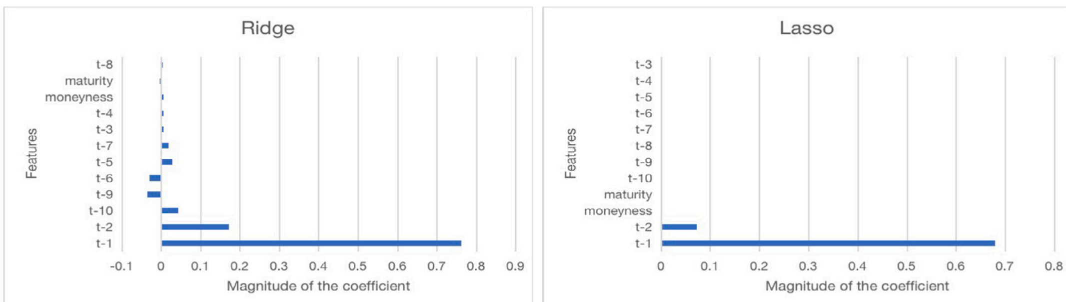
Also, our findings indicate that Ridge linear models with L2 regularization consistently demonstrate superior predictive performance, explaining up to 80% of the variance in out-of-sample. While advanced nonlinear models, such as deep neural networks, explain about 78% of the variance, their performance is comparable to, or even slightly lower than, that of linear models. Our results further affirm that simpler models in financial forecasting can yield unexpectedly better outcomes.

Our study presents financial practitioners with a reference strategy for predicting implied volatility. This strategy involves linear interpolation of implied volatility when necessary, employing a basic linear model with L2 regularization, and incorporating preceding volatilities spanning at least 20 days. This approach is highly likely to achieve a high R^2 value. Additionally, our findings highlight the significant impact of the previous moment's implied volatility on predictions, an aspect crucial for practical trading. Our empirical results hold consistently across both liquid, mature markets and less liquid, emerging markets. Furthermore, a trading strategy based on our forecasts yielded notable returns and a Sharpe ratio, demonstrating the profitability of our modeling approach in trading. However, our strategy is relatively simple and does not account for some real-market complexities.

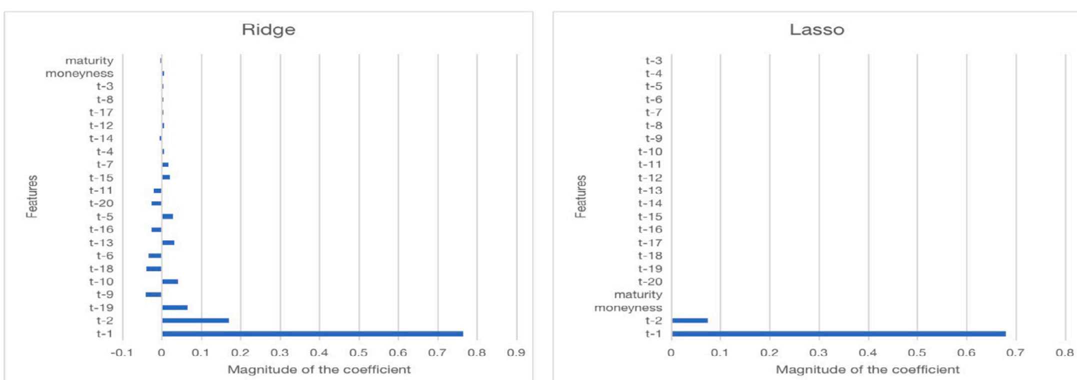
With AI's evolution, nonlinear models have been anticipated and shown to outperform linear ones in financial data forecasting. However, our research, alongside (Guyon & Lekeufack, 2023), converges to the conclusion that linear models excel in predicting implied volatility. This superiority could stem from a stronger linear relationship between implied volatility and its features, which nonlinear models may fail to capture effectively. Additionally, due to their simplicity, linear models can significantly reduce time and computational costs for market practitioners in real-world forecasting. On another note, machine learning broadens research avenues, with tools like XGBoost models offering direct rankings of features' predictive importance. Nonetheless, our study has limitations, such as focusing on simpler models and not considering aspects like static arbitrage in constructing implied volatility surfaces



(a) Input historical sequence = 5 (b) Input historical sequence = 5

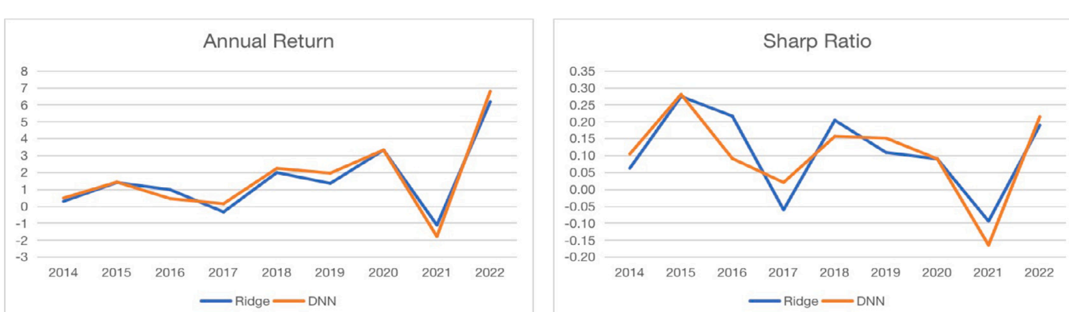


(c) Input historical sequence = 10 (d) Input historical sequence = 10



(e) Input historical sequence = 20 (f) Input historical sequence = 20

Fig. 17. SSE 50 ETF put option magnitude of the coefficient. This figure displays the magnitude of the coefficient for put options derived from Ridge and Lasso models. We predict the implied volatility on day t, and the historical sequence assumes there are l days. We use $t-1$ through $t-l$ to represent the features. For example, $t-1$ represents the implied volatility feature for one day before day t. The length of each bar reflects the magnitude of the coefficient, indicating its influence on the forecasted implied volatility. The longer the bar of a feature, the more significant its role in aligning with the model's current implied volatility prediction. If there is no corresponding bar for a feature in the figure, it indicates that the Lasso model has reduced the coefficient of that feature to 0.



(a) Input historical sequence = 5 (b) Input historical sequence = 5

Fig. 18. Trading strategy with S&P500 option. These figures display the annual returns and Sharpe ratios for S&P 500 options, following the application of trading strategies based on implied volatility predictions using Ridge and DNN models. We exclusively utilize results from the DVF interpolation method. Additionally, we conducted simulated trades for all out-of-sample periods, encompassing all call and put options from 2014 to 2022.

Table A.1
Linear models hyperparameters.This table presents the parameter configurations for three linear models.

Hyperparameters	Lasso	Ridge	Elastic
alpha	0.001	0.5	0.001
tol	0.001	0.0001	0.0001
max_iter	1000		1000
l1_ratio			0.5

Table A.2
The XGboost model hyperparameters.This table presents the parameter configurations for the XGboost model.

Hyperparameters	Choice
importance_type	Gain
learning_rate	0.3
max_depth	6
min_child_weight	1
subsample	0.5
early_stopping	10

Table A.3
The LSTM and DNN Models Hyperparameters.This table presents the parameter configurations for the LSTM and DNN models.

Hyperparameters	LSTM	DNN
Learning rate	0.001	0.001
Batch-size	32	128
Layer	2	1
Hidden neurons	10	10
Epochs	100	100
Optimizer	Adam	Adam
Activation function	ReLU	ReLU
Loss function	MSE	MSE

and feature matrices. Our findings do not suggest linear models are universally the best choice for all forecasting scenarios. Given the financial market's complexity and variability, exploring linear and nonlinear models across different market conditions and datasets remains crucial.

Data availability

Data will be made available on request.

Acknowledgment

This work was supported by the National Natural Science Foundation of China, Young Scientists Fund [grant number 72101207].

Appendix A. Models hyperparameters

This section provides details on the relevant parameters of the models listed in Section 4 (see Tables A.1–A.3).

Appendix B. Forecast performance across years

In this section, we present the yearly R^2 performance of models not reported in Section 5.3. Specifically, for S&P500 call options, we supplement the results for Lasso, XGBoost, and LSTM-DNN under the DVF method. For put options, we add the results of all models except Ridge under the DVF method and all models except DNN under the Kernel method. Regarding the SSE 50 ETF sample, we include the results of all models for both call and put options using the DVF interpolation method (see Figs. B.1–B.4).

Table C.1
Forecasting S&P500 R^2 performance of call option term structure with 5-historical sequence. This table gives R^2 of the implied volatility predictions under different m and τ combinations. The input for these models consists of historical sequence lengths of five, ten, and twenty days. we provides the models which R^2 values are not reported in Section 5.4. Specifically, it includes the Lasso, XGBoost, and LSTM-DNN models under the DVF method, as well as all models employing the kernel method. The evaluation utilizes implied volatility data from S&P500 index call options spanning from 2012 to 2022. Model training encompasses data ranging from 2012 to 2013, with the remaining data reserved for out-of-sample evaluation.

Interpolation method	Models	m	Short-term	Medium-term	Long-term
DVF	Lasso	ATM	−3.7243	−14.4790	−5.2309
		DOTM	−1.6249	−9.1649	−4.5970
		ITM	−3.0061	−5.0385	−1.6577
		OTM	−2.2364	−7.7459	−3.4636
		VDOTM	−1.1034	−3.2627	−2.6772
	XGBoost	ATM	−0.2432	−1.5758	−0.0599
		DOTM	−0.3507	−2.1268	−0.9659
		ITM	0.1201	0.0472	0.2066
		OTM	−0.1589	−0.9821	−0.1692
		VDOTM	−0.4348	−1.2786	−1.2225
	LSTM-DNN	ATM	−0.6458	−0.3498	−0.0678
		DOTM	−0.6644	−4.8752	−3.0547
		ITM	−0.1516	0.1255	0.1856
		OTM	−0.3399	−0.8084	−0.4356
		VDOTM	−0.9813	−2.7436	−2.7491
Kernel	Lasso	ATM	−103.9123	−64.3899	−17.6855
		DOTM	−27.8629	−18.0084	−2.5983
		ITM	−92.8814	−70.1386	−12.5427
		OTM	−94.1185	−50.6120	−14.5130
		VDOTM	−9.0367	−6.6344	−1.2314
	Ridge	ATM	−5.8936	−2.6359	−0.2917
		DOTM	−0.1469	0.1450	0.2886
		ITM	−10.4398	−4.5088	−0.2922
		OTM	−5.2905	−2.0927	−0.1603
		VDOTM	−5.5549	−3.6510	−0.7125
	XGBoost	ATM	−27.4016	−10.1526	−1.7948
		DOTM	−4.0680	−2.3776	−0.5242
		ITM	−14.9814	−5.5913	−0.9802
		OTM	−28.4969	−11.3296	−1.9912
		VDOTM	−14.0094	−6.8430	−1.4741
	DNN	ATM	−8.4043	−4.0825	−0.4799
		DOTM	−4.7188	−2.6538	−0.4116
		ITM	−5.1449	−2.7169	−0.2308
		OTM	−8.7661	−3.5773	−0.4144
		VDOTM	−2.3433	−2.3886	−0.7384
	LSTM-DNN	ATM	−8.4043	−4.0825	−0.4799
		DOTM	−4.7188	−2.6538	−0.4116
		ITM	−5.1449	−2.7169	−0.2308
		OTM	−8.7661	−3.5773	−0.4144
		VDOTM	−2.3433	−2.3886	−0.7384

Appendix C. Term-structure analysis

In this section, we present the yearly R^2 performance of models not reported in Section 5.4. Specifically, for S&P500 call options, we supplement the results for Lasso, XGBoost, and LSTM-DNN under the DVF method. For put options, we add the results of all models except Ridge under the DVF method and all models except DNN under the Kernel method. Regarding the SSE 50 ETF sample, we include the results of all models for both call and put options using the DVF interpolation method. Subsequently, our tables list the results of different models across various historical sequences and compare two interpolation methods within these tables (see Tables C.1–C.12).

Fig. B.1. Forecasting S&P500 call options R^2 performance across years. This figure presents the R^2 of the forecasting performance of implied volatility for Call options on the S&P 500 Index. we provides the models which R^2 values are not reported in Section 5.3. Specifically, it includes the Lasso, XGBoost, and LSTM-DNN models under the DVF method, as well as all models employing the kernel method. The evaluation encompasses out-of-sample time periods spanning from 2014 to 2022. The models are fed with historical sequence of varying durations, specifically five, ten, and twenty days. A comparative examination of R^2 values is conducted across diverse historical sequence lengths, with a specific focus on implied volatility for Call options over the specified time frame.

Fig. B.2. Forecasting S&P500 put options R^2 performance across years. This figure presents the R^2 of the forecasting performance of implied volatility for Put options on the S&P 500 Index. we provides the models which R^2 values are not reported in Section 5.3. Specifically, it includes the Ridge, Lasso, XGBoost, and LSTM-DNN models under the DVF method, as well as all models except DNN employing the kernel method. The evaluation encompasses out-of-sample time periods spanning from 2014 to 2022. The models are fed with historical sequence of varying durations, specifically five, ten, and twenty days. A comparative examination of R^2 values is conducted across diverse historical sequence lengths, with a specific focus on implied volatility for Call options over the specified time frame.

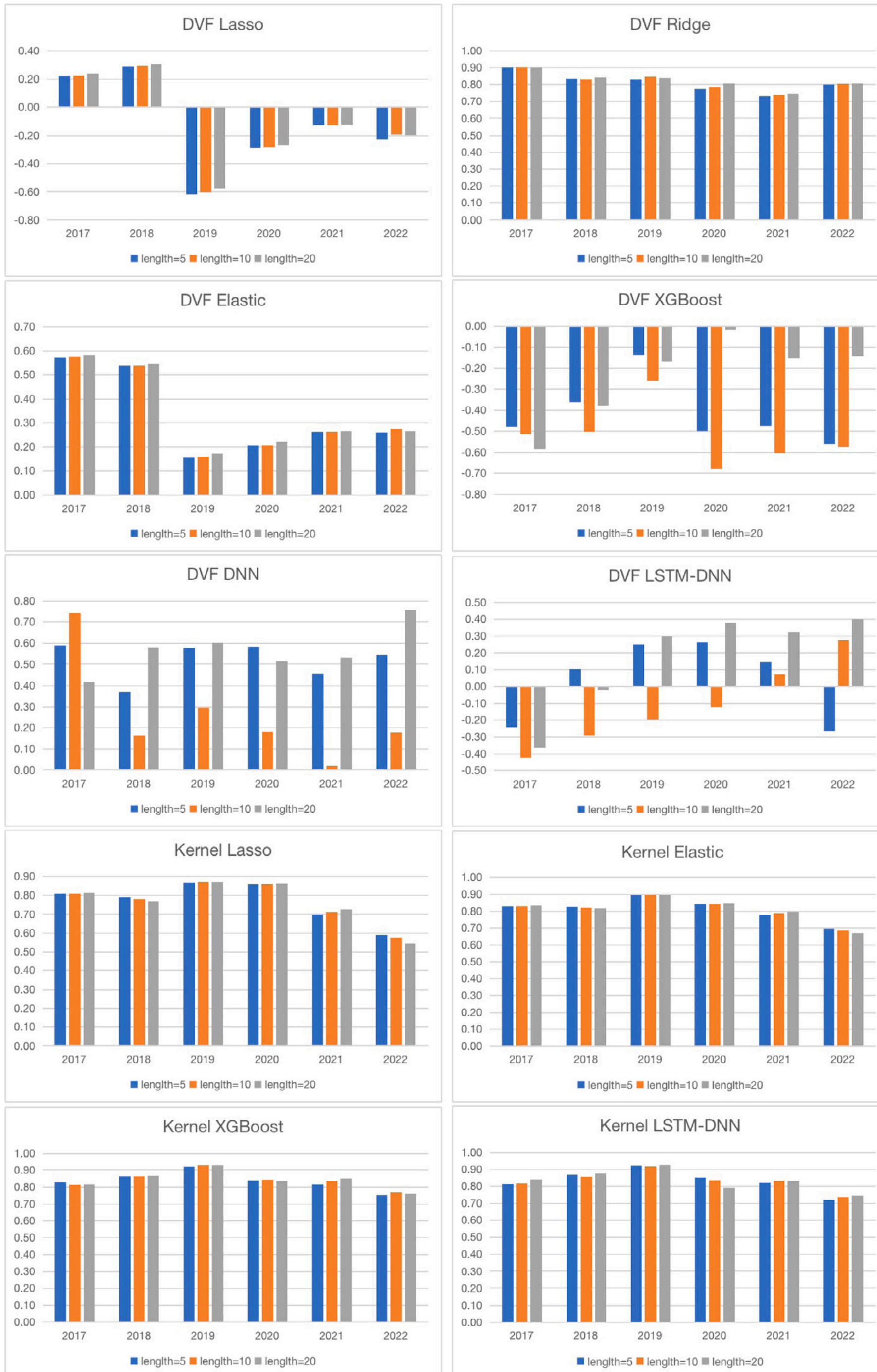


Fig. B.3. Forecasting SSE 50 ETF call options R^2 performance across years. This figure presents the R^2 of the forecasting performance of implied volatility for Call options on the SSE 50 ETF Call options. we provides the models which R^2 values are not reported in Section 5.3. We include the results of all models for call options using the DVF interpolation method, and models except Ridge and DNN under kernel method. The evaluation encompasses out-of-sample time periods spanning from 2017 to 2022. The models are fed with historical sequence of varying durations, specifically five, ten, and twenty days. A comparative examination of R^2 values is conducted across diverse historical sequence lengths, with a specific focus on implied volatility for Call options over the specified time frame.

Fig. B.4. Forecasting SSE 50 ETF call options R^2 performance across years. This figure presents the R^2 of the forecasting performance of implied volatility for Call options on the SSE 50 ETF Call options. we provides the models which R^2 values are not reported in Section 5.3. We include the results of all models for call options using the DVF interpolation method, and models except Ridge and DNN under kernel method. The evaluation encompasses out-of-sample time periods spanning from 2017 to 2022. The models are fed with historical sequence of varying durations, specifically five, ten, and twenty days. A comparative examination of R^2 values is conducted across diverse historical sequence lengths, with a specific focus on implied volatility for Call options over the specified time frame.

Table C.2
Forecasting S&P500 R^2 performance of call option term structure with 10-historical sequence. This table gives R^2 of the implied volatility predictions under different m and τ combinations. The input for these models consists of historical sequence lengths of 20 days. we provides the models which R^2 values are not reported in Section 5.4. Specifically, it includes the Lasso, XGBoost, and LSTM-DNN models under the DVF method, as well as all models employing the kernel method. The evaluation utilizes implied volatility data from S&P500 index call options spanning from 2012 to 2022. Model training encompasses data ranging from 2012 to 2013, with the remaining data reserved for out-of-sample evaluation.

Interpolation method	Models	m	Short-term	Medium-term	Long-term
DVF	Lasso	ATM	−3.5058	−14.3109	−5.0968
		DOTM	−1.5220	−8.8888	−4.4054
		ITM	−2.8042	−4.9693	−1.6188
		OTM	−2.0898	−7.4897	−3.3228
		VDOTM	−1.0430	−3.1706	−2.5914
	XGBoost	ATM	−0.1492	−1.1678	−0.1755
		DOTM	−0.3953	−1.6684	−0.8352
		ITM	0.1446	0.0743	0.2443
		OTM	−0.1407	−0.8004	−0.1661
		VDOTM	−0.4882	−1.1698	−1.0898
	LSTM-DNN	ATM	−1.6260	−0.5178	0.1197
		DOTM	−0.5943	−2.6325	−1.3230
		ITM	−0.3603	0.2412	0.3253
		OTM	−0.6915	−0.2722	0.1407
		VDOTM	−0.8426	−2.0556	−1.9517
Kernel	Lasso	ATM	−105.0529	−65.6424	−17.8018
		DOTM	−9.3800	−5.8666	−0.9996
		ITM	−173.4107	−82.8636	−11.7116
		OTM	−94.9654	−51.3885	−14.6266
		VDOTM	−98.4052	−87.4390	−27.2170
	Ridge	ATM	−3.2936	−1.4241	0.0549
		DOTM	0.1271	0.3233	0.3284
		ITM	−6.5221	−2.5024	0.0088
		OTM	−2.9167	−1.0311	0.1405
		VDOTM	−3.1072	−2.1108	−0.2126
	XGBoost	ATM	−25.5912	−9.1243	−1.2721
		DOTM	−3.1800	−1.6240	−0.3101
		ITM	−19.2576	−6.0430	−0.6729
		OTM	−25.4304	−9.3323	−1.4131
		VDOTM	−16.2113	−7.1587	−1.0587
	DNN	ATM	−13.1472	−2.7585	−0.5230
		DOTM	−1.3589	−0.5106	−0.0308
		ITM	−16.5329	−8.3800	−1.9247
		OTM	−12.5796	−2.5279	−0.4076
		VDOTM	−10.2510	−3.6531	−1.1023
	LSTM-DNN	ATM	−12.5311	−3.9033	−6.5720
		DOTM	−9.3037	−5.7012	−1.3424
		ITM	−3.7765	−2.1484	−11.7197
		OTM	−14.0326	−4.4246	−5.3107
		VDOTM	−6.3882	−4.2437	−0.8862

Table C.3
Forecasting S&P500 R^2 performance of call option term structure with 20-historical sequence. This table gives R^2 of the implied volatility predictions under different m and τ combinations. The input for these models consists of historical sequence lengths of 10 days. we provides the models which R^2 values are not reported in Section 5.4. Specifically, it includes the Lasso, XGBoost, and LSTM-DNN models under the DVF method, as well as all models employing the kernel method. The evaluation utilizes implied volatility data from S&P500 index call options spanning from 2012 to 2022. Model training encompasses data ranging from 2012 to 2013, with the remaining data reserved for out-of-sample evaluation.

Interpolation method	Models	m	Short-term	Medium-term	Long-term
DVF	Lasso	ATM	−3.2529	−13.8152	−4.8166
		DOTM	−1.4157	−8.8503	−4.2668
		ITM	−2.6337	−5.0292	−1.5722
		OTM	−1.9334	−7.2231	−3.1116
		VDOTM	−0.9886	−3.1259	−2.5357
	XGBoost	ATM	−0.1492	−1.1678	−0.1755
		DOTM	−0.3953	−1.6684	−0.8352
		ITM	0.1446	0.0743	0.2443
		OTM	−0.1407	−0.8004	−0.1661
		VDOTM	−0.4882	−1.1698	−1.0898
	LSTM-DNN	ATM	−1.6260	−0.5178	0.1197
		DOTM	−0.5943	−2.6325	−1.3230
		ITM	−0.3603	0.2412	0.3253
		OTM	−0.6915	−0.2722	0.1407
		VDOTM	−0.8426	−2.0556	−1.9517
Kernel	Lasso	ATM	−108.2438	−67.5252	−18.0729
		DOTM	−27.7071	−18.2592	−2.5849
		ITM	−96.2325	−74.6371	−12.7171
		OTM	−96.9950	−52.4475	−14.8611
		VDOTM	−8.8321	−6.6533	−1.2226
	Ridge	ATM	−2.6300	−1.1378	0.1501
		DOTM	0.1798	0.3643	0.3351
		ITM	−5.4584	−2.0761	0.0858
		OTM	−2.2837	−0.7391	0.2315
		VDOTM	−2.5148	−1.7858	−0.1030
	XGBoost	ATM	−26.7045	−10.2951	−2.3247
		DOTM	−2.8328	−1.4629	−0.2418
		ITM	−11.7951	−5.3848	−1.2024
		OTM	−27.3127	−11.3481	−2.4512
		VDOTM	−15.3385	−6.3205	−1.9235
	DNN	ATM	−21.3445	−0.6270	0.3535
		DOTM	−2.6856	−0.3697	−0.2093
		ITM	−39.8436	−2.3745	−0.1558
		OTM	−18.2943	−0.5882	0.3436
		VDOTM	−23.0736	−0.7509	0.3338
	LSTM-DNN	ATM	−11.5301	−6.4186	−1.0929
		DOTM	−6.1903	−4.9359	−0.7685
		ITM	−7.3332	−4.9241	−0.6379
		OTM	−11.1246	−5.4385	−0.9695
		VDOTM	−3.8296	−3.7668	−0.9128

Table C.4
Forecasting S&P500 R^2 performance of put option term structure with 5-historical sequence. This table gives R^2 of the implied volatility predictions under different m and τ combinations. The input for these models consists of historical sequence lengths of 5 days. we provides the models which R^2 values are not reported in Section 5.4. Specifically, it includes the Ridge, Lasso, XGBoost, and LSTM-DNN models under the DVF method, as well as all models except DNN employing the kernel method. The evaluation utilizes implied volatility data from S&P500 index call options spanning from 2012 to 2022. Model training encompasses data ranging from 2012 to 2013, with the remaining data reserved for out-of-sample evaluation.

Interpolation method	Models	m	Short-term	Medium-term	Long-term
DVF	Lasso	ITM	−1.0339	−2.3828	−2.1156
		ATM	−4.2144	−6.7809	−2.5162
		OTM	−1.2941	−3.7226	−2.6395
		DOTM	−3.1395	−10.1910	−4.6265
		VDOTM	−5.2973	−19.6606	−7.2850
	Ridge	ITM	0.1802	0.1871	0.2889
		ATM	0.1692	0.1816	0.2328
		OTM	−0.0035	−0.3400	0.1934
		DOTM	−0.1904	−0.4916	0.0280
		VDOTM	−0.2838	−1.3881	−0.0752
	XGBoost	ITM	−0.0552	−0.1383	0.1666
		ATM	−0.1221	−0.7138	0.1168
		OTM	−0.2265	−1.0598	−0.0196
		DOTM	−0.4823	−1.2624	−0.8586
		VDOTM	−1.3967	−3.2984	−3.0694
	LSTM-DNN	ITM	−4.8479	−0.4926	0.0764
		ATM	−8.3139	−2.1674	−0.5970
		OTM	−8.6074	−1.7132	−0.7061
		DOTM	−10.1051	−5.6491	−4.0672
		VDOTM	−8.0573	−6.1379	−4.6900
Kernel	DNN	ITM	−6.4383	−2.1467	−0.2688
		ATM	−1.4927	−0.0538	0.3049
		OTM	−5.6153	−2.5145	−0.5496
		DOTM	−4.7124	−1.0149	−0.0064
		VDOTM	−0.9056	−0.1308	−0.2394
	Lasso	ITM	−57.0978	−32.0073	−9.2976
		ATM	−22.2786	−13.9376	−3.1785
		OTM	−19.8890	−10.6732	−1.7817
		DOTM	−45.6504	−24.4589	−7.9284
		VDOTM	−11.3581	−9.4384	−3.1017
	XGBoost	ITM	−12.6751	−4.7865	−0.5627
		ATM	−3.8583	−2.2019	0.0598
		OTM	−1.5283	−0.7173	−0.0503
		DOTM	−12.4665	−4.6198	−0.5796
		VDOTM	−1.5170	−1.1240	−1.2417
	LSTM-DNN	ITM	−7.9047	−4.3267	−0.7580
		ATM	−3.3970	−2.0691	−0.1496
		OTM	−3.7141	−2.2581	−0.4778
		DOTM	−6.5636	−2.8713	−0.4378
		VDOTM	−2.4879	−2.6962	−1.0997

Table C.5
Forecasting S&P500 R^2 performance of put option term structure with 10-historical sequence. This table gives R^2 of the implied volatility predictions under different m and τ combinations. The input for these models consists of historical sequence lengths of 10 days. we provides the models which R^2 values are not reported in Section 5.4. Specifically, it includes the Ridge, Lasso, XGBoost, and LSTM-DNN models under the DVF method, as well as all models except DNN employing the kernel method. The evaluation utilizes implied volatility data from S&P500 index call options spanning from 2012 to 2022. Model training encompasses data ranging from 2012 to 2013, with the remaining data reserved for out-of-sample evaluation.

Interpolation method	Models	m	Short-term	Medium-term	Long-term
DVF	Lasso	ITM	−0.9619	−2.2713	−2.0138
		ATM	−3.8888	−6.5928	−2.4041
		OTM	−1.1965	−3.5638	−2.5026
		DOTM	−2.8934	−9.7238	−4.3850
		VDOTM	−4.9106	−19.1808	−7.0200
	Ridge	ITM	0.2188	0.2928	0.4657
		ATM	0.2781	0.4071	0.4443
		OTM	0.2501	0.3257	0.3568
		DOTM	0.0832	−0.2321	0.3171
		VDOTM	0.1255	0.0104	0.2143
	XGBoost	ITM	−0.0346	−0.1428	0.0862
		ATM	−0.3003	−1.5789	−0.3060
		OTM	−0.3258	−2.3763	−0.3873
		DOTM	−0.1591	−0.5879	−0.4023
		VDOTM	−0.8174	−2.0511	−2.0755
	LSTM-DNN	ITM	0.0097	0.1922	0.2649
		ATM	−0.1713	−0.7289	−0.2056
		OTM	−0.3054	−1.0153	−0.4192
		DOTM	−0.6870	−2.0182	−1.6168
		VDOTM	−0.8769	−1.9683	−1.9077
Kernel	DNN	ITM	−8.1031	−0.5347	0.1279
		ATM	−1.7435	0.2941	0.4546
		OTM	−3.5656	−0.4251	−0.2742
		DOTM	−6.1261	−0.1058	0.3055
		VDOTM	−0.8631	0.1107	−0.0663
	Lasso	ITM	−60.6359	−34.3785	−9.4904
		ATM	−22.4288	−14.0264	−3.2018
		OTM	−20.7484	−10.8663	−1.7783
		DOTM	−47.4442	−25.6696	−8.0562
		VDOTM	−11.4244	−9.5001	−3.1268
	XGBoost	ITM	−7.5762	−3.0687	−0.1159
		ATM	−1.2630	−0.2960	0.3981
		OTM	−0.2661	−0.1600	0.0341
		DOTM	−7.5157	−2.8387	−0.1085
		VDOTM	−0.4032	−0.7493	−0.9303
	LSTM-DNN	ITM	−4.8210	−3.1653	−1.0179
		ATM	−0.7688	−0.8548	−0.0646
		OTM	−3.1694	−3.2362	−0.9273
		DOTM	−3.3055	−1.9914	−0.5894
		VDOTM	−0.5697	−1.1022	−1.2171

Table C.6
Forecasting S&P500 R^2 performance of put option term structure with 20-historical sequence. This table gives R^2 of the implied volatility predictions under different m and τ combinations. The input for these models consists of historical sequence lengths of 20 days. we provides the models which R^2 values are not reported in Section 5.4. Specifically, it includes the Ridge, Lasso, XGBoost, and LSTM-DNN models under the DVF method, as well as all models except DNN employing the kernel method. The evaluation utilizes implied volatility data from S&P500 index call options spanning from 2012 to 2022. Model training encompasses data ranging from 2012 to 2013, with the remaining data reserved for out-of-sample evaluation.

Interpolation method	Models	m	Short-term	Medium-term	Long-term
DVF	Lasso	ITM	−0.8937	−2.1799	−1.9264
		ATM	−3.6068	−6.5797	−2.2938
		OTM	−1.0970	−3.4506	−2.3827
		DOTM	−2.6253	−9.2118	−4.0486
		VDOTM	−4.4628	−18.2113	−6.5511
	Ridge	ITM	0.2845	0.5245	0.5612
		ATM	0.2988	0.4752	0.4895
		OTM	0.2044	0.2231	0.4619
		DOTM	0.2653	0.3660	0.3901
		VDOTM	0.2180	0.1572	0.2422
	XGBoost	ITM	−0.0550	−0.1387	0.1453
		ATM	−0.0965	−0.7326	−0.0908
		OTM	−0.1160	−1.2110	−0.2550
		DOTM	−0.1139	−0.7649	−0.6298
		VDOTM	−0.6963	−2.0575	−1.9565
	LSTM-DNN	ITM	−0.2912	−0.0814	0.0808
		ATM	−1.4568	−2.5746	−0.7264
		OTM	−1.0118	−1.9837	−0.7973
		DOTM	−0.6776	−1.7284	−1.4629
		VDOTM	−0.6523	−1.4880	−1.5620
Kernel	DNN	ITM	−2.7685	−1.1238	−0.5669
		ATM	0.4453	0.3737	0.3878
		OTM	−5.0760	−3.8010	−0.9842
		DOTM	−1.3714	−0.3747	−0.2050
		VDOTM	0.6037	0.3650	−0.0128
	Lasso	ITM	−64.8897	−37.5209	−9.9435
		ATM	−22.8088	−14.3777	−3.3082
		OTM	−22.0669	−11.9295	−1.8367
		DOTM	−49.6408	−27.1303	−8.3733
		VDOTM	−11.6777	−9.8303	−3.1907
	XGBoost	ITM	−4.8213	−1.9308	0.1742
		ATM	−0.3035	0.2867	0.6910
		OTM	−0.1024	−0.0494	0.0546
		DOTM	−4.9635	−1.5526	0.1593
		VDOTM	0.3434	0.3186	−0.0609
	LSTM-DNN	ITM	−18.7342	−11.2490	−1.8896
		ATM	−5.3038	−3.4138	−0.1194
		OTM	−9.1859	−4.7780	−1.0465
		DOTM	−14.6746	−6.6815	−1.2995
		VDOTM	−2.8495	−2.3509	−0.7267

Table C.7
Forecasting SSE 50 ETF R^2 performance of call option term structure with 5-historical sequence. This table gives R^2 of the implied volatility predictions under different m and τ combinations. The input for these models consists of historical sequence lengths of 5 days. we provides the models which R^2 values are not reported in Section 5.4. we include the results of all models for call options using the DVF interpolation method. The evaluation utilizes implied volatility data from SSE 50 ETF index call options spanning from 2015 to 2022. Model training encompasses data ranging from 2015 to 2016, with the remaining data reserved for out-of-sample evaluation.

Input historical sequence length	Models	m	Short-term	Medium-term	Long-term
DVF	Lasso	ATM	0.0279	−13.3743	−11.5627
		DITM	−0.2978	−1.3448	−1.4701
		DOTM	−0.4048	−5.6138	−4.7249
		ITM	0.0159	−15.8090	−12.1497
		OTM	−0.0527	−18.4104	−12.3996
	Ridge	ATM	0.7563	0.1816	0.1589
		DITM	0.7594	0.8039	0.7885
		DOTM	0.7509	0.5359	0.5555
		ITM	0.7516	0.0594	0.1224
		OTM	0.7621	−0.0859	0.0833
	Elastic	ATM	0.3927	−6.9692	−6.1270
		DITM	0.2184	−0.3322	−0.4059
		DOTM	0.1661	−2.7339	−2.2734
		ITM	0.3843	−8.3028	−6.5109
		OTM	0.3537	−9.6946	−6.5554
	XGBoost	ATM	−0.1778	−1.0100	−1.3091
		DITM	−0.8368	−1.7521	−1.7701
		DOTM	−0.8004	−4.4389	−4.3955
		ITM	−0.1953	−1.4673	−1.5302
		OTM	−0.2838	−2.1274	−1.7949
	DNN	ATM	0.5799	0.1546	−0.0043
		DITM	0.3676	0.0959	0.0017
		DOTM	0.4779	−0.5778	−0.7658
		ITM	0.5670	−0.0039	−0.0372
		OTM	0.5799	−0.2665	−0.4876
	LSTM-DNN	ATM	0.3128	−0.0798	−0.8026
		DITM	−0.2317	−1.0077	−1.1541
		DOTM	−0.1037	−2.2741	−2.8109
		ITM	0.3005	−0.3425	−1.0148
		OTM	0.2485	−1.9405	−2.2216
Kernel	Lasso	ATM	0.7556	0.7908	0.6965
		DITM	0.8393	0.8272	0.5086
		DOTM	0.6385	0.6152	0.6569
		ITM	0.7971	0.8182	0.6773
		OTM	0.6542	0.7027	0.6513
	Elastic	ATM	0.8072	0.8556	0.7542
		DITM	0.8616	0.8536	0.5552
		DOTM	0.7552	0.7926	0.7416
		ITM	0.8342	0.8682	0.7306
		OTM	0.7515	0.8221	0.7499
	XGBoost	ATM	0.9017	0.9123	0.7823
		DITM	0.8896	0.8786	0.5184
		DOTM	0.8866	0.8849	0.8076
		ITM	0.9050	0.9154	0.7411
		OTM	0.8934	0.9004	0.8106
	LSTM-DNN	ATM	0.8787	0.9146	0.7947
		DITM	0.8818	0.8815	0.5057
		DOTM	0.8474	0.9024	0.8120
		ITM	0.8881	0.9193	0.7520
		OTM	0.8554	0.9033	0.8215

Table C.8
Forecasting SSE 50 ETF R^2 performance of call option term structure with 10-historical sequence. This table gives R^2 of the implied volatility predictions under different m and τ combinations. The input for these models consists of historical sequence lengths of 10 days. we provides the models which R^2 values are not reported in Section 5.4. we include the results of all models for call options using the DVF interpolation method. The evaluation utilizes implied volatility data from SSE 50 ETF index call options spanning from 2015 to 2022. Model training encompasses data ranging from 2015 to 2016, with the remaining data reserved for out-of-sample evaluation.

Input historical sequence length	Models	m	Short-term	Medium-term	Long-term
DVF	Lasso	ATM	0.0352	−13.5748	−11.6069
		DITM	−0.2842	−1.3236	−1.4476
		DOTM	−0.4061	−5.5931	−4.6786
		ITM	0.0239	−15.9678	−12.2813
		OTM	−0.0480	−18.7513	−12.3862
	Ridge	ATM	0.7589	0.3032	0.2644
		DITM	0.7652	0.8217	0.8067
		DOTM	0.7541	0.5879	0.6027
		ITM	0.7548	0.2038	0.2412
		OTM	0.7642	0.0707	0.1953
	Elastic	ATM	0.3973	−7.1076	−6.1763
		DITM	0.2249	−0.3256	−0.3983
		DOTM	0.1649	−2.7342	−2.2602
		ITM	0.3895	−8.4223	−6.6150
		OTM	0.3565	−9.9022	−6.5735
	XGBoost	ATM	−0.2384	−4.5574	−4.0975
		DITM	−0.9023	−2.2005	−2.1912
		DOTM	−0.8882	−6.2286	−5.5028
		ITM	−0.2611	−5.4559	−4.5862
		OTM	−0.3609	−9.7998	−6.9573
	DNN	ATM	0.4262	−1.0116	−5.0091
		DITM	0.1801	−0.9692	−1.1450
		DOTM	0.2228	−4.9736	−4.1903
		ITM	0.4121	−2.1997	−6.3488
		OTM	0.4092	−7.7843	−8.1618
	LSTM-DNN	ATM	0.2402	−4.8525	−5.2014
		DITM	−0.3288	−1.4788	−1.8367
		DOTM	−0.2529	−3.8717	−4.0373
		ITM	0.2322	−6.4258	−6.0933
		OTM	0.1598	−13.5758	−11.1609
Kernel	Lasso	ATM	0.7438	0.7853	0.6861
		DITM	0.8353	0.8231	0.5044
		DOTM	0.6282	0.6086	0.6508
		ITM	0.7883	0.8135	0.6670
		OTM	0.6386	0.6962	0.6397
	Elastic	ATM	0.7998	0.8521	0.7455
		DITM	0.8592	0.8513	0.5537
		DOTM	0.7494	0.7883	0.7371
		ITM	0.8285	0.8652	0.7228
		OTM	0.7423	0.8181	0.7407
	XGBoost	ATM	0.8963	0.9069	0.7849
		DITM	0.8846	0.8744	0.5420
		DOTM	0.8816	0.8793	0.8160
		ITM	0.9033	0.9150	0.7455
		OTM	0.8893	0.8991	0.8192
	LSTM-DNN	ATM	0.8890	0.9059	0.7845
		DITM	0.8900	0.8759	0.4831
		DOTM	0.8628	0.8711	0.8332
		ITM	0.8982	0.9122	0.7432
		OTM	0.8671	0.8866	0.8093

Table C.9
Forecasting SSE 50 ETF R^2 performance of call option term structure with 20-historical sequence. This table gives R^2 of the implied volatility predictions under different m and τ combinations. The input for these models consists of historical sequence lengths of 20 days. we provides the models which R^2 values are not reported in Section 5.4. we include the results of all models for call options using the DVF interpolation method. The evaluation utilizes implied volatility data from SSE 50 ETF index call options spanning from 2015 to 2022. Model training encompasses data ranging from 2015 to 2016, with the remaining data reserved for out-of-sample evaluation.

Input historical sequence length	Models	m	Short-term	Medium-term	Long-term
DVF	Lasso	ATM	0.0407	−14.0000	−11.7159
		DITM	−0.2714	−1.2993	−1.4160
		DOTM	−0.4040	−5.5221	−4.6041
		ITM	0.0301	−16.3158	−12.4239
		OTM	−0.0439	−19.2970	−12.5823
	Ridge	ATM	0.7653	0.3623	0.3514
		DITM	0.7725	0.8308	0.8171
		DOTM	0.7591	0.6227	0.6338
		ITM	0.7616	0.2754	0.3270
		OTM	0.7700	0.1415	0.2711
	Elastic	ATM	0.4015	−7.4058	−6.2675
		DITM	0.2311	−0.3191	−0.3872
		DOTM	0.1659	−2.7115	−2.2336
		ITM	0.3940	−8.6833	−6.7353
		OTM	0.3595	−10.2328	−6.7048
	XGBoost	ATM	−0.1145	−1.7662	−0.9670
		DITM	−0.6495	−0.9698	−0.8832
		DOTM	−0.6918	−2.7319	−2.4397
		ITM	−0.1286	−1.8154	−0.9527
		OTM	−0.2283	−6.7692	−2.9035
	DNN	ATM	0.4815	0.4234	0.2544
		DITM	0.4308	0.6655	0.6281
		DOTM	0.4180	0.2297	0.2430
		ITM	0.4684	0.3522	0.3470
		OTM	0.4863	−0.3709	−0.1743
	LSTM-DNN	ATM	0.3753	−2.2454	−1.9029
		DITM	−0.0197	−0.7034	−0.9582
		DOTM	0.0251	−2.3426	−2.7234
		ITM	0.3750	−2.7282	−1.6885
		OTM	0.3090	−4.5414	−3.5487
Kernel	Lasso	ATM	0.7189	0.7772	0.6760
		DITM	0.8271	0.8182	0.4977
		DOTM	0.6059	0.6029	0.6434
		ITM	0.7696	0.8062	0.6574
		OTM	0.6056	0.6874	0.6305
	Elastic	ATM	0.7842	0.8486	0.7391
		DITM	0.8530	0.8490	0.5499
		DOTM	0.7369	0.7873	0.7341
		ITM	0.8165	0.8619	0.7173
		OTM	0.7230	0.8152	0.7358
	XGBoost	ATM	0.8997	0.9030	0.7722
		DITM	0.8881	0.8709	0.5482
		DOTM	0.8893	0.8844	0.8150
		ITM	0.9051	0.9089	0.7320
		OTM	0.8935	0.9032	0.8110
	LSTM-DNN	ATM	0.8970	0.9105	0.7860
		DITM	0.8849	0.8598	0.5013
		DOTM	0.8541	0.8785	0.8283
		ITM	0.9031	0.9145	0.7503
		OTM	0.8782	0.8993	0.8089

Table C.10
Forecasting SSE 50 ETF R^2 performance of put option term structure with 5-historical sequence. This table gives R^2 of the implied volatility predictions under different m and τ combinations. The input for these models consists of historical sequence lengths of 5 days. we provides the models which R^2 values are not reported in Section 5.4. we include the results of all models for call options using the DVF interpolation method. The evaluation utilizes implied volatility data from SSE 50 ETF index call options spanning from 2015 to 2022. Model training encompasses data ranging from 2015 to 2016, with the remaining data reserved for out-of-sample evaluation.

Input historical sequence length	Models	m	Short-term	Medium-term	Long-term
DVF	Lasso	ATM	0.4653	−3.9377	−3.1586
		DOTM	0.3948	0.2657	0.2318
		ITM	0.2080	−1.0114	−0.6908
		OTM	0.4672	−4.6147	−2.5102
		VDOTM	0.3874	−5.6607	−2.9088
	Ridge	ATM	0.7707	0.7061	0.6367
		DOTM	0.7875	0.8860	0.8747
		ITM	0.7743	0.7773	0.7659
		OTM	0.7670	0.6760	0.6702
		VDOTM	0.7764	0.5896	0.5625
	Elastic	ATM	0.5665	−2.3662	−1.9202
		DOTM	0.5298	0.4925	0.4659
		ITM	0.3930	−0.3687	−0.1696
		OTM	0.5672	−2.8089	−1.4515
		VDOTM	0.5145	−3.5127	−1.7141
	XGBoost	ATM	−0.0485	0.1965	0.0549
		DOTM	−0.6373	−1.8995	−1.9684
		ITM	−0.6054	−5.2042	−4.6855
		OTM	−0.0650	0.0699	0.0326
		VDOTM	−0.1472	−1.8972	−1.3406
	DNN	ATM	0.6853	0.6953	0.5969
		DOTM	0.6371	0.6261	0.5487
		ITM	0.6200	0.6869	0.6620
		OTM	0.6839	0.6754	0.6186
		VDOTM	0.5755	0.4643	0.4041
	LSTM-DNN	ATM	0.3122	0.6895	0.2088
		DOTM	0.0582	−0.3076	−0.6005
		ITM	0.0186	−0.3254	−0.6885
		OTM	0.3054	0.6655	0.3221
		VDOTM	−0.4323	−1.2920	−1.4927
Kernel	Lasso	ATM	0.3001	0.3428	0.5570
		DOTM	0.3473	0.3667	0.5469
		ITM	0.3667	0.3845	0.5403
		OTM	0.3300	0.3433	0.4947
		VDOTM	0.3514	0.3507	0.4404
	Elastic	ATM	0.6135	0.6403	0.6920
		DOTM	0.6470	0.6587	0.6866
		ITM	0.6592	0.6703	0.6753
		OTM	0.6379	0.6572	0.6381
		VDOTM	0.6533	0.6593	0.6053
	XGBoost	ATM	0.8623	0.8702	0.8055
		DOTM	0.8822	0.8832	0.7502
		ITM	0.8888	0.8917	0.7326
		OTM	0.8757	0.8883	0.7263
		VDOTM	0.8604	0.8676	0.6143
	LSTM-DNN	ATM	0.8901	0.9049	0.8592
		DOTM	0.8960	0.9127	0.8432
		ITM	0.8934	0.9012	0.8213
		OTM	0.8978	0.9040	0.8018
		VDOTM	0.8974	0.9238	0.7947

Table C.11
Forecasting SSE 50 ETF R^2 performance of put option term structure with 10-historical sequence. This table gives R^2 of the implied volatility predictions under different m and τ combinations. The input for these models consists of historical sequence lengths of 10 days. we provides the models which R^2 values are not reported in Section 5.4. we include the results of all models for call options using the DVF interpolation method. The evaluation utilizes implied volatility data from SSE 50 ETF index call options spanning from 2015 to 2022. Model training encompasses data ranging from 2015 to 2016, with the remaining data reserved for out-of-sample evaluation.

Input historical sequence length	Models	m	Short-term	Medium-term	Long-term
DVF	Lasso	ATM	0.4694	−3.8999	−3.1107
		DOTM	0.3989	0.2691	0.2328
		ITM	0.2009	−1.0067	−0.6789
		OTM	0.4719	−4.6383	−2.4818
		VDOTM	0.3879	−5.6488	−2.8592
	Ridge	ATM	0.7786	0.6917	0.6302
		DOTM	0.7945	0.8855	0.8737
		ITM	0.7786	0.7809	0.7683
		OTM	0.7755	0.6594	0.6729
		VDOTM	0.7828	0.5870	0.5644
	Elastic	ATM	0.5767	−2.2711	−1.8252
		DOTM	0.5401	0.5043	0.4762
		ITM	0.3982	−0.3315	−0.1346
		OTM	0.5779	−2.7459	−1.3734
		VDOTM	0.5231	−3.4003	−1.6193
	XGBoost	ATM	−0.0488	−0.3814	0.0607
		DOTM	−0.6116	−1.5195	−1.5009
		ITM	−0.6007	−4.3631	−4.4004
		OTM	−0.0651	−0.5156	0.1154
		VDOTM	−0.1471	−5.9029	−3.0477
	DNN	ATM	0.6762	0.1551	−0.0105
		DOTM	0.6399	0.6197	0.5132
		ITM	0.6448	0.6491	0.5695
		OTM	0.6725	0.1229	0.2259
		VDOTM	0.6179	0.5430	0.4742
	LSTM-DNN	ATM	0.3259	−0.0602	0.0122
		DOTM	0.1238	0.2942	0.2065
		ITM	0.1145	0.2687	0.0376
		OTM	0.3149	−0.2368	0.1779
		VDOTM	−0.4424	−1.1231	−1.1139
Kernel	Lasso	ATM	0.1556	0.2135	0.5075
		DOTM	0.2209	0.2521	0.4887
		ITM	0.2509	0.2798	0.4835
		OTM	0.2405	0.2400	0.4455
		VDOTM	0.2531	0.2515	0.3750
	Elastic	ATM	0.5440	0.5768	0.6658
		DOTM	0.5874	0.6034	0.6562
		ITM	0.6047	0.6200	0.6458
		OTM	0.5975	0.6091	0.6141
		VDOTM	0.6077	0.6124	0.5729
	XGBoost	ATM	0.8817	0.8828	0.7634
		DOTM	0.8795	0.8839	0.7470
		ITM	0.8729	0.8747	0.7359
		OTM	0.8443	0.8495	0.6959
		VDOTM	0.8542	0.8691	0.6463
	LSTM-DNN	ATM	0.8928	0.8982	0.8419
		DOTM	0.8944	0.9028	0.8161
		ITM	0.8972	0.8969	0.7933
		OTM	0.9005	0.8990	0.7693
		VDOTM	0.8838	0.8795	0.7268

Table C.12
Forecasting SSE 50 ETF R^2 performance of put option term structure with 20-historical sequence. This table gives R^2 of the implied volatility predictions under different m and τ combinations. The input for these models consists of historical sequence lengths of 20 days. we provides the models which R^2 values are not reported in Section 5.4. we include the results of all models for call options using the DVF interpolation method. The evaluation utilizes implied volatility data from SSE 50 ETF index call options spanning from 2015 to 2022. Model training encompasses data ranging from 2015 to 2016, with the remaining data reserved for out-of-sample evaluation.

Input historical sequence length	Models	m	Short-term	Medium-term	Long-term
DVF	Lasso	ATM	0.4714	−3.7620	−3.0363
		DOTM	0.4005	0.2765	0.2405
		ITM	0.1858	−0.9862	−0.6576
		OTM	0.4746	−4.4667	−2.4175
		VDOTM	0.3838	−5.6494	−2.8178
	Ridge	ATM	0.7922	0.7510	0.6981
		DOTM	0.8041	0.8899	0.8790
		ITM	0.7888	0.7991	0.7848
		OTM	0.7893	0.7267	0.7274
		VDOTM	0.7954	0.6613	0.6284
	Elastic	ATM	0.5879	−2.0793	−1.6886
		DOTM	0.5515	0.5208	0.4936
		ITM	0.4045	−0.2853	−0.0945
		OTM	0.5893	−2.5152	−1.2636
		VDOTM	0.5324	−3.2637	−1.5145
	XGBoost	ATM	0.1216	0.1324	−0.0093
		DOTM	−0.3889	−0.7357	−0.6862
		ITM	−0.3109	−1.9139	−2.0350
		OTM	0.1090	−0.0811	0.0756
		VDOTM	0.0519	−1.1645	−0.5972
	DNN	ATM	0.6009	0.8115	0.5422
		DOTM	0.5394	0.6657	0.5627
		ITM	0.5364	0.6409	0.5333
		OTM	0.5955	0.8155	0.6527
		VDOTM	0.4765	0.4825	0.3902
	LSTM-DNN	ATM	0.3577	−0.0570	0.1082
		DOTM	0.1463	0.2734	0.1171
		ITM	0.1184	0.3383	0.2023
		OTM	0.3471	−0.1160	0.2800
		VDOTM	−0.3197	−0.4512	−0.4913
Kernel	Lasso	ATM	−0.2079	−0.0898	0.4277
		DOTM	−0.0986	−0.0222	0.3784
		ITM	−0.0361	0.0336	0.3724
		OTM	0.0467	0.0254	0.3389
		VDOTM	0.0320	0.0400	0.2387
	Elastic	ATM	0.4046	0.4639	0.6395
		DOTM	0.4699	0.5066	0.6324
		ITM	0.4993	0.5331	0.6234
		OTM	0.5325	0.5392	0.5900
		VDOTM	0.5277	0.5380	0.5420
	XGBoost	ATM	0.8709	0.8752	0.7982
		DOTM	0.8657	0.8796	0.7753
		ITM	0.8633	0.8712	0.7059
		OTM	0.7648	0.6701	0.5796
		VDOTM	0.8310	0.7707	0.4162
	LSTM-DNN	ATM	0.8175	0.8503	0.8021
		DOTM	0.8227	0.8392	0.7886
		ITM	0.8084	0.8536	0.7866
		OTM	0.8194	0.8355	0.6990
		VDOTM	0.8176	0.8370	0.6254

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